

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a195a4aa3eb04c735.jsonl`  
- SHA-256: `5f9c906eae41d04afd5fdd8821f737c6e8ddd20ed2012abe974feed7cc93daac`  
- Source modified: `2026-06-10T06:10:10+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-10T05:59:28.042Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

TASK: deep ENGINEERING + METHODOLOGY audit of the indexer's eval/calibration/learning stack.

Read docs/CODE_MAP.md section 2.3 first, then deep-read: backend/eval/ (metrics.py, annotations.py, calibration.py, harness.py, batch.py, regression.py, training.py, classifier.py, calibrate_wasb.py, calibrate_local.py, distill_local.py, export_wasb_segments.py, gemini_refine.py, eval_partial_labels.py, rally_seq_proto.py, audio/e1_probe.py), backend/tools/rally_slicer.py, run_eval.py, run_phase3_eval.py, run_regression.py, run_process_and_stitch.py.

ASSESS: (1) statistical/methodology soundness ? greedy IoU matching, LOVO honesty, partial-label window restriction, overfit guards, the numpy-only LogisticRegression, rally_seq_proto's feature design (incl. the known fps-non-normalization), threshold-transfer problem; (2) is the regression gate (run_regression.py) actually a usable CI gate ? semantics, silent-pass paths, baseline handling; (3) code quality ? duplication of constants (BASELINE/TUNED/paddings), shared de-facto APIs, fragile imports; (4) correctness bugs in metrics/scoring (off-by-one, empty-set handling, division by zero); (5) whether this stack can honestly support the owned-model Phase-0 plan (Gen-0 TAL harness): what is missing or untrustworthy.

Severity-rate honestly. New findings beat re-reporting CODE_MAP gotchas. Quantified claims (e.g. 'LOVO with n=6 means $\pm X$ noise') should be reasoned, not hand-waved.

2. user (2026-06-10T05:59:33.536Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-255 of 330 total (27423 tokens, cap 25000). Call Read with offset=256 limit=255 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1      # CODE_MAP.md
2
3      **Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-
hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a
segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.
4
5      **Regenerate:** rerun the `build-code-map` workflow (parallel subsystem readers ? this
doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure
(this revision follows #40).
6
7      **LEGEND**
8      - `@path` = file (always repo-relative)
9      - `::sym` = symbol (class/func/const) in the nearest preceding `@path`
10     - `->` = dataflow / call / "produces"
11     - `$` = data contract (canonical shape)
12     - `?` = gotcha / footgun
13     - `?' = registry / plugin extension point
14     - `WSL` = runs inside WSL2 conda env, NOT the Windows Python process
15
16     ---
17
18     ## 1. PIPELINE
19
20     ### 1A. Runtime: video file -> highlight reel
21     ```
22     typed config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-
defaults, never fatal)
23                                     @backend/config/loader.py::load_config ->
@backend/config/models.py::AppConfig
24     -> video_id = MD5(whole file)          @backend/utils/hashing.py::compute_video_id
(single shared helper, 64KB chunks; imported by main + all run_*.py ? no more per-entrypoint
copies)
25     -> validate
@backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)
26     FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity
-> SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see
@backend/pipeline/validators/__init__.py)
27     [validation FAIL -> persist status="failed", return early; report STILL emitted
in finally]
28     -> db.add_video(video_id, filepath, status, [r.dict()])
@backend/infrastructure/database.py
29     -> seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback
'motion')
30     -> seg = segmenter_registry.get(seg_name)(db, global_config) ?
@backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)
31     -> segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)
32     [STOP POINT: segments-only ? predictions now in DB; eval/regression operate here
without stitching]
33     -> finally: emit_indexer_report(...) @backend/reporting/__init__.py (? ALWAYS runs
? even on validation failure / exception; NEVER raises; grafts response["reports"])
34     -> select segment ids -> [(start,end)] (+ stitching padding)
35     -> VideoCompiler.compile_highlights(filepath, time_pairs, out_name)
@backend/pipeline/stitcher/compiler.py
36     per-clip ffmpeg re-encode (libx264/superfast/crf22) -> concat -c copy ->
output/<name>.mp4
37     ```
38     Hybrid segmenter internal 4-phase (yolo_hybrid / tracknet_hybrid / wasb_hybrid ? same
shape):
39     ```
40     P1 chunk video (yolo only; tracknet/wasb = whole-video single pass, chunk_index=0)
41     -> P2 candidate "action windows" from motion/trajectory [STOP: candidates persisted
via db.add_candidate_segment]
42     -> P3 pad clip (ai_handoff_padding) -> ThreadPoolExecutor ->
provider.analyze_video(clip, prompt) ? provider_registry
43     -> P4 db.add_play_segment + db.link_candidate_to_segment

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44     ``
45     Pure-AI path `gemini`: no P2; chunk-or-single -> provider -> heavy validate
(clamp/dedup) -> add_play_segment.
46     Legacy `motion`: pixel-diff -> segments + **fabricated** metadata/events (? not ground
truth).
47
48     ### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
49     ``
50     WasbHybridSegmenter.process_video
@backend/pipeline/segmenters/wasb_hybrid.py
51     -> WasbRunner.healthcheck() (gate; not ok -> return [])
@backend/pipeline/detectors/wasb_runner.py
52     -> WasbRunner.run_predict(video, out_dir)
53     stage video + wasb_infer.py into WSL fs -> `wsl bash -c 'source conda.sh && python
wasb_infer.py ...'`
54     -> @backend/pipeline/detectors/wasb_infer.py::run (WSL): sliding windows -> GPU
detector pass (CHECKPOINTED det_raw.jsonl) -> CPU tracker (always full re-run) -> trajectory CSV
55     -> TrackNetRunner.parse_trajectory_csv(csv)
@backend/pipeline/detectors/tracknet_runner.py (shared, re-exported by WasbRunner)
56     -> TrackNetRunner.trajectory_to_action_windows(points, fps, frame_width, chunk_start,
velocity_thresh, merge_gap, min_window_duration, chunk_index) [the model-agnostic WINDOWING
BRAIN]
57     ``
58     (`tracknet_hybrid` identical, swapping WasbRunner->TrackNetRunner; `_probe_fps_width`
fallback (30.0, 1280.0).)
59
60     ### 1C. Calibration / eval flow (NO pipeline run unless told)
61     ``
62     GT CSV -> annotations.load_annotations / calibrate_wasb.load_golden_gts ->
List[Interval]
63     DB predictions -> harness.get_predictions(db, video_id, stage) -> List[Interval] stage
? {candidates, final}
64     -> metrics.match_intervals (greedy temporal-IoU) -> metrics.score -> Score
65     -> harness.evaluate -> EvalReport -> report_to_dict
66     -> batch.aggregate (corpus micro/macro) @backend/eval/batch.py
67     -> regression.regress(_with_provider) vs baseline @backend/eval/regression.py
[CI gate]
68     Calibration: calibrate_wasb.run -> calibration.{select_best, improvement_report,
cross_validate} @backend/eval/calibration.py
69     Distill Gemini->local: calibrate_local (thresholds) OR distill_local (learned
classifier) @backend/eval/{calibrate_local,distill_local}.py
70     Learned segmenter (offline): training.build_training_set(X,y) ->
classifier.LogisticRegression.fit/save
71     Gemini refinement driver (tuning, standalone):
gemini_refine.{refine_window,full_video_rallies} @backend/eval/gemini_refine.py
72     Audio probe (diagnostic only, NOT in live pipeline): audio/e1_probe.run
@backend/eval/audio/e1_probe.py
73     ``
74     Entrypoints: `@run_eval.py` (single video score), `@run_phase3_eval.py` (manifest batch,
can segment), `@run_regression.py` (baseline gate, exit 1=regression),
`@backend/eval/calibrate_wasb.py` + `@backend/eval/export_wasb_segments.py` (WASB tuning
drivers).
75
76     ---
77
78     ## 2. SUBSYSTEMS
79
80     ### 2.0 Orchestration & config
81     - `@backend/main.py` ? Thin CLI + FastAPI bootstrap (orchestrator; modes via FLAGS).
Static UI mounts `/static` from `backend/api/static`. Logging is initialized with `basicConfig`
(INFO, timestamped format); backend modules use `logger = logging.getLogger(__name__)` (replaces
stray prints in hot paths like motion, stitcher, wasb_infer, config loader). User-facing CLI
summaries may still use `print` for direct output.
82     - `::app` FastAPI; `::db_instance`, `::global_config` (`Optional[AppConfig]`) module
globals set ONLY in `main()`. ? importing `app` for ASGI without `main()` -> both `None` ->

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every endpoint NPEs.

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83     - `::process_video` `@POST /api/process` -> validate -> add_video -> segment ->
`finally:` always `emit_indexer_report` (never raises). `::compile_highlights` `@POST
/api/compile`. `::query_play_segments` `@POST /api/query`. `::get_config` `@GET /api/config`.
`::run_cli_diagnose_clip` `--debug-clip`: lazy `import cv2`, samples  $\pm 3s$  vs
`indexing.motion_threshold` (dual model/dict read).
84     - `::load_config` ? ? legacy thin wrapper returning a dict
(`load_app_config(path).model_dump(...)`); kept for transition ? new code imports
`load_app_config`/`AppConfig` directly. `::ProcessRequest`/`::QueryRequest`/`::CompileRequest`
pydantic bodies.
85     - `main()` sets `global_config.output.output_dir = args.output_dir` (typed field on
`OutputConfig`); `db_path = os.path.join(global_config.output.output_dir, output.db_name)`.
`compile_highlights` reads the SAME `global_config.output.output_dir`, so `--output-dir` is
honored end-to-end (the old write-only `_runtime_overrides` hack is gone).
86     - ? `--segmenter` argparse `choices` frozen at build time from
`segmenter_registry.list_available()` ? a lazily-registered segmenter wouldn't appear. CLI
default `segmenter` fallback `motion`.
87     - `@run_process_and_stitch.py` ? one-shot demo; ? hardcoded paths, interactive `input()`
(blocks automation), skips validation. Config via `backend.config.load_config` (typed);
segmenter, padding, and db path all read from the model (no literal fallbacks).
88     - `@run_eval.py::main` ? score one video's DB preds vs GT CSV.
`--stage{candidates,final,both}`. Exit 2=arg/IO. ? pure scorer; errors if video never processed.
`::default_db_path` resolves via `backend.config.load_config`
(`output.output_dir`/`output.db_name`). `get_file_md5` is an alias for `compute_video_id`.
89     - `@run_phase3_eval.py::main` ? walk manifest, segment-if-needed, score, aggregate.
`load_dotenv()` at import. ? DB key = file MD5, NOT manifest's `video_id` (kept as label);
config via `backend.config.load_config`; `segmenter_cls(db, cfg)` receives the typed `AppConfig`
(same object the live pipeline feeds segmenters).
90     - `@run_regression.py::main` ? `regress_with_provider` vs baseline. Exit codes semantic:
0=ok, 1=REGRESSION (CI gate), 2=missing DB**. ? `--update-baseline` runs AFTER compare
(snapshots even a regressing run); `::default_db_path` resolves via
`backend.config.load_config`.
91
92     - `@backend/config/__init__.py` ? ? canonical config import surface. Re-exports
`load_config`, `save_default_config` (from `.loader`) + `AppConfig`, `Config`, `IndexingConfig`,
`ValidationConfig` (from `.models`); `__all__` = exactly those 6. Use `from backend.config
import load_config, AppConfig`.
93     - `@backend/config/models.py` ? Pydantic v2 models; single source of truth for
defaults**.
94     - `::AppConfig` ? root. Scalars `sport='badminton'`, `ai_provider='gemini'`,
`ai_model='gemini-2.5-flash'`; sections `validation`, `indexing`, `output`, `stitching`
(`default_factory`). `model_config = {extra:'allow', validate_assignment:True}`. ?
`extra='allow'` -> unknown/typo'd config.json keys silently kept (legacy tolerance).
95     - ? `::AppConfig.get` ? legacy dict-compat shim: returns nested sections as plain
dicts* (`model_dump(mode='json')`, NOT the model) so chained
`config.get("indexing",{}).get(...)` keeps working; falls back to searching a leaf key in any
section (sections discovered dynamically from `model_fields`). ? "bit every validator on real
runs" per docstring ? returning the model breaks `.get`. `::AppConfig.__getitem__` raises
`KeyError` for unknown top-level keys else delegates to `.get`.
96     - `::IndexingConfig` ? segmenter/detector knobs (`motion_threshold=0.08`,
`min_segment_duration=3.0`, `dead_time_merge_gap=2.0`, `chunk_duration=90.0`,
`chunk_overlap=10.0`, `yolo_velocity_threshold=0.15`, `yolo_frame_skip=3`,
`ai_handoff_padding=2.0`, `ai_max_workers=5`, `log/store_yolo_candidates=True`, nested
`tracknet`). ? `default_segmenter='yolo_hybrid'` HERE but config.json overrides to `gemini`. ?
`::IndexingConfig.segmenter_must_be_registered` `@field_validator` is a no-op pass-through ?
a typo'd segmenter validates fine, only fails later at `segmenter_registry.get(...)`.
97     - `::ValidationConfig` (`min_resolution_width=1280`, `min_resolution_height=720`, `min_fps=29.9`,
`min_blur_threshold=20.0`, `max_scene_cuts_per_minute=0.5`, nested `relevance`);
`::ValidationRelevanceConfig` (court HSV gates, `court_pixels_min_ratio=0.05`).
`::TrackNetConfig` (WSL invocation contract: `wsl_distro='Ubuntu'`, `repo_dir`,
`conda_env='TrackNetV4'`, `weights_path='', `queue_length=5`, `velocity_threshold=0.02`).
`::OutputConfig` (`default_highlights_name`, `db_name='sports_indexer.db'`,
`output_dir='output'` ? CLI `--output-dir` overrides it on the typed model in `main()`).
`::StitchingConfig`
(`begin_padding=0.5`, `end_padding=0.5`, `use_cache=False`, `min_shot_count=1`). `::Sport` Literal
```

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of 5 sports. `::Config = AppConfig` alias.
98     - `@backend/config/loader.py` ? load/save with defaults-merge.
99     - `::DEFAULT_CONFIG_PATH = Path("config.json")` (? CWD-relative, NOT repo-root).
`::get_builtin_defaults` = `AppConfig().model_dump(...)`
100    - `::load_config` ? precedence built-in-defaults -> file overrides. ? **shallow top-
level merge** (`{**defaults, **file_data}`): a section in config.json REPLACES the whole
defaults dict for that section (Pydantic then re-applies field defaults for missing leaves).
Missing/unreadable/parse-error file -> fully-defaulted AppConfig + `[CONFIG WARNING]` (non-
fatal).
101    - `::save_default_config` ? writes fresh `config.json`, **overwrites** if present
(used by setup.py/--setup). `::get_default_config` ? ? transitional plain-dict alias.
102    - `@config.json` ? on-disk config; mirrors `AppConfig`. ?
**indexing.default_segmenter='gemini'` ** diverges from model default `yolo_hybrid` ? file
wins at runtime.
103    - `@setup.py` ? `::init_default_config` ? delegates to
`backend.config.save_default_config` (single-source defaults; skips if config.json exists).
`::run_pip_install` GPU-aware (`nvidia-smi` -> cu124 else cpu). `::run_diagnostics` (Python?3.8,
ffmpeg/ffprobe on PATH, optional torch+CUDA).
104
105    ### 2.1 Segmenters (under pipeline/segmenters) ?
106    - `@backend/pipeline/segmenters/base.py`
107    - `::BasePlaySegmenter` ABC ? `__init__(db, config)`; abstract
`name`, `description`, `process_video`. & `process_video(video_id, video_path, player_pool=None,
max_frames=None) -> List[dict{id, start_time, end_time, duration, metadata}]`.
108    - `::SegmenterRegistry`, `::segmenter_registry` (singleton). ? **register a new
segmenter**: subclass `BasePlaySegmenter`, call `segmenter_registry.register(MyCls)` at module
bottom, AND import the module in `@backend/pipeline/segmenters/__init__.py` (registration is an
IMPORT side-effect). ? `register()` builds `cls(None, {})` just to read `name` ->
name/description/`__init__` MUST NOT touch `self.db`/`self.config`. ? `list_available()` re-
instantiates + swallows exceptions into "No description available." (a throwing `description` is
masked).
109    - `@backend/pipeline/segmenters/__init__.py` ? imports all 5
(`motion, gemini, yolo_hybrid, tracknet_hybrid, wasb_hybrid`) = the de-facto registration manifest;
re-exports them + `segmenter_registry` via `__all__`.
110    - `@backend/pipeline/segmenters/wasb_hybrid.py::WasbHybridSegmenter` (name
`wasb_hybrid`) ? WSL WASB substrate (MIT; ships pretrained badminton weights, the LIVE
default). Imports `WasbRunner`/`WasbConfig` from `pipeline/detectors/wasb_runner`. Config
`indexing.wasb.velocity_threshold(0.02)` + shared
`dead_time_merge_gap/min_action_window_duration/ai_handoff_padding/ai_max_workers`. ? whole-
video single pass; reuses YOLO-named keys `log_yolo_candidates/store_yolo_candidates`; needs
`{PROVIDER}_API_KEY` (missing -> []); `detection_params` has NO `queue_length` (unlike
tracknet). ? divergent shot_count: `int(shot_count)` inside try with no `is None` short-circuit
(copy-paste drift).
111    - `@backend/pipeline/segmenters/tracknet_hybrid.py::TrackNetHybridSegmenter` (name
`tracknet_hybrid`) ? copy-paste twin of wasb_hybrid; imports `TrackNetRunner`/`TrackNetConfig`
from `pipeline/detectors/tracknet_runner`; `indexing.tracknet.velocity_threshold(0.02)`;
`detection_params` adds `queue_length`. ? any P3/P4 fix must be applied to BOTH (and arguably
yolo_hybrid).
112    - `@backend/pipeline/segmenters/yolo_hybrid.py::HybridYoloSegmenter` (name
`yolo_hybrid`, no YOLO remains ? torchvision detector + supervision ByteTrack; ADR-005).
`::PERSON_CLASS_ID=1` ? (ultralytics used 0; wrong value -> zero detections).
`::lazy_load_model` (lazy `torchvision` import so module/tests load without it),
`::extract_action_windows_from_chunk`, `::make_tracker` (per-chunk ByteTrack ? track IDs
chunk-local), `::DETECTOR_FACTORIES`/`::DEFAULT_DETECTOR='fasterrcnn_resnet50_fpn'`. ?
velocity at `effective_fps = fps/frame_skip` -> `yolo_velocity_threshold` coupled to
`frame_skip`; pipelined-vs-sequential P2 (prefetch ffmpeg overlaps sequential GPU);
`yolo_tracker`/`bytetrack.yaml` is dead/back-compat. ? `gemini_*` config keys are deprecated
aliases still honored.
113    - `@backend/pipeline/segmenters/gemini.py::GeminiPlaySegmenter` (name `gemini`) ?
pure-AI, no P2. ? chunks only if `duration>chunk_duration`; `max_workers=5` hardcoded;
`max_segment_duration=90.0` hardcoded; `debug_duration` silently truncates; `parse_time` accepts
colon timestamps (warns); does NOT persist candidate segments (only hybrids feed the fine-tuning
table).
114    - `@backend/pipeline/segmenters/motion.py::MotionPlaySegmenter` (name `motion`) ?
pixel absdiff baseline. ? **metadata + serve/smash/foul/winner events are RANDOM**

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(random.sample/randint/uniform/choice); only segmenter using `db.add_event` and passing
server/receiver; uses `print()`; honors `max_frames`.
115
116     ### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
117     - `@backend/pipeline/detectors/base.py` ? DetectorRunner ABC (healthcheck() ->
Tuple[bool,str] never-raises; run_predict(...) -> Optional[str] CSV path). Contract deliberately
matches the pre-existing tuple returns and call sites in the hybrids. Rich module docstring
includes "how to add a new runner", Linux-native path notes (the de-risk goal), and guidance on
threading health into reports. Re-exported from `detectors/__init__.py`. This finishes the WSL
de-risk seam (low-regret 6 / review item 1).
118     - `@backend/pipeline/detectors/tracknet_runner.py`
119     - `::TrackNetConfig` (+`.from_indexing_cfg`; ? key drift `wsl_distro` JSON -> `distro`
field), `::TrackNetRunner` ? runs TrackNet `predict.py` in WSL.
120     - `::to_wsl_mnt_path` (`C:\x`->`/mnt/c/x`; ? POSIX/`~` passthrough, UNC unhandled),
`::TrajectoryPoint` (frame,visible,x,y).
121     - `::parse_trajectory_csv` @staticmethod** + `::trajectory_to_action_windows`
@staticmethod** = MODEL-AGNOSTIC brain, re-exported verbatim by WasbRunner. ? `weights_path`
defaults `` -> `run_predict` returns None; predict.py only `.mp4/.avi` + hard-names
`<stem>_predict.csv`; `visible==False` resets prev (occlusion gap breaks track); `fps<=0->30`,
`frame_width<=0->1280` silent; `track_id_count` hardcoded 1; `timeout_sec=0` -> NO timeout
(hangs forever). `::stage_video` "OK" sentinel.
122     - `@backend/pipeline/detectors/wasb_runner.py::WasbRunner` (inherits DetectorRunner)
(+`::WasbConfig`, from `indexing.wasb`; ? `weights_path` HAS a real default
`wasb_badminton_best.pth.tar` ? no required guard) ? stages `wasb_infer.py` (`::INFER_WIN`) +
video into WSL, runs in `wasb` env. `::parse_trajectory_csv`/`::trajectory_to_action_windows` =
`staticmethod(TrackNetRunner.*)` (explicit rebind = the plugin-equivalence point). Output hard-
named `<stem>_wasb.csv`. ? `wasb_infer.py` re-staged each run (editing Windows copy inert until
staged); `wsl_bash` duplicated (drift risk). Both runners now satisfy the DetectorRunner ABC
for future swappability.
123     - `@backend/pipeline/detectors/wasb_infer.py` (WSL) ? `::run` core. ? imports WASB-SBDT
repo modules + torch ? NOT importable on Windows. Reboot-resumable detector cache:
`::DET_FILE='det_raw.jsonl'`, `::MANIFEST_FILE='manifest.jsonl'`, `::CACHE_VERSION=1`,
`::load_and_clean_cache` (reads longest CONTIGUOUS prefix where `w==line index`, rewrites to
heal crash-truncated tail), `::manifest_compatible` (? `frames_dir` abspath'd but `weights`
compared raw), `::atomic_write_text/_atomic_write_trajectory`, `::dets_to_tracker` (xy MUST be
np.array), `::build_cfg` (`runner.gpus=[0]` hardcoded), `::extract_frames` (cv2-PNG SLOW),
`::default_cache_dir` (`<stem>_wasbcache` next to out). ? GPU pass resumable; CPU tracker
STATEFUL -> always fully re-run; `--fresh` ignores cache.
124     - `@backend/pipeline/detectors/rally_gate.py` ? Gate A net-crossing post-filter (pure,
no I/O/model). `::estimate_play_axis` (larger-spread axis; ? ties -> 'x'),
`::count_net_crossings` (sign changes vs median net; deadband jitter zone),
`::gate_windows`/`::apply_gate` (`min_crossings=2` default; kills single-toss service feeds).
`::GatedWindow` (start,end,crossings,visible_points,kept). ? window contract here is
`(start,end)` tuples NOT the rich window dict ? caller must adapt. Consumed by calibrate_local /
distill_local / export_wasb_segments.
125
126     ### 2.3 Eval & calibration (all pure core; I/O in CLI drivers)
127     - `@backend/eval/metrics.py` ? scoring primitive (THE spine). `::interval_iou`,
`::match_intervals` (greedy, deterministic), `::score`->`::Score`(`.as_dict`) = wire shape),
`::pr_curve`, `::Match`, `::MatchResult`. ? greedy not Hungarian; empty matches -> 0.0 (not NaN)
for mean_iou/boundary errors.
128     - `@backend/eval/annotations.py` ? generic GT loader. `::load_annotations` (`start,end`
CSV; RAISES on bad rows/`start>=end`), `::parse_timestamp` (decimal or MM:SS/HH:MM:SS),
`::write_scaffold`.
129     - `@backend/eval/calibration.py` ? overfit guard. `::cross_validate` (within-video
k-fold; ? defaults `metric='recall` deliberately), `::leave_one_video_out` (honest cross-video;
needs 22 labelled videos), `::improvement_report` (OPTIMISTIC ? selected on full GT),
`::select_best`, `::kfold_indices`, `::pct_improvement` (? from-zero -> `inf`), `::evaluate`. $
`PredictFn = Callable[[Config], List[Interval]]` = the plug-in seam. ? `generalization_gap
>~0.1` = overfit alarm.
130     - `@backend/eval/calibrate_wasb.py` ? WASB tuning CLI. `::BASELINE=(0.02,2.0,2.0)`,
`::default_grid` (45 cfgs), `::make_predict_fn` (parses trajectory once, closes over
`trajectory_to_action_windows`), `::load_golden_gts` ($`start_time,end_time` cols; ? SILENTLY
skips bad rows ? opposite of annotations.load_annotations). ? `{load_golden_gts,
make_predict_fn, BASELINE}` imported by 4 downstream drivers (calibrate_local,

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export_wasb_segments, gemini_refine, audio/e1_probe).
131 - `@backend/eval/calibrate_local.py` ? distill Gemini->local thresholds (windowing +
net-crossing gate, no cloud at runtime). `::local_grid` (180 cfts), `::make_local_predict_fn`
(adds `rally_gate.apply_gate` when `min_crossings>0`), `::build_videos` (? `SystemExit` if a
labels file yields no rallies), `::run`/`::main`. $
`LocalConfig=(velocity,merge,min_dur,min_crossings)`, `BASELINE_LOCAL=(0.02,2.0,2.0,0)`;
manifest JSON shared with distill_local.
132 - `@backend/eval/export_wasb_segments.py` ? tuned cfg -> golden/slicer CSV + comparison
+ optional clip cut. `::TUNED=(0.05,1.0,1.0)` (? from ONE video),
`::SEG_FIELDS`, `::COMP_FIELDS`, `::build_comparison`, `::seconds_to_mmss`, `::maybe_slice`
(lazy-imports rally_slicer). `--slice` requires `--video`. ? `write_segments_csv` output loads
straight into `rally_slicer.load_rally_labels` AND re-imported by gemini_refine ? schema is
load-bearing.
133 - `@backend/eval/batch.py` ? corpus aggregation (pure; orchestration is in
run_phase3_eval). `::aggregate` (micro pool TP/FP/FN, TP-weighted mean_iou, macro F1),
`::default_stages_for` (`auto`; `::FINAL_ONLY_SEGMENTERS={motion,gemini}`),
`::resolve_video_path` (normalizes collector `./data`..` Windows paths), `::format_aggregate`.
134 - `@backend/eval/harness.py` ? DB-pred scorer (no re-run).
`::STAGES=('candidates','final')`, `::get_predictions`
(candidates->`query_candidate_segments(detector=)`; final->`query_play_segments({})`); ? unknown
stage raises ValueError; reused by regression.py),
`::evaluate`->`::EvalReport`(`::StageReport`)->`::report_to_dict` (input contract for
batch.aggregate), `::format_report_text`.
135 - `@backend/eval/training.py` ? learned-segmenter data layer. `::YOLO_FEATURE_NAMES`
(5-d), `::extract_yolo_features` (? reads `row['stats']` dict from DB; missing fields default
0.0), `::label_candidates` (same IoU `match_intervals` as evaluator),
`::build_training_set`->(X,y,intervals); `feature_fn` is the substrate plug-in seam.
136 - `@backend/eval/classifier.py::LogisticRegression` ? numpy-only logreg (no sklearn).
`fit/predict_proba/predict/to_dict/from_dict/save/load`. ? `class_weight='balanced'` default;
stores feature standardization (mean/std) ? inference MUST use same feature ORDER; `sigmoid`
clips z to [-30,30]; `lr=0.1,epochs=1000,l2=1e-3`; JSON "model":"logreg" tag.
137 - `@backend/eval/eval_partial_labels.py` ? score pipeline vs **PARTIAL human labels**
(PR #60). `::restrict_to_window` (drop preds outside `[0, last GT end]` so un-labeled-tail
detections aren't FPs ? THE partial-label methodology), `::run` (baseline/+gate/tuned/+gate ops
table + per-rally diff via `export_wasb_segments.build_comparison`), `::sweep`/`--sweep` (grid
vs labels; ? fit-on-self = optimistic), `::DEFAULT_GRID`, `::TUNED=(0.05,1.0,1.0)`. ?
`window_end` inferred from `max(GT end)` is WRONG if a video is labelled PAST its last rally ?
P1 fix = collector persists `labeled_window` in the manifest. Reuses calibrate_wasb + rally_gate
+ metrics.
138 - `@backend/eval/rally_seq_proto.py` ? **PROTOTYPE** learned per-FRAME rally segmenter =
owned-model **Gen-0 seed** (PR #61). 6 features over the WASB trajectory (`vis_rate, spd_mean,
spd_max, cross_rate, x_std, y_std`) -> `classifier.LogisticRegression`, honest LOVO.
`::build_frame_arrays`, `::features` (scipy `uniform/maximum_filter1d` windowed; ? speed NOT
fps-normalized yet ? 30 vs 60fps), `::labels_for`, `::probs_to_segments`
(smooth->threshold->merge/min-dur), `::roc_auc` (rank-based, no sklearn), `::run` (per-video AUC
+ LOVO-threshold F1 + oracle-threshold F1). ? DIRECTIONAL only (n?5, linear, threshold-transfer
unsolved); held-out per-frame AUC 0.83?0.92, **beats the heuristic on multi-court footage**
(Kushagra/gBaaddy). Manifest `output/human_lovo_manifest.json` (the n=5 human corpus) shared w/
calibrate_local.
139 - ? **MULTI-COURT / background-game confound (2026-06-08 finding):** WASB tracks EVERY
shuttle in frame incl. other courts; videos with background games (Kushagra, gBaaddy) have high
dead-time shuttle visibility (41?57% vs 18?22% clean) ? the velocity-windowing heuristic can't
find clean rally boundaries (F1 0.16/0.28 vs ~0.75 on single-court). Diagnostic = in-rally÷dead-
time visibility "contrast" (?3.6× heuristic works, ?1.9× fails). Lever = **foreground-court
isolation** (spatial gate to the prominent court, needs real court-region detection). Academy =
multi-court = the target env. See `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
140 - `@backend/eval/distill_local.py` ? distill Gemini -> a LEARNED classifier (beyond
calibrate_local's threshold ceiling). `::FEATURE_NAMES` (5 fps/res-INVARIANT:
duration,peak,mean,active_ratio,net_crossings), `::build_candidates` (recall-first proposal),
`::predict_rallies` (proba filter then `gemini_refine.merge_overlaps(gap_tol=1.0)`),
`::run`/`::main`. consts `PROPOSAL=(0.005,1.0,0.5)`, `BASELINE_LOCAL=(0.02,2.0,2.0)`. ? leave-
one-video-out needs ?2 videos; final model trained on all.
141 - `@backend/eval/gemini_refine.py` ? refine WASB candidates with Gemini over SHORT
padded clips (precision lever; eval/tuning, NOT the live pipeline).
`::merge_overlaps(gap_tol=0.0)` (reused by distill_local with gap_tol=1.0), `::refine_window` (?

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FRESH genai client per worker ? shared throws socket errors), `::full_video_rallies`,
`::run`/`::main` (`--full-video` skips `--trajectory`). ? requires `GEMINI_API_KEY` (raises
SystemExit); imports `BASELINE` from calibrate_wasb +
`TUNED`/`write_segments_csv`/`seconds_to_mmss` from export_wasb_segments.
142     - `@backend/eval/regression.py` ? CI gate. `::DEFAULT_TOLERANCE=0.05`,
`::DEFAULT_BASELINE_PATH=output/regression_baseline.json`, `::regress`,
`::regress_with_provider` (GT via ? `annotation_registry.get(tier)`;
"file"->FileProvider/"vendor"->HumanVendor/"auto"->oracle),
`::save_baseline`/`::load_baselines`, `::RegressionResult`. ? imports `annotation_registry` from
`**backend.annotations` NOT `backend.eval.annotations`; gate fires on precision OR recall drop
(F1 reported, not gated); no baseline -> `regressed=False` (silent pass).
143
144     ### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)
145     - `@backend/pipeline/audio/hit_detector.py` ? classical-onset shuttle-hit detector. Pure
numpy + stdlib `wave` (NO scipy); ffmpeg only for extraction. `::HitEvent`(t,strength),
`::extract_audio` (ffmpeg -> mono 16k PCM WAV), `::load_wav`, `::onset_envelope` (pure-numpy
STFT spectral flux; `band=(lo,hi)` band-limit), `::detect_hits` (adaptive `mean+k.std`;
`k`=sensitivity knob), `::detect_hits_in_video`. ? separate signal ? WASB never modified by
this; per audio-e1-findings recall 100% but discrimination ~2.2x, band-pass null -> NOT adopted
(PR #35).
146     - `@backend/eval/audio/e1_probe.py` ? E1 GO/NO-GO diagnostic (no training/fusion).
`::cluster_hits` (groups thwacks; ? `min_hits>=2` so a lone service-feed hit isn't a rally),
`::run` (discrimination ratio + recall recovery + audio-only P/R/F1), `::main`. reuses
`hit_detector` + `calibrate_wasb.{load_golden_gts,make_predict_fn,BASELINE}` + `metrics`.
147
148     ### 2.4 Stitcher / tools / utils
149     - `@backend/pipeline/stitcher/compiler.py::VideoCompiler` ? `compile_highlights(raw,
segments:[(start,end)], out_name)`. Per-clip re-encode -> `-f concat -c copy`. `::_parse_time`
(mirrors gemini.parse_time; ? unparseable -> 0.0), `::get_video_duration`. ? if duration probe
fails (0.0) NO boundary validation; `begin_padding`(0.5)/`end_padding`(1.0) defaults; temp clips
in a TemporaryDirectory under output_dir; raises ValueError/FileNotFoundError/RuntimeError; uses
`print()`. ? two-stage codec contract: clips re-encoded with IDENTICAL settings precisely so
final concat can stream-copy ? change a preset and concat may break.
150     - `@backend/tools/rally_slicer.py` ? standalone CLI. `::compute_clip_windows` (pure,
unit-tested), `::load_rally_labels` (golden schema; ? silently skips degenerate rows),
`::slice_rallies`, `::ClipWindow` (rally:str), `::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (?
duplicate compiler's by copy). `read_time` -> backend/eval/annotations.parse_timestamp.
151     - `@backend/utils/video.py` ? `::get_video_duration` (ffprobe; ? 0.0 on failure =
"unknown"), `::extract_video_chunk(...,reencode=False,scale_width=None)`. ? copy mode cuts at
nearest keyframe (drift) ? pass `reencode=True` for short/accurate clips; `scale_width` requires
`reencode=True`.
152     - `@backend/utils/geometry.py` ? `::get_velocity_normalised` (fraction of frame-width/s;
? raw-px fallback if width<=0; used by yolo_hybrid), `::calculate_iou` (? +1 px convention
inflates IoU), `::get_center/get_distance/get_velocity`.
153
154     ### 2.5 Validators ?
155     - `@backend/pipeline/validators/base.py::VideoValidator` ABC, `::ValidationResult`
(pydantic: validator_name,passed,message,details), `::ValidationRegistry` (ordered LIST ?
registration order = exec order; ? no dedup), `::registry` (singleton). § `validate(video_path,
config, context) -> ValidationResult`. `::run_all_with_context -> (results, context)` (wraps
each validate in try/except -> failed result not abort); `::run_all` discards context. ? **add a
validator**: subclass + `registry.register(MyValidator())`.
156     - `@backend/pipeline/validators/__init__.py` ? registers in EXEC order: FormatValidator,
ResolutionFPSValidator, PTSContinuityValidator, SceneCutDensityValidator, BlurValidator,
RelevanceValidator (? producers before consumers ? format_check first; ? import mutates global
registry as side effect).
157     - `@backend/pipeline/validators/format.py::FormatValidator` (`format_check`) ?
**PRODUCER** of `context['ffprobe_meta']`; ? MUST run before resolution_fps; container check is
substring `mp4` (so MOV-family passes); external `ffprobe`.
158     - `@backend/pipeline/validators/resolution_fps.py::ResolutionFPSValidator`
(`resolution_fps_check`) ? CONSUMES `ffprobe_meta` (? hard-fails "Run format_check first" if
absent). `min_fps` 29.9; min res 1280x720.
159     - `@backend/pipeline/validators/pts_continuity.py::PTSContinuityValidator`
(`pts_continuity_check`) ? re-probes (no context). ? 10.0s keyframe-gap threshold hardcoded
(comment says 8.0s ? drift, NOT config-driven); any backward jump fails; `<2` keyframes PASSES

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(too short).
160     - `@backend/pipeline/validators/scene_cut.py::SceneCutDensityValidator`
(`scene_cut_check`) ? OpenCV. ? `cut_threshold=0.6` hardcoded (NOT config); samples ~2fps, caps
at 100 SAMPLED frames (~first 50s); `max_scene_cuts_per_minute` (0.5) is the config fail
threshold.
161     - `@backend/pipeline/validators/blur.py::BlurValidator` (`blur_check`) ? Laplacian var <
`min_blur_threshold`(default 100.0 in-file, 20.0 in models/config) = fail; samples 15 frames. ?
magic absolute, not resolution-normalized.
162     - `@backend/pipeline/validators/relevance.py::RelevanceValidator` (`relevance_check`) ?
HSV court-green ratio (? purely color, no geometry despite name); lazy `import backend.sports`;
3-tier config precedence (`validation.relevance` -> sport profile -> hardcoded); ? unknown sport
-> empty cfg, falls back to defaults, does NOT fail.
163
164     ### 2.6 Sports / Providers / Annotations ?
165     - `@backend/sports/base.py::SportProfile` ABC
(`name`, `get_prompt`, `get_relevance_config`);
`@backend/sports/badminton.py::BadmintonSportProfile` (the Gemini prompt + HSV relevance cfg);
`@backend/sports/__init__.py::sport_registry` (+`::SportRegistry`, auto-registers badminton; ?
`register` instantiates profile to read `.name`). ? **add a sport**: subclass `SportProfile`,
`sport_registry.register(MyProfile)`. $ prompt emits JSON array `{start_time(float DECIMAL SEC),
end_time, shuttle_exchange_count(int), shot_count_confidence('Low'|'Medium'|'High')}`. ? prompt
actively defends against MM:SS; `get()` returns a fresh instance each call.
166     - `@backend/providers/base.py::AIVideoProvider` ABC (`__init__(api_key, model_name)`);
`@backend/providers/gemini.py::GeminiProvider` (lazy `::client` property; `::MAX_UPLOAD_MB=500`;
upload->poll->generate JSON temp 0->delete); `@backend/providers/__init__.py::provider_registry`
(+`::ProviderRegistry`, auto-registers `gemini`). ? **add a provider**: subclass,
`provider_registry.register('name', MyCls)` (key passed EXPLICITLY); `get(name, api_key,
model_name)`. $ `analyze_video(path, prompt, chunk_duration=0.0, label='')` -> (segments:list,
metrics:dict{upload_time,process_time,gen_time}). ? blocking 10s poll; returns `([], metrics)`
on ANY failure (empty != error); oversize clip silently skipped; orphaned remote file possible
if process dies mid-run.
167     - `@backend/annotations/base.py::AnnotationProvider` ABC (`submit/poll/fetch/annotate`,
`::Interval` (= metrics.Interval), `::QUEUED/RUNNING/DONE/FAILED`/`::STATUSES`,
`::AnnotationProviderRegistry` (decorator-friendly `register` returns the class; ? keys off
`provider_cls().name`, falls back to `__name__` if no-arg ctor throws; `list_available()` ->
LIST not dict), `::annotation_registry`. ? **add a tier**: subclass (define `name`),
`annotation_registry.register(MyCls)` at module bottom + import in `__init__.py`. $ `fetch` ->
List[Interval]` sorted; contract says RAISE (not `[]`) on unavailable.
168     - `@backend/annotations/file_provider.py::FileProvider` (`file`; job_id IS the
resolved CSV path; `_resolve` -> `/.csv`; ->
`backend.eval.annotations.load_annotations`; ? `guideline`/`sport` ignored).
169     - `@backend/annotations/automated_provider.py::AutomatedProvider` (`auto`; GS-4 stub;
injectable `::Labeler`, in-process `jobs` store, `::warn_if_same_lineage` ORACLE RULE ? only
warns, does not block; default `_not_configured` RAISES NotImplementedError). ? get via
`annotation_registry.get('auto', labeler=fn, lineage='...')`; registered as "auto" only
because no-arg ctor succeeds.
170     - `@backend/annotations/__init__.py` ? ? `noqa F401` imports of
file_provider/automated_provider are load-bearing (removing de-registers providers).
171
172     ### 2.7 Infrastructure (DB) ?
173     - `@backend/infrastructure/database.py::Database` ? SQLite, 4 tables
(`videos/play_segments/events/candidate_segments`), `PRAGMA user_version` migration ladder in
`::init_db` (currently =1, `version<1` creates `candidate_segments`). `::get_connection` (?
re-issues `PRAGMA foreign_keys=ON` per connection ? else CASCADE/SET NULL silently skipped).
Methods: `add_video`, `add_play_segment`, `add_event`, `add_candidate_segment`, `link_candidate_to_s
egment`, `query_candidate_segments(detector=,only_unlabeled=)`, `query_play_segments`, `get_video`,
`clear_video_data`. ? candidate_segments is the detector-agnostic fine-tuning extension point
(common fields columns, detector-specifics in `stats`/`detection_params` JSON; index
`idx_candidates_video_detector`). ? `add_video` is `INSERT OR REPLACE` -> cascade-DELETES
dependent segments on re-ingest; events column is `type` not `event_type`; `query_play_segments`
filters use truthiness so `duration_min=0`/empty-string = "no filter"; write helpers swallow
exceptions.
174
175     ### 2.8 Reporting / observability ? (per-video, soft-dep `obsreport`)
176     - `@backend/reporting/__init__.py` ? `::emit_indexer_report(*, db, video_id, video_path,
```

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config, val_results, val_context=None, segmenter_requested=, segmenter_actual=,
was_reingest=False, segmentation_ran=True)` ? single entrypoint, keyword-only. Normalizes typed
config (`config.model_dump(...)` if `hasattr model_dump`); short-circuits to None if
unavailable/disabled; pulls play_segments/candidates from `db` (read-only
`query_play_segments(vid,{})`/`query_candidate_segments(vid)`); delegates to
`harvest.record_run`; writes via `rec.write(output_dir=config["report"]["output_dir"])`. ? wraps
everything in bare `except Exception: return None` ? **NEVER raises** (designed for `main.py`'s
`finally:`); writes `

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receiver, metadata:dict, events:[...]]`. Hybrid P4 metadata `{shot_count,
shot_count_confidence:'Unknown', detector:f'{family}-hybrid-{model}'}`; gemini detector = bare
`model_name`; motion adds random `intensity/avg_speed_kmh`.
197
198     $ **AI segment dict** (provider output, consumed by hybrid P3 / gemini): `{start_time,
end_time, shuttle_exchange_count?, shot_count_confidence?}` (offset by chunk/window start;
`_candidate_id` stamped internally).
199
200     $ **GT / golden CSVs** (TWO conventions ? do not conflate):
201     - generic (`annotations.load_annotations`): `start,end` 2-col; optional header; decimal
or MM:SS/HH:MM:SS; RAISES on bad row.
202     - golden/slicer (`calibrate_wasb.load_golden_gts`, `rally_slicer.load_rally_labels`,
`export.SEG_FIELDS`): `rally_number,start_time,end_time[,duration,start_mmss,end_mmss]`;
SILENTLY skips bad rows.
203
204     $ **det_raw.jsonl** (one window/line): `{ "w":<window_index>,
"f":[[frame_id,[[x,y,score,scale],...], ...]]`. Lines MUST be contiguous (line N has `w==N`);
first violation/JSONDecodeError -> truncate-heal.
205     $ **manifest.json** (cache): `{version, frames_dir(abspath), weights, sport, frames_in,
frames_total, windows_total, windows_done}`. Reuse only if ALL match + `version==CACHE_VERSION`.
206     $ **collector manifest.json** (eval): `{eval_sets:[{video_id(label), local_path, csv},
...]}`; ? DB key = file MD5, manifest `video_id` is a label.
207
208     $ **eval primitives**: `Interval=Tuple[float,float]` (start<end).
`Match{pred_index,gt_index,iou}`. `MatchResult{matches, unmatched_pred_indices(=FP),
unmatched_gt_indices(=FN)}`. `Score{iou_threshold,num_pred,num_gt,true_positives,false_positives
,false_negatives,precision,recall,f1,mean_iou,mean_start_error,mean_end_error}.as_dict()`.
209
210     $ **WSL detector candidate** (in-WSL, model<->tracker): `{ 'xy':np.array([x,y]),
'score':float, 'scale':int}`; plain JSON form `[x,y,score,scale]`. ? `_dets_to_tracker` MUST
rebuild `xy` as np.array (tracker does `np.linalg.norm`).
211
212     $ **HitEvent** (audio, `pipeline/audio/hit_detector`): `{t:float(sec), strength:float}`
? onset-envelope peak.
213
214     $ **observability report** (`reporting/harvest.record_run` -> `RunRecorder`): stages
`validation/segmentation/ai_handoff` + highlights + verdict ? {pass,pass_with_warnings,fail};
flags `{code,faq_ref}`; `video_meta.fps_source` ? {probed,fallback}`. ? stage/metric/detail names
are coupled to `spec.yaml::rules[].when[].path` ? pairs that MUST stay in sync:
`ai_handoff.details.{api_key_present,ai_based}`, `segmentation.metrics.segment_count`, `segmenta
tion.details.{segmenter_actual,matches_config_default,segmenter_explicitly_requested,was_reinges
t}`, `validation.status`, `video_meta.fps_source`.
215
216     ---
217
218     ## 4. WASB WSL CONTRACT (external dep `~/models/WASB-SBDT`, MIT; NOT in this repo ?
claims below are unverified against external source, only the repo-side caller wasb_infer.py was
checked)
219     - Env: WSL conda `wasb`, cwd `~/models/WASB-SBDT/src`. Hydra
`compose(config_name='eval', overrides=[dataset=<sport>, model=wasb,
detector.model_path=<weights>, runner.device=cuda, runner.gpus=[0]])`. `gpus=[0]` because box is
single-GPU. (overrides VERIFIED in @backend/pipeline/detectors/wasb_infer.py::build_cfg.)
220     - `model=wasb` -> `name=hrnet`, `frames_in=3`, `frames_out=3`, input 512x288 (WxH),
ImageNet normalize; affine resize maps heatmap coords -> original-frame coords. (model dims read
from cfg["model"] at runtime in wasb_infer.run.)
221     - `detectors/detector.py::TracknetV2Detector.run_tensor(imgs, affine_mats)` -> (results,
hms_vis). `results[bid][eid]` = list of `{xy:np.array (ORIGINAL coords), score, scale}`.
(caller uses `detector.run_tensor(imgs, trans)` returning `(batch_results, _)` ? consistent.)
222     - `trackers/online.py::OnlineTracker` ? STATEFUL, sequential per `update()` (assumes
every frame fed once, in order, no gaps). `update(frame_dets)->{x,y,visi,score}`; `refresh()`
resets. `det['xy']` MUST be np.array. Full ordered re-run is deterministic. (caller calls
`build_tracker(cfg)`, `tracker.refresh()`, `tracker.update(...)` returning dict with `visi/x/y`
? consistent.)
223     - `dataloaders/dataset_loader.py::ImageDataset(cfg, samples, input_wh, output_wh,
transform, seq_transform, is_train)`; `samples=[{frames:[paths], annos:[{frame_path,
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center:Center(is_visible,x,y))]]]`. (caller constructs exactly this ? VERIFIED in
wasb_infer.run.)
224     - Weights `pretrained_weights/wasb_badminton_best.pth.tar` ? NOT committed, academic-
data-trained -> ? confirm terms before COMMERCIAL deploy. monotrack weights = noncommercial,
avoided.
225
226     ---
227
228     ## 5. CROSS-REPO (sibling private repo `sports-data-collector` ? external, unverified
here)
229     - Indexer keys video by **FILE MD5**; collector keys by human `video_id`. ? re-
encoding/re-downloading changes bytes -> MD5; acquisition MUST precede processing; manifest
tracks the exact file used.
230     - `python -m src.cli.curate export` (collector) -> `//.csv`
(indexer `start,end` format) + `manifest.json` -> consumed by `@backend/eval/batch.py` +
`@run_phase3_eval.py`.
231     - Collector also emits golden rally labels (`rally_number,start_time,end_time,...`) ->
consumed by `@backend/eval/calibrate_wasb.py` / `export_wasb_segments.py` /
`@backend/tools/rally_slicer.py`.
232
233     ---
234
235     ## 6. GOTCHAS (cross-cutting footguns)
236     - **MD5 keying:** `video_id` = MD5 of WHOLE file via the single
`@backend/utils/hashing.py::compute_video_id` helper (64KB chunks), imported by main + all
run_*.py. Manifest `video_id` is a label, NOT the DB key.
237     - **Config access:** `backend/main.py` and all four `run_*.py` load via
`backend.config.load_config` (typed `AppConfig`). The legacy `.get()` dict-compat shim remains
for validators/segmenters. Remaining literal: a defensive `getattr(..., 'motion')` fallback in
`main.py` that only fires if `global_config`.`indexing` is `None` (the ASGI-import edge case).
238     - **Segmenter default:** single source of truth = model
`IndexingConfig.default_segmenter='yolo_hybrid'`; `config.json` may override at runtime (ships
`gemini`). run_*.py read the model value (no literals); `main.py` keeps the defensive `motion`
getattr fallback noted above.
239     - **`AppConfig.get` must return dict sections, not models:** the legacy
`.get("indexing",{}).get(...)` chains everywhere depend on it; the `extra="allow"` model also
silently keeps typo'd config keys (no rejection).
240     - **`output_dir`:** a real field on `OutputConfig` (default `"output"`); the CLI
`--output-dir` overrides it on the typed model in `main()`, and `compile_highlights` reads the
same `global_config.output.output_dir` ? honored end-to-end.
241     - **Gemini free-tier / API-key dead path:** hybrids + gemini return `[]` (not error) if
`{PROVIDER}_API_KEY` missing OR WSL healthcheck fails ? a silent no-op. `analyze_video` also
returns `[]` on real failure -> empty != "no rallies". The reporting
`silent_provider_noop`/`ai_empty_vs_no_rallies` rules exist precisely to disambiguate this in
the report.
242     - **Reporting never breaks a run:** `emit_indexer_report` swallows all exceptions
(designed for `main.py` `finally:`) and is emitted even on validation failure. `obsreport` is a
SOFT dep ? absent -> no-op.
243     - **spec.yaml <-> harvest.py coupling:** rule `path` strings dotted-navigate harvest's
stage/metric/detail names; rename a detail in harvest.py and the matching rule silently stops
firing. Add rules in spec.yaml, not code.
244     - **fps / frame_width must be PROBED not assumed:** `_probe_fps_width` / runner
fallbacks (30.0, 1280.0) silently mis-scale velocity thresholds; calibrate_wasb CLI bakes
`59.94/1920`. The report's `fps_fallback_used` flag surfaces this.
245     - **WSL `/mnt` can't see Google Drive / network mounts:** stage video into native WSL fs
(`~/clips`) ? `/mnt/c` reads are slow and Drive-backed paths invisible.
246     - **ffmpeg NOT in WSL** by design: all ffmpeg/ffprobe run Windows-side (must be on
PATH); WSL only runs the GPU model. `extract_frames` uses cv2 PNG (slow) ? prefer host ffmpeg +
`--frames_dir`. ? Don't import `wasb_infer` from Windows code (it imports torch + WASB repo
modules).
247     - **GPU float nondeterminism:** detector pass is checkpointed/resumable but not bit-
reproducible; the CPU tracker re-run over the ordered cache IS deterministic.
248     - **`output/` is gitignored:** DB, baselines, cache, reels live there. Predictions MUST
exist in `output/<db_name>` before any eval/regression.
249     - **Validator ordering coupling:** exec order is Format -> ResolutionFPS ->

```

```

PTSContinuity -> SceneCut -> Blur -> Relevance; `resolution_fps_check` hard-fails unless
`format_check` ran earlier (shared `context['ffprobe_meta']`). PTS 10.0s gap + scene-cut 0.6
correlation are NOT config-overridable.
250     - **`add_video` INSERT OR REPLACE + FK CASCADE** -> re-ingesting a `video_id` cascade-
deletes its segments. Use `clear_video_data` deliberately. `_get_connection` MUST re-issue
`PRAGMA foreign_keys=ON` per connection.
251     - **Twin segmenters:** `wasb_hybrid.py` / `tracknet_hybrid.py` are copy-paste; fixes to
P3/P4 must land in both. ? subtle drift: wasb's shot_count `int(None)`-via-TypeError fallback vs
tracknet's `is None` check.
252     - **Two "annotations" modules:** `backend.eval.annotations` (GT CSV loader) vs
`backend.annotations` (provider registry). regression.py imports the latter.
253     - **`calibrate_wasb.{load_golden_gts, make_predict_fn, BASELINE}` is a de-facto shared
API** for calibrate_local, export_wasb_segments, gemini_refine, e1_probe ? changing its golden-
CSV schema or `BASELINE` ripples widely. `gemini_refine.merge_overlaps` reused by
`distill_local`; `harness.get_predictions` reused by `regression.regress`.
254     - **Hardcoded tuned constants:** `calibrate_wasb.BASELINE`, `export.TUNED`,
slicer/compiler padding ? duplicated by copy, not import; re-tuning won't propagate.
255     - **`timeout_sec=0` = no timeout:** a hung WSL/GPU call blocks forever.

```

3. user (2026-06-10T05:59:34.784Z)

```
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/:
```

```

__init__.py
__pycache__
annotations.py
audio
batch.py
calibrate_local.py
calibrate_wasb.py
calibration.py
classifier.py
distill_local.py
eval_partial_labels.py
export_wasb_segments.py
gemini_refine.py
harness.py
metrics.py
rally_seq_proto.py
regression.py
training.py

```

```
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/audio/:
```

```

__init__.py
__pycache__
e1_probe.py

```

4. user (2026-06-10T05:59:41.614Z)

```

1      """Pure temporal interval-matching metrics for rally evaluation.
2
3      No I/O, no DB, no global state ? every function here is deterministic and unit
4      tested. An "interval" is a `(start, end)` tuple of floats in seconds with
5      start < end. We score a list of *predicted* rally intervals against a list of
6      *ground-truth* intervals via temporal IoU and one-to-one greedy matching.
7      """
8
9      from dataclasses import dataclass, field
10     from typing import List, Tuple, Optional
11
12     Interval = Tuple[float, float]
13
14
15     def interval_iou(a: Interval, b: Interval) -> float:
16         """Temporal intersection-over-union of two intervals.
17

```

```

18     Returns 0.0 when they don't overlap (or either is degenerate). IoU is
19     overlap / union, both measured as durations on the timeline.
20     """
21     a_start, a_end = a
22     b_start, b_end = b
23     if a_end <= a_start or b_end <= b_start:
24         return 0.0
25     inter = max(0.0, min(a_end, b_end) - max(a_start, b_start))
26     if inter <= 0.0:
27         return 0.0
28     union = (a_end - a_start) + (b_end - b_start) - inter
29     return inter / union if union > 0 else 0.0
30
31
32 @dataclass
33 class Match:
34     """One matched predicted?ground-truth pair."""
35     pred_index: int
36     gt_index: int
37     iou: float
38
39
40 @dataclass
41 class MatchResult:
42     """Outcome of matching predictions to ground truth at a given IoU threshold."""
43     matches: List[Match] = field(default_factory=list)
44     unmatched_pred_indices: List[int] = field(default_factory=list) # false positives
45     unmatched_gt_indices: List[int] = field(default_factory=list)   # false negatives
46 (misses)
47
48 def match_intervals(preds: List[Interval], gts: List[Interval],
49                    iou_threshold: float = 0.5) -> MatchResult:
50     """Greedy one-to-one matching of predictions to ground truth by descending IoU.
51
52     Every candidate (pred, gt) pair with IoU >= threshold is considered; pairs
53     are claimed highest-IoU first, and each prediction/GT can be used once. This
54     keeps a single prediction from being credited for two ground-truth rallies
55     (and vice versa).
56     """
57     candidates = []
58     for pi, p in enumerate(preds):
59         for gi, g in enumerate(gts):
60             iou = interval_iou(p, g)
61             if iou >= iou_threshold:
62                 candidates.append((iou, pi, gi))
63     # Highest IoU first; ties broken by indices for determinism.
64     candidates.sort(key=lambda c: (-c[0], c[1], c[2]))
65
66     used_pred, used_gt = set(), set()
67     matches: List[Match] = []
68     for iou, pi, gi in candidates:
69         if pi in used_pred or gi in used_gt:
70             continue
71         used_pred.add(pi)
72         used_gt.add(gi)
73         matches.append(Match(pred_index=pi, gt_index=gi, iou=iou))
74
75     unmatched_pred = [i for i in range(len(preds)) if i not in used_pred]
76     unmatched_gt = [i for i in range(len(gts)) if i not in used_gt]
77     return MatchResult(matches=matches,
78                        unmatched_pred_indices=unmatched_pred,
79                        unmatched_gt_indices=unmatched_gt)
80
81

```

```

82     @dataclass
83     class Score:
84         """Aggregate metrics for one (predictions vs ground-truth) evaluation."""
85         iou_threshold: float
86         num_pred: int
87         num_gt: int
88         true_positives: int
89         false_positives: int
90         false_negatives: int
91         precision: float
92         recall: float
93         f1: float
94         mean_iou: float          # over matched pairs (0.0 if none)
95         mean_start_error: float  # mean |pred_start - gt_start| over matches, seconds
96         mean_end_error: float    # mean |pred_end - gt_end| over matches, seconds
97
98         def as_dict(self) -> dict:
99             return self.__dict__.copy()
100
101
102     def score(preds: List[Interval], gts: List[Interval],
103             iou_threshold: float = 0.5,
104             match_result: Optional[MatchResult] = None) -> Score:
105         """Compute precision/recall/F1, mean matched IoU, and boundary errors.
106
107         Pass a precomputed `match_result` to avoid re-matching (e.g. when also
108         extracting the failure lists); otherwise matching is done here.
109         """
110         mr = match_result if match_result is not None else match_intervals(preds, gts,
111 iou_threshold)
112         tp = len(mr.matches)
113         fp = len(mr.unmatched_pred_indices)
114         fn = len(mr.unmatched_gt_indices)
115
116         precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
117         recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
118         f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0
119 else 0.0
120
121         if mr.matches:
122             mean_iou = sum(m.iou for m in mr.matches) / tp
123             mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
124 mr.matches) / tp
125             mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in
126 mr.matches) / tp
127         else:
128             mean_iou = mean_start_err = mean_end_err = 0.0
129
130         return Score(
131             iou_threshold=iou_threshold,
132             num_pred=len(preds), num_gt=len(gts),
133             true_positives=tp, false_positives=fp, false_negatives=fn,
134             precision=precision, recall=recall, f1=f1,
135             mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
136         )
137
138     def pr_curve(preds: List[Interval], gts: List[Interval],
139             thresholds: Optional[List[float]] = None) -> List[Score]:
140         """Score across a sweep of IoU thresholds (default 0.1..0.9 step 0.1).
141
142         Lets you see strict-vs-lenient behaviour rather than a single operating point.
143         """
144         if thresholds is None:

```

```

143         thresholds = [round(0.1 * i, 1) for i in range(1, 10)]
144         return [score(preds, gts, iou_threshold=t) for t in thresholds]
145

```

5. user (2026-06-10T05:59:42.062Z)

```

1     """Load hand-annotated ground-truth rally timestamps from CSV.
2
3     Annotation file format (one file per video):
4
5         start,end
6         00:01:12,00:01:39
7         102.5,131.0
8
9     - A header row `start,end` is optional (auto-detected and skipped).
10    - Each timestamp is either decimal seconds (`102.5`) or colon-separated
11      `MM:SS` / `HH:MM:SS` (`00:01:12`). Both forms may be mixed in one file.
12    - Blank lines and `#` comment lines are ignored.
13    """
14
15    import csv
16    import logging
17    from typing import List, Tuple
18
19    logger = logging.getLogger(__name__)
20
21    Interval = Tuple[float, float]
22
23
24    def parse_timestamp(value: str) -> float:
25        """Parse a timestamp string to float seconds.
26
27        Accepts decimal seconds or colon-separated MM:SS / HH:MM:SS.
28        Raises ValueError on anything unparseable so bad annotations fail loudly.
29        """
30        value = value.strip()
31        if not value:
32            raise ValueError("empty timestamp")
33        if ":" in value:
34            parts = value.split(":")
35            if len(parts) > 3:
36                raise ValueError(f"too many ':' segments in timestamp '{value}'")
37            total = 0.0
38            for part in parts:
39                total = total * 60 + float(part)
40            return total
41        return float(value)
42
43
44    def load_annotations(csv_path: str) -> List[Interval]:
45        """Load and validate ground-truth rally intervals from a CSV file.
46
47        Returns intervals sorted by start time. Raises ValueError on malformed rows
48        (bad timestamp, start >= end) so labeling mistakes surface immediately rather
49        than silently skewing scores.
50        """
51        intervals: List[Interval] = []
52        with open(csv_path, newline="") as f:
53            reader = csv.reader(f)
54            for lineno, row in enumerate(reader, start=1):
55                if not row:
56                    continue
57                first = row[0].strip()
58                if not first or first.startswith("#"):
59                    continue

```

```

60         # Skip an optional header row.
61         if first.lower() in ("start", "start_time"):
62             continue
63         if len(row) < 2:
64             raise ValueError(f"{csv_path}:{lineno}: expected 'start,end', got
{row!r}")
65         try:
66             start = parse_timestamp(row[0])
67             end = parse_timestamp(row[1])
68         except ValueError as e:
69             raise ValueError(f"{csv_path}:{lineno}: {e}") from e
70         if start >= end:
71             raise ValueError(f"{csv_path}:{lineno}: start ({start}) >= end ({end})")
72         intervals.append((start, end))
73
74     intervals.sort(key=lambda iv: iv[0])
75     logger.info("Loaded %d ground-truth rallies from %s", len(intervals), csv_path)
76     return intervals
77
78
79 def write_scaffold(csv_path: str) -> None:
80     """Write an empty annotation file with just the header, for hand-filling."""
81     with open(csv_path, "w", newline="") as f:
82         writer = csv.writer(f)
83         writer.writerow(["start", "end"])
84

```

6. user (2026-06-10T05:59:46.614Z)

```

1     """Quantify rally-segmentation improvement ? and guard against overfitting.
2
3     This is the measurement layer for per-video / per-corpus calibration. It answers
4     two questions honestly:
5
6     1. """Did precision improve, by how much?""" ? `improvement_report` scores a
7         baseline config vs a tuned config on the same ground truth and reports the
8         before/after metrics and the % delta.
9
10    2. """Is that gain real or overfit?""" ? calibration must be measured on data the
11        tuning never saw, otherwise you're just memorising. Two protocols:
12        - `cross_validate` ? k-fold over one video's rallies: pick the config on the
13          train folds, score on the held-out fold. The **generalization_gap**
14          (train ? test) is the overfit alarm.
15        - `leave_one_video_out` ? the protocol that matters as more videos arrive:
16          pick the config on all *other* videos, score on the held-out video. Reports
17          per-video held-out scores + mean±std. This is the number to trust.
18
19    Everything is segmenter-agnostic: callers pass a `predict_fn(config) ->
20    List[Interval]` (e.g. WASB trajectory ? windows at those thresholds), so the same
21    harness works for WASB, yolo_hybrid, or anything else. Pure + unit-tested; no I/O.
22    """
23    from __future__ import annotations
24
25    import statistics as _st
26    from typing import Callable, Dict, List, Sequence, Tuple, Any
27
28    from backend.eval.metrics import score, Score, Interval
29
30    Config = Any                                # opaque to this module (e.g. a thresholds
dict)
31    PredictFn = Callable[[Config], List[Interval]]
32
33
34    def pct_improvement(before: float, after: float) -> float:
35        """Percentage change from `before` to `after` (e.g. 0.6 -> 0.75 == +25.0)."""

```

```

36         if before <= 0:
37             return float("inf") if after > 0 else 0.0
38         return (after - before) / before * 100.0
39
40
41     def evaluate(preds: List[Interval], gts: List[Interval], iou_threshold: float = 0.5) ->
Score:
42         """Precision / recall / F1 / mean-IoU of predicted vs ground-truth intervals."""
43         return score(preds, gts, iou_threshold)
44
45
46     def kfold_indices(n: int, k: int) -> List[Tuple[List[int], List[int]]]:
47         """Deterministic k-fold split of indices 0..n-1 into (train_idx, test_idx)."""
48         if n <= 0:
49             return []
50         k = max(2, min(k, n)) # need >=2 folds; cap at n
51         folds = [list(range(i, n, k)) for i in range(k)] # round-robin ? balanced,
deterministic
52         splits = []
53         for f in range(k):
54             test = folds[f]
55             train = [i for i in range(n) if i not in set(test)]
56             if test and train:
57                 splits.append((train, test))
58         return splits
59
60
61     def select_best(predict_fn: PredictFn, configs: Sequence[Config], gts: List[Interval],
62                     metric: str = "f1", iou_threshold: float = 0.5) -> Tuple[Config, float]:
63         """Pick the config whose predictions maximise `metric` against `gts`."""
64         best_cfg, best_val = None, -1.0
65         for cfg in configs:
66             val = getattr(score(predict_fn(cfg), gts, iou_threshold), metric)
67             if val > best_val:
68                 best_cfg, best_val = cfg, val
69         return best_cfg, best_val
70
71
72     def cross_validate(predict_fn: PredictFn, configs: Sequence[Config], gts:
List[Interval],
73                       k: int = 5, metric: str = "recall", iou_threshold: float = 0.5) ->
Dict[str, Any]:
74         """k-fold threshold calibration on ONE video's rallies (within-video overfit guard).
75
76         Per fold: choose the best config on the train rallies, score it on the held-out
77         rallies. A large `generalization_gap` (train >> test) means the thresholds are
78         overfitting that video's quirks.
79
80         Default metric is recall on purpose: predictions are global (whole-video
81         windows), so scoring them against a fold subset of GT would unfairly count the
82         other rallies' windows as false positives ? precision/F1 are not well-defined
83         per-fold. Recall ("did the tuned thresholds still detect the held-out rallies?")
84         is the honest within-video generalization signal. For precision/F1, use
85         `leave_one_video_out` (each video scored whole) or `improvement_report` on a
86         held-out video.
87         """
88         splits = kfold_indices(len(gts), k)
89         if not splits:
90             return {"error": "not enough rallies for cross-validation", "n": len(gts)}
91         train_scores, test_scores, picked = [], [], []
92         for train_idx, test_idx in splits:
93             gts_tr = [gts[i] for i in train_idx]
94             gts_te = [gts[i] for i in test_idx]
95             cfg, tr = select_best(predict_fn, configs, gts_tr, metric, iou_threshold)
96             te = getattr(score(predict_fn(cfg), gts_te, iou_threshold), metric)

```

```

97         train_scores.append(tr)
98         test_scores.append(te)
99         picked.append(cfg)
100     mean_test = _st.mean(test_scores)
101     mean_train = _st.mean(train_scores)
102     return {
103         "metric": metric,
104         "folds": len(splits),
105         "mean_test": mean_test,
106         "std_test": _st.pstdev(test_scores) if len(test_scores) > 1 else 0.0,
107         "mean_train": mean_train,
108         "generalization_gap": mean_train - mean_test,    # >~0.1 = likely overfitting
109         "picked_configs": picked,
110     }
111
112
113 def leave_one_video_out(videos: List[Dict[str, Any]], configs: Sequence[Config],
114                         metric: str = "f1", iou_threshold: float = 0.5) -> Dict[str,
115 Any]:
116     """Leave-one-video-out CV ? the honest metric as the corpus grows.
117
118     `videos`: list of {"name": str, "predict_fn": PredictFn, "gts": [Interval]}.
119     For each held-out video: pick the config that's best *pooled across the other
120     videos*, then score it on the held-out one. The mean held-out score is the
121     "precision on unseen videos" number; per-video spread shows stability.
122     """
123     if len(videos) < 2:
124         return {"error": "need >=2 videos for leave-one-video-out", "n": len(videos)}
125     per_video = []
126     for i, held in enumerate(videos):
127         others = [v for j, v in enumerate(videos) if j != i]
128
129         def pooled_score(cfg: Config) -> float:
130             vals = [getattr(score(v["predict_fn"])(cfg), v["gts"], iou_threshold),
131                     for v in others]
132             return _st.mean(vals)
133
134         best_cfg, best_train = max(((c, pooled_score(c)) for c in configs), key=lambda
135 t: t[1])
136         held_score = getattr(score(held["predict_fn"])(best_cfg), held["gts"],
137 iou_threshold), metric)
138         per_video.append({"video": held.get("name", f"video_{i}"),
139                          "held_out": held_score, "train_pooled": best_train, "config":
140 best_cfg})
141     held_scores = [p["held_out"] for p in per_video]
142     train_scores = [p["train_pooled"] for p in per_video]
143     return {
144         "metric": metric,
145         "mean_held_out": _st.mean(held_scores),
146         "std_held_out": _st.pstdev(held_scores) if len(held_scores) > 1 else 0.0,
147         "generalization_gap": _st.mean(train_scores) - _st.mean(held_scores),
148         "per_video": per_video,
149     }
150
151
152 def improvement_report(predict_fn: PredictFn, baseline_config: Config, tuned_config:
153 Config,
154                         gts: List[Interval], iou_threshold: float = 0.5) -> Dict[str,
155 Any]:
156     """The "+X% precision" headline: baseline vs tuned config on the same GT.
157
158     NB: for an *honest* headline, pass GT the tuning never saw (a held-out video),
159     or read the cross-validated `mean_test` from `cross_validate` instead.
160     """

```

```

155         b = score(predict_fn(baseline_config), gts, iou_threshold)
156         t = score(predict_fn(tuned_config), gts, iou_threshold)
157         return {
158             "baseline": {"precision": b.precision, "recall": b.recall, "f1": b.f1,
159 "mean_iou": b.mean_iou},
160             "tuned": {"precision": t.precision, "recall": t.recall, "f1": t.f1, "mean_iou":
161 t.mean_iou},
162             "precision_delta_pts": t.precision - b.precision,
163             "precision_delta_pct": pct_improvement(b.precision, t.precision),
164             "f1_delta_pct": pct_improvement(b.f1, t.f1),
165         }

```

7. user (2026-06-10T05:59:47.106Z)

```

1     """Score the pipeline's detected rallies against ground-truth annotations.
2
3     Reads predictions straight from the SQLite DB (no pipeline re-run, no API cost),
4     so scoring is fast, deterministic, and free. Two prediction "stages" can be
5     scored:
6
7     - ``candidates`` ? raw detector windows (``candidate_segments`` table). What
8       the shuttle/person detector found, before AI confirmation.
9     - ``final`` ? confirmed play segments (``play_segments`` table). After the
10       AI handoff trimmed/validated boundaries.
11
12     Comparing the two tells you whether the weak link is *detection* (candidates
13     already miss rallies) or *segmentation* (candidates catch them but final is
14     wrong) ? see docs/EVALUATION.md.
15     """
16
17     import logging
18     from dataclasses import dataclass, field
19     from typing import List, Dict, Optional
20
21     from backend.infrastructure.database import Database
22     from backend.eval import metrics
23     from backend.eval.metrics import Interval, Score, MatchResult
24
25     logger = logging.getLogger(__name__)
26
27     STAGES = ("candidates", "final")
28
29
30     @dataclass
31     class StageReport:
32         stage: str
33         pred_intervals: List[Interval]
34         score: Score
35         match_result: MatchResult
36         pr_curve: List[Score] = field(default_factory=list)
37
38
39     @dataclass
40     class EvalReport:
41         video_id: str
42         iou_threshold: float
43         gt_intervals: List[Interval]
44         stages: Dict[str, StageReport] = field(default_factory=dict)
45
46
47     def _rows_to_intervals(rows: List[dict]) -> List[Interval]:
48         """Pull (start_time, end_time) out of DB segment rows, sorted by start."""
49         ivs = [(float(r["start_time"]), float(r["end_time"])) for r in rows]
50         ivs.sort(key=lambda iv: iv[0])

```

```

51         return ivs
52
53
54     def get_predictions(db: Database, video_id: str, stage: str,
55                        detector: Optional[str] = None) -> List[Interval]:
56         """Fetch predicted rally intervals for one stage from the DB."""
57         if stage == "candidates":
58             rows = db.query_candidate_segments(video_id, detector=detector)
59         elif stage == "final":
60             rows = db.query_play_segments(video_id, {})
61         else:
62             raise ValueError(f"unknown stage {stage!r}; expected one of {STAGES}")
63         return _rows_to_intervals(rows)
64
65
66     def evaluate(db: Database, video_id: str, gt_intervals: List[Interval],
67                stages: List[str], iou_threshold: float = 0.5,
68                detector: Optional[str] = None,
69                with_pr_curve: bool = False) -> EvalReport:
70         """Score the requested stages for one video against ground truth."""
71         report = EvalReport(video_id=video_id, iou_threshold=iou_threshold,
72                             gt_intervals=gt_intervals)
73         for stage in stages:
74             preds = get_predictions(db, video_id, stage, detector=detector)
75             mr = metrics.match_intervals(preds, gt_intervals, iou_threshold)
76             s = metrics.score(preds, gt_intervals, iou_threshold, match_result=mr)
77             curve = metrics.pr_curve(preds, gt_intervals) if with_pr_curve else []
78             report.stages[stage] = StageReport(stage=stage, pred_intervals=preds,
79                                                score=s, match_result=mr, pr_curve=curve)
80         return report
81
82
83     # ----- reporting
84
85     def _fmt_ts(seconds: float) -> str:
86         seconds = max(0, int(seconds))
87         h, rem = divmod(seconds, 3600)
88         m, s = divmod(rem, 60)
89         return f"{h:02d}:{m:02d}:{s:02d}"
90
91
92     def report_to_dict(report: EvalReport) -> dict:
93         """JSON-serialisable view of an EvalReport."""
94         out = {
95             "video_id": report.video_id,
96             "iou_threshold": report.iou_threshold,
97             "num_ground_truth": len(report.gt_intervals),
98             "stages": {},
99         }
100         for stage, sr in report.stages.items():
101             out["stages"][stage] = {
102                 "score": sr.score.as_dict(),
103                 "pr_curve": [s.as_dict() for s in sr.pr_curve],
104             }
105         return out
106
107
108     def format_report_text(report: EvalReport, show_failures: bool = False) -> str:
109         """Human-readable console report for one video."""
110         lines = [
111             "=" * 64,
112             f" Rally evaluation ? video '{report.video_id}'",
113             f" Ground-truth rallies: {len(report.gt_intervals)} | IoU threshold: {report.iou_threshold}",
114             "=" * 64,

```

```

115         f" {'Stage':<12} {'Pred':>5} {'TP':>4} {'FP':>4} {'FN':>4} "
116         f"{'Prec':>6} {'Rec':>6} {'F1':>6} {'mIoU':>6} {'dStart':>7} {'dEnd':>7}",
117         "-" * 64,
118     ]
119     for stage in report.stages:
120         s = report.stages[stage].score
121         lines.append(
122             f" {stage:<12} {s.num_pred:>5} {s.true_positives:>4} {s.false_positives:>4}
123             f"{'s.false_negatives:>4} {s.precision:>6.2f} {s.recall:>6.2f} {s.f1:>6.2f} "
124             f"{'s.mean_iou:>6.2f} {s.mean_start_error:>6.1f}s {s.mean_end_error:>6.1f}s"
125         )
126     lines.append("=" * 64)
127
128     if show_failures:
129         for stage, sr in report.stages.items():
130             misses = [report.gt_intervals[i] for i in
131 sr.match_result.unmatched_gt_indices]
132             fps = [sr.pred_intervals[i] for i in sr.match_result.unmatched_pred_indices]
133             lines.append(f"\n [{stage}] missed rallies (false negatives):
134 {len(misses)}")
135             for iv in misses:
136                 lines.append(f"    - {_fmt_ts(iv[0])}?{_fmt_ts(iv[1])}")
137             lines.append(f" [{stage}] spurious detections (false positives):
138 {len(fps)}")
139             for iv in fps:
140                 lines.append(f"    - {_fmt_ts(iv[0])}?{_fmt_ts(iv[1])}")
141
142     return "\n".join(lines)
143
144
145

```

8. user (2026-06-10T05:59:55.798Z)

```

1     """Batch evaluation across a collector export manifest.
2
3     Phase 3 of the collector->indexer bridge: walk the `manifest.json` emitted by
4     sports-data-collector's `curate export`, (optionally) process each video through
5     a segmenter so detections land in the indexer DB, then score every video's
6     detected rallies against its ground-truth CSV and aggregate the results.
7
8     This module holds the *pure* pieces (path resolution, stage selection, metric
9     aggregation) so they can be unit-tested without touching video files, the GPU,
10    or any API. The orchestration/side-effects live in `run_phase3_eval.py`.
11    """
12
13    import os
14    from typing import Dict, List, Optional
15
16
17    # Segmenters that only ever populate the `final` play_segments table (no raw
18    # candidate stage), so scoring `candidates` for them would be meaninglessly empty.
19    _FINAL_ONLY_SEGMENTERS = {"motion", "gemini"}
20
21
22    def resolve_video_path(local_path: Optional[str], videos_root: str) -> Optional[str]:
23        """Resolve a manifest `local_path` to an absolute path.
24
25        The collector stores paths like ``./data/videos\\shuttleset_1.mp4`` relative
26        to its own repo root and with Windows separators. Normalise separators, strip
27        a leading ``./``, and resolve relative paths against `videos_root` (the
28        collector root). Returns None for a missing/empty path.
29        """
30        if not local_path:
31            return None
32        p = local_path.replace("\\", "/")

```

```

33     if p.startswith("./"):
34         p = p[2:]
35     # Cross-platform: treat Windows drive-letter absolute paths (C:/...) as absolute
36     # even when running on Linux (os.path.isabs("C:/...") == False on Unix).
37     # This keeps collector manifest paths working in tests and mixed environments.
38     if (len(p) > 1 and p[1] == ":" and p[0].isalpha()) or os.path.isabs(p):
39         return os.path.normpath(p)
40     return os.path.normpath(os.path.join(videos_root, p))
41
42
43 def default_stages_for(segmenter: str, requested: str = "auto") -> List[str]:
44     """Pick which prediction stages to score.
45
46     ``auto`` scores both stages for two-stage detectors (yolo_hybrid/tracknet_*)
47     but only ``final`` for segmenters that don't emit candidates. An explicit
48     request ('candidates' | 'final' | 'both') overrides the heuristic.
49     """
50     if requested == "both":
51         return ["candidates", "final"]
52     if requested in ("candidates", "final"):
53         return [requested]
54     # auto
55     if segmenter in _FINAL_ONLY_SEGMENTERS:
56         return ["final"]
57     return ["candidates", "final"]
58
59
60 def _safe_div(num: float, den: float) -> float:
61     return num / den if den else 0.0
62
63
64 def aggregate(per_video: List[dict], stages: List[str]) -> Dict[str, dict]:
65     """Aggregate per-video stage scores into corpus-level metrics.
66
67     `per_video` is a list of dicts shaped like ``report_to_dict`` output (each has
68     ``stages[stage]["score"]`` with tp/fp/fn/mean_iou/...). For each stage we
69     compute **micro** precision/recall/F1 (pooling TP/FP/FN across all videos) and
70     a TP-weighted mean IoU, plus the macro mean of per-video F1.
71     """
72     out: Dict[str, dict] = {}
73     for stage in stages:
74         tp = fp = fn = 0
75         iou_weighted = 0.0
76         f1s: List[float] = []
77         n_videos = 0
78         for rep in per_video:
79             sr = rep.get("stages", {}).get(stage)
80             if not sr:
81                 continue
82             s = sr["score"]
83             tp += s["true_positives"]
84             fp += s["false_positives"]
85             fn += s["false_negatives"]
86             iou_weighted += s["mean_iou"] * s["true_positives"]
87             f1s.append(s["f1"])
88             n_videos += 1
89         precision = _safe_div(tp, tp + fp)
90         recall = _safe_div(tp, tp + fn)
91         out[stage] = {
92             "videos": n_videos,
93             "tp": tp, "fp": fp, "fn": fn,
94             "micro_precision": precision,
95             "micro_recall": recall,
96             "micro_f1": _safe_div(2 * precision * recall, precision + recall),
97             "macro_f1": _safe_div(sum(f1s), len(f1s)),

```

```

98         "mean_iou": _safe_div(iou_weighted, tp),
99     }
100     return out
101
102
103 def format_aggregate(agg: Dict[str, dict]) -> str:
104     """Human-readable corpus-level summary table."""
105     lines = [
106         "=" * 78,
107         " CORPUS AGGREGATE",
108         "=" * 78,
109         f" {'Stage':<12} {'Vids':>4} {'TP':>5} {'FP':>5} {'FN':>5} "
110         f"{'mPrec':>6} {'mRec':>6} {'microF1':>8} {'macroF1':>8} {'mIoU':>6}",
111         "-" * 78,
112     ]
113     for stage, a in agg.items():
114         lines.append(
115             f" {stage:<12} {a['videos']:>4} {a['tp']:>5} {a['fp']:>5} {a['fn']:>5} "
116             f"{a['micro_precision']:>6.2f} {a['micro_recall']:>6.2f} "
117             f"{a['micro_f1']:>8.2f} {a['macro_f1']:>8.2f} {a['mean_iou']:>6.2f}"
118         )
119     lines.append("=" * 78)
120     return "\n".join(lines)
121

```

9. user (2026-06-10T05:59:56.288Z)

```

1     """Regression runner ? verify the production tool against golden-set ground truth.
2
3     Implements the explicit ask: *"given a video, get a golden set created and verify
4     its precision and recall against the production tool."* This is deliberately a thin
5     wrapper over machinery that already exists:
6
7     ground truth ? AnnotationProvider (file / vendor / oracle) [backend.annotations]
8     predictions  ? the production pipeline's stored DB rows    [backend.eval.harness]
9     comparison   ? interval matching + P/R/F1                  [backend.eval.metrics]
10    baseline     ? a per-video snapshot of "known good" numbers [this module]
11
12    A "regression" is a precision/recall drop beyond `tolerance` versus the snapshotted
13    baseline. Tiers map to annotation providers: e.g. tier "file" ? FileProvider (CI),
14    later tier "vendor" ? HumanVendorProvider (authoritative), tier "auto" ? an
15    independent-lineage oracle (fast smoke alarm). See docs/GOLDEN_SET_IMPLEMENTATION.md.
16    """
17    import os
18    import json
19    import logging
20    from dataclasses import dataclass, asdict
21    from typing import Dict, List, Optional
22
23    from backend.infrastructure.database import Database
24    from backend.eval import metrics
25    from backend.eval.harness import get_predictions
26    from backend.annotations import annotation_registry
27
28    logger = logging.getLogger(__name__)
29
30    DEFAULT_BASELINE_PATH = os.path.join("output", "regression_baseline.json")
31    DEFAULT_TOLERANCE = 0.05
32
33
34    @dataclass
35    class RegressionResult:
36        video_id: str
37        stage: str
38        iou_threshold: float

```

```

39         precision: float
40         recall: float
41         f1: float
42         # Baseline values compared against (None when no baseline existed yet).
43         baseline_precision: Optional[float]
44         baseline_recall: Optional[float]
45         tolerance: float
46         regressed: bool
47         reasons: List[str]
48
49     def as_dict(self) -> dict:
50         return asdict(self)
51
52
53     # ----- baseline store
54
55     def load_baselines(path: str = DEFAULT_BASELINE_PATH) -> Dict[str, dict]:
56         """Load the per-video baseline snapshot file (returns {} if absent)."""
57         if not os.path.isfile(path):
58             return {}
59         with open(path) as f:
60             return json.load(f)
61
62
63     def _baseline_key(video_id: str, stage: str) -> str:
64         return f"{video_id}::{stage}"
65
66
67     def save_baseline(video_id: str, stage: str, precision: float, recall: float,
68                     f1: float, path: str = DEFAULT_BASELINE_PATH) -> None:
69         """Snapshot a video/stage's current numbers as the new 'known good' baseline.
70
71         Use this to (re)baseline after a *legitimate* production improvement, so old
72         baselines don't raise false regressions. Stored as plain JSON for easy diffing.
73         """
74         baselines = load_baselines(path)
75         baselines[_baseline_key(video_id, stage)] = {
76             "video_id": video_id, "stage": stage,
77             "precision": precision, "recall": recall, "f1": f1,
78         }
79         os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
80         with open(path, "w") as f:
81             json.dump(baselines, f, indent=2, sort_keys=True)
82         logger.info("Saved baseline for %s (precision=%.3f recall=%.3f)",
83 _baseline_key(video_id, stage), precision, recall)
84
85     # ----- the runner
86
87     def regress(db: Database, video_id: str, ground_truth: List[metrics.Interval],
88               stage: str = "final", iou_threshold: float = 0.5,
89               detector: Optional[str] = None,
90               tolerance: float = DEFAULT_TOLERANCE,
91               baseline_path: str = DEFAULT_BASELINE_PATH) -> RegressionResult:
92         """Score stored predictions for one video against ground truth and compare to
93         baseline.
94
95         `ground_truth` is a list of (start, end) intervals ? typically obtained from an
96         AnnotationProvider (see `regress_with_provider`). A regression is flagged when
97         precision OR recall falls more than `tolerance` below the snapshotted baseline.
98         If no baseline exists yet, `regressed` is False (nothing to compare to) and the
99         caller can choose to snapshot the current numbers via `save_baseline`.
100         """
101         preds = get_predictions(db, video_id, stage, detector=detector)
102         s = metrics.score(preds, ground_truth, iou_threshold)

```

```

102
103     baselines = load_baselines(baseline_path)
104     b = baselines.get(_baseline_key(video_id, stage))
105
106     reasons: List[str] = []
107     regressed = False
108     base_p = base_r = None
109     if b is not None:
110         base_p, base_r = b["precision"], b["recall"]
111         if s.precision < base_p - tolerance:
112             regressed = True
113             reasons.append(f"precision {s.precision:.3f} < baseline {base_p:.3f} -
{tolerance}")
114         if s.recall < base_r - tolerance:
115             regressed = True
116             reasons.append(f"recall {s.recall:.3f} < baseline {base_r:.3f} -
{tolerance}")
117     else:
118         reasons.append("no baseline recorded for this video/stage ? nothing to compare")
119
120     return RegressionResult(
121         video_id=video_id, stage=stage, iou_threshold=iou_threshold,
122         precision=s.precision, recall=s.recall, f1=s.f1,
123         baseline_precision=base_p, baseline_recall=base_r,
124         tolerance=tolerance, regressed=regressed, reasons=reasons,
125     )
126
127
128 def regress_with_provider(db: Database, video_id: str, video_ref: str,
129                          tier: str = "file", stage: str = "final",
130                          iou_threshold: float = 0.5,
131                          detector: Optional[str] = None,
132                          tolerance: float = DEFAULT_TOLERANCE,
133                          baseline_path: str = DEFAULT_BASELINE_PATH,
134                          guideline: Optional[str] = None,
135                          sport: str = "badminton",
136                          provider_kwargs: Optional[dict] = None) -> RegressionResult:
137     """End-to-end: obtain ground truth from the `tier` annotation provider, then
regress.
138
139     `tier` is an annotation-provider registry key ("file" today; "vendor"/"auto"
140     once those providers land). `video_ref` is whatever that provider expects (a CSV
141     path for FileProvider, a video_id/URL for a vendor). `video_id` keys the stored
142     predictions in the DB (the file MD5).
143     """
144     provider = annotation_registry.get(tier, **(provider_kwargs or {}))
145     gt = provider.annotate(video_ref, guideline=guideline, sport=sport)
146     return regress(db, video_id, gt, stage=stage, iou_threshold=iou_threshold,
147                  detector=detector, tolerance=tolerance, baseline_path=baseline_path)
148
149
150 def format_result(r: RegressionResult) -> str:
151     verdict = "REGRESSED" if r.regressed else "ok"
152     head = (f"[{verdict}] {r.video_id} ({r.stage}) "
153            f"precision={r.precision:.3f} recall={r.recall:.3f} f1={r.f1:.3f}")
154     if r.baseline_precision is not None:
155         head += f" | baseline p={r.baseline_precision:.3f} r={r.baseline_recall:.3f}
tol={r.tolerance}"
156     return head + " ".join(f"\n    - {why}" for why in r.reasons)
157

```

10. user (2026-06-10T06:00:01.126Z)

```

1     """Build a training set for the offline, API-free learned rally segmenter.
2

```

```

3 Phase 4 of the collector->indexer bridge (see docs/ROADMAP.md, ADR-004). The idea:
4 a detector (YOLO player-motion today, optionally WASB shuttle-motion later)
5 produces candidate rally windows with features; the collector's ground-truth
6 rally intervals tell us which windows are real. A lightweight classifier learns
7 that mapping, replacing the paid Gemini refinement with a local model.
8
9 This module is the *data layer* ? pure and unit-tested:
10
11 - `label_candidates` ? generic: label windows positive/negative by whether they
12   match a ground-truth rally (same IoU matching the evaluator uses, so training
13   labels are consistent with how we score).
14 - `extract_yolo_features` ? YOLO-specific feature vector from a candidate row's
15   persisted `stats`. Pluggable: a WASB substrate supplies its own feature_fn.
16 - `build_training_set` ? glue: rows + GT -> (X, y, intervals).
17
18 No model training or I/O here; that lives in the trainer/segmenter once the
19 substrate is chosen from the Phase 3 candidate-recall result.
20 """
21
22 from typing import Callable, Dict, List, Tuple
23
24 from backend.eval.metrics import Interval, match_intervals
25
26 # YOLO Stage-1 persists these per candidate window in `candidate_segments.stats`
27 # (see HybridYoloSegmenter). Duration is derived from the window bounds.
28 YOLO_FEATURE_NAMES = [
29     "duration", "peak_velocity", "mean_velocity", "active_frame_count",
30     "track_id_count",
31 ]
32
33 def extract_yolo_features(row: Dict) -> List[float]:
34     """Feature vector for one YOLO candidate row (dict from query_candidate_segments).
35
36     `stats` is already JSON-deserialized by the DB layer. Missing fields default
37     to 0.0 so a sparse/old row still yields a well-formed vector.
38     """
39     stats = row.get("stats") or {}
40     duration = float(row["end_time"]) - float(row["start_time"])
41     return [
42         duration,
43         float(stats.get("peak_velocity", 0.0)),
44         float(stats.get("mean_velocity", 0.0)),
45         float(stats.get("active_frame_count", 0)),
46         float(stats.get("track_id_count", 0)),
47     ]
48
49
50 def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
51                     iou_threshold: float = 0.5) -> List[int]:
52     """Label each candidate window 1 (real rally) or 0 (spurious).
53
54     A window is positive iff it is matched one-to-one to a ground-truth rally at
55     `iou_threshold` ? the *same* matching the evaluator uses for true positives,
56     so the classifier is trained on exactly what we score. Substrate-agnostic:
57     works for YOLO or WASB-derived windows alike.
58     """
59     mr = match_intervals(candidate_intervals, gt_intervals, iou_threshold)
60     positive = {m.pred_index for m in mr.matches}
61     return [1 if i in positive else 0 for i in range(len(candidate_intervals))]
62
63
64 def build_training_set(
65     candidate_rows: List[Dict],
66     gt_intervals: List[Interval],

```

```

67     iou_threshold: float = 0.5,
68     feature_fn: Callable[[Dict], List[float]] = extract_yolo_features,
69 ) -> Tuple[List[List[float]], List[int], List[Interval]]:
70     """Turn one video's candidate rows + ground truth into (X, y, intervals).
71
72     `feature_fn` is pluggable so a WASB substrate can supply shuttle-motion
73     features without changing the labeling logic. Returns the feature matrix `X`,
74     binary labels `y`, and the candidate intervals (handy for downstream merging).
75     """
76     intervals: List[Interval] = [
77         (float(r["start_time"]), float(r["end_time"])) for r in candidate_rows
78     ]
79     y = label_candidates(intervals, gt_intervals, iou_threshold)
80     X = [feature_fn(r) for r in candidate_rows]
81     return X, y, intervals
82

```

11. user (2026-06-10T06:00:01.692Z)

```

1     """A tiny, dependency-free logistic-regression classifier (numpy only).
2
3     The offline learned segmenter (ADR-004) must run locally with no external API and
4     no heavy ML stack ? the project already depends on numpy, and sklearn is not
5     installed. This implements just enough: standardized features, balanced-class
6     gradient descent, sigmoid probabilities, and JSON (de)serialization so a trained
7     model is a small, human-readable, version-controllable file.
8
9     It deliberately stays simple: the rally-window features are few and roughly
10    linearly separable (high player motion + sustained activity => real rally), so a
11    well-regularized linear model is a sensible, inspectable baseline. Swap in a
12    richer model later behind the same fit/predict_proba/save/load interface.
13    """
14
15    import json
16    from typing import List, Optional
17
18    import numpy as np
19
20
21    class LogisticRegression:
22        def __init__(self, lr: float = 0.1, epochs: int = 1000, l2: float = 1e-3):
23            self.lr = lr
24            self.epochs = epochs
25            self.l2 = l2
26            self.w: Optional[np.ndarray] = None
27            self.b: float = 0.0
28            self.mean: Optional[np.ndarray] = None
29            self.std: Optional[np.ndarray] = None
30            self.feature_names: Optional[List[str]] = None
31
32        @staticmethod
33        def _sigmoid(z: np.ndarray) -> np.ndarray:
34            # Clip for numerical stability (avoid overflow in exp).
35            return 1.0 / (1.0 + np.exp(-np.clip(z, -30, 30)))
36
37        def _standardize(self, X: np.ndarray) -> np.ndarray:
38            return (X - self.mean) / self.std
39
40        def fit(self, X, y, feature_names: Optional[List[str]] = None,
41               class_weight: str = "balanced") -> "LogisticRegression":
42            """Fit with optional balanced class weighting (rally windows are imbalanced
43            ? many spurious candidates per real rally)."""
44            X = np.asarray(X, dtype=float)
45            y = np.asarray(y, dtype=float)
46            if X.ndim != 2 or X.shape[0] == 0:

```

```

47         raise ValueError("X must be a non-empty 2D array")
48     self.feature_names = feature_names
49
50     self.mean = X.mean(axis=0)
51     self.std = X.std(axis=0)
52     self.std[self.std == 0] = 1.0 # guard constant features
53     Xs = self._standardize(X)
54     n, d = Xs.shape
55
56     if class_weight == "balanced":
57         n_pos = max(1, int(y.sum()))
58         n_neg = max(1, int((1 - y).sum()))
59         w_pos, w_neg = n / (2.0 * n_pos), n / (2.0 * n_neg)
60     else:
61         w_pos = w_neg = 1.0
62     sw = np.where(y == 1, w_pos, w_neg)
63
64     self.w = np.zeros(d)
65     self.b = 0.0
66     for _ in range(self.epochs):
67         p = self._sigmoid(Xs @ self.w + self.b)
68         err = (p - y) * sw
69         grad_w = Xs.T @ err / n + self.l2 * self.w
70         grad_b = float(err.mean())
71         self.w -= self.lr * grad_w
72         self.b -= self.lr * grad_b
73     return self
74
75     def predict_proba(self, X) -> np.ndarray:
76         if self.w is None:
77             raise RuntimeError("model is not fitted")
78         Xs = self._standardize(np.asarray(X, dtype=float))
79         return self._sigmoid(Xs @ self.w + self.b)
80
81     def predict(self, X, threshold: float = 0.5) -> np.ndarray:
82         return (self.predict_proba(X) >= threshold).astype(int)
83
84     # --- persistence (small JSON, no pickle/joblib) -----
85
86     def to_dict(self) -> dict:
87         return {
88             "model": "logreg",
89             "lr": self.lr, "epochs": self.epochs, "l2": self.l2,
90             "w": self.w.tolist() if self.w is not None else None,
91             "b": self.b,
92             "mean": self.mean.tolist() if self.mean is not None else None,
93             "std": self.std.tolist() if self.std is not None else None,
94             "feature_names": self.feature_names,
95         }
96
97     @classmethod
98     def from_dict(cls, d: dict) -> "LogisticRegression":
99         m = cls(lr=d.get("lr", 0.1), epochs=d.get("epochs", 1000), l2=d.get("l2", 1e-3))
100         m.w = np.asarray(d["w"], dtype=float) if d.get("w") is not None else None
101         m.b = float(d.get("b", 0.0))
102         m.mean = np.asarray(d["mean"], dtype=float) if d.get("mean") is not None else
None
103         m.std = np.asarray(d["std"], dtype=float) if d.get("std") is not None else None
104         m.feature_names = d.get("feature_names")
105         return m
106
107     def save(self, path: str) -> None:
108         with open(path, "w") as f:
109             json.dump(self.to_dict(), f, indent=2)
110

```

```

111     @classmethod
112     def load(cls, path: str) -> "LogisticRegression":
113         with open(path) as f:
114             return cls.from_dict(json.load(f))
115

```

12. user (2026-06-10T06:00:06.088Z)

```

1     """Per-video calibration driver: WASB trajectory + golden rallies -> tuned thresholds +
"+X%".
2
3     Wires the already-tested pieces together:
4         WASB trajectory CSV (Frame,Visibility,X,Y)
5         -> TrackNetRunner.trajectory_to_action_windows(cfg)    [predict_fn]
6         -> backend.eval.calibration (improvement_report / cross_validate)
7
8     Run (after a full-video WASB pass exists):
9         python -m backend.eval.calibrate_wasb --trajectory full_traj.csv \
10             --labels "golden_labels_badminton.csv" --fps 59.94 --frame-width 1920
11
12     Honesty note: the grid is *selected* on the full GT, so `improvement_report` on that
13     same GT is an **optimistic** upper bound. The trustworthy number is the
14     cross-validated held-out **recall** (and, once a 2nd labelled video exists,
15     `leave_one_video_out` for precision/F1 generalization).
16     """
17     from __future__ import annotations
18
19     import csv
20     import argparse
21     from typing import List, Tuple, Callable
22
23     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
24     from backend.eval.calibration import (
25         select_best, improvement_report, cross_validate, evaluate,
26     )
27
28     Interval = Tuple[float, float]
29     Config = Tuple[float, float, float]    # (velocity_thresh, merge_gap,
min_window_duration)
30
31     BASELINE: Config = (0.02, 2.0, 2.0)    # the WasbConfig / trajectory_to_action_windows
defaults
32
33
34     def load_golden_gts(csv_path: str) -> List[Interval]:
35         """Rally [start,end] seconds from the golden CSV (start_time/end_time cols)."""
36         gts: List[Interval] = []
37         with open(csv_path, newline="", encoding="utf-8") as f:
38             reader = csv.DictReader(f)
39             reader.fieldnames = [(n or "").strip().lower() for n in (reader.fieldnames or
[])]
40             for row in reader:
41                 row = {(k or "").strip().lower(): v for k, v in row.items()}
42                 try:
43                     s = float((row.get("start_time") or "").strip())
44                     e = float((row.get("end_time") or "").strip())
45                 except (TypeError, ValueError):
46                     continue
47                 if e > s:
48                     gts.append((s, e))
49             return sorted(gts)
50
51
52     def make_predict_fn(trajectory_csv: str, fps: float, frame_width: float) ->
Callable[[Config], List[Interval]]:

```

```

53     """predict_fn(cfg) -> rally windows, from a cached WASB trajectory (parsed once)."""
54     points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
55
56     def predict_fn(cfg: Config) -> List[Interval]:
57         vt, mg, mwd = cfg
58         wins = TrackNetRunner.trajectory_to_action_windows(
59             points, fps=fps, frame_width=frame_width,
60             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd,
61         )
62         return [(w["start"], w["end"]) for w in wins]
63
64     return predict_fn
65
66
67     def default_grid() -> List[Config]:
68         return [(vt, mg, mwd)
69                 for vt in (0.005, 0.01, 0.02, 0.03, 0.05)
70                 for mg in (1.0, 2.0, 3.0)
71                 for mwd in (1.0, 2.0, 3.0)]
72
73
74     def run(trajectory_csv: str, labels_csv: str, fps: float, frame_width: float,
75            iou: float = 0.5) -> dict:
76         gts = load_golden_gts(labels_csv)
77         if not gts:
78             raise SystemExit(f"no usable golden rallies in {labels_csv}")
79         predict_fn = make_predict_fn(trajectory_csv, fps, frame_width)
80         grid = default_grid()
81
82         base = evaluate(predict_fn(BASELINE), gts, iou)
83         best_cfg, best_f1 = select_best(predict_fn, grid, gts, metric="f1",
84 iou_threshold=iou)
85         fit = improvement_report(predict_fn, BASELINE, best_cfg, gts, iou) # OPTIMISTIC
86         (fit on full GT)
87         cv = cross_validate(predict_fn, grid, gts, k=5) # HONEST held-
88 out recall
89
90         print("=== WASB rally-segmentation calibration ===")
91         print(f"golden rallies: {len(gts)} | IoU thresh: {iou}")
92         print(f"baseline cfg {BASELINE}: precision={base.precision:.3f}
93 recall={base.recall:.3f} "
94 f"f1={base.f1:.3f} meanIoU={base.mean_iou:.3f}")
95         print(f"tuned cfg {best_cfg}: f1={best_f1:.3f}")
96         print(f" precision {fit['baseline']['precision']:.3f} ->
97 {fit['tuned']['precision']:.3f} "
98 f"({fit['precision_delta_pct']:+.1f}% , fit-on-full = optimistic)")
99         print(f" f1 {fit['baseline']['f1']:.3f} -> {fit['tuned']['f1']:.3f} "
100 f"({fit['f1_delta_pct']:+.1f}%)")
101         print(f"HONEST (cross-val, {cv.get('folds','?')} folds): held-out recall "
102 f"{cv.get('mean_test',float('nan')):.3f} ± {cv.get('std_test',0):.3f}, "
103 f"generalization_gap {cv.get('generalization_gap',float('nan')):+.3f}")
104         print("(For precision/F1 generalization, add a 2nd labelled video ->
105 leave_one_video_out.)")
106         return {"baseline": base.as_dict(), "best_cfg": best_cfg, "fit": fit, "cv": cv}
107
108
109     def main():
110         ap = argparse.ArgumentParser()
111         ap.add_argument("--trajectory", required=True, help="WASB trajectory CSV
112 (Frame,Visibility,X,Y)")
113         ap.add_argument("--labels", required=True, help="golden labels CSV
114 (start_time/end_time cols)")
115         ap.add_argument("--fps", type=float, default=59.94)
116         ap.add_argument("--frame-width", type=float, default=1920.0)
117         ap.add_argument("--iou", type=float, default=0.5)

```

```

110     args = ap.parse_args()
111     run(args.trajectory, args.labels, args.fps, args.frame_width, args.iou)
112
113
114     if __name__ == "__main__":
115         main()
116

```

13. user (2026-06-10T06:00:10.754Z)

```

1     """Distill Gemini (teacher) into the LOCAL-ONLY pipeline (student) ? calibrate the
2     on-device WASB+heuristic knobs to match Gemini's rally labels, so production runs
3     100% local (no cloud, privacy lever intact, ~zero marginal cost).
4
5     Pipeline being tuned (all local, no Gemini at runtime):
6         WASB trajectory CSV -> trajectory_to_action_windows(velocity, merge_gap, min_dur)
7                             -> rally_gate (min_crossings)                -> rally windows
8
9     Ground truth = Gemini full-context silver labels (one CSV per video). With >=2
10    videos we use leave-one-video-out (tune on the others, score the held-out one) ?
11    the honest "how precise is the local pipeline on an unseen video" number. The
12    output is a single local config + the precision it buys, with NO Gemini dependency
13    shipped.
14
15    Reuses: TrackNetRunner windowing, rally_gate, calibrate_wasb.load_golden_gts,
16    calibration.{select_best, leave_one_video_out, improvement_report}.
17
18    Run:
19        python -m backend.eval.calibrate_local --manifest videos.json
20    where videos.json = [{"name","trajectory","fps","frame_width","labels"}, ...]
21    (labels = a Gemini full-context CSV; trajectory = the WASB CSV).
22    """
23    from __future__ import annotations
24
25    import json
26    import argparse
27    from typing import Callable, Dict, List, Tuple
28
29    from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
30    from backend.pipeline.detectors import rally_gate
31    from backend.eval.calibrate_wasb import load_golden_gts
32    from backend.eval.calibration import select_best, leave_one_video_out,
improvement_report, evaluate
33
34    # (velocity_thresh, merge_gap, min_window_duration, min_crossings)
35    LocalConfig = Tuple[float, float, float, int]
36    BASELINE_LOCAL: LocalConfig = (0.02, 2.0, 2.0, 0) # today's local default (windowing,
no gate)
37
38
39    def make_local_predict_fn(trajectory_csv: str, fps: float, frame_width: float) ->
Callable[[LocalConfig], list]:
40        """predict_fn(cfg) -> rally windows from a cached WASB trajectory, LOCAL ONLY
41        (windowing thresholds + the shuttle net-crossing gate). Parsed once."""
42        points = TrackNetRunner.parse_trajectory_csv(trajectory_csv)
43
44        def predict_fn(cfg: LocalConfig) -> List[Tuple[float, float]]:
45            vt, mg, mwd, mc = cfg
46            wins = TrackNetRunner.trajectory_to_action_windows(
47                points, fps=fps, frame_width=frame_width,
48                velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd,
49            )
50            wins = [(w["start"], w["end"]) for w in wins]
51            if mc > 0:
52                wins = rally_gate.apply_gate(points, wins, fps, min_crossings=mc)

```

```

53         return wins
54
55     return predict_fn
56
57
58     def local_grid() -> List[LocalConfig]:
59         return [(vt, mg, mwd, mc)
60                 for vt in (0.005, 0.01, 0.02, 0.03, 0.05)
61                 for mg in (1.0, 2.0, 3.0)
62                 for mwd in (1.0, 2.0, 3.0)
63                 for mc in (0, 1, 2, 3)]
64
65
66     def build_videos(specs: List[Dict]) -> List[Dict]:
67         videos = []
68         for s in specs:
69             pf = make_local_predict_fn(s["trajectory"], float(s["fps"]),
float(s["frame_width"]))
70             gts = load_golden_gts(s["labels"])
71             if not gts:
72                 raise SystemExit(f"no usable labels in {s['labels']} for {s.get('name')}")
73             videos.append({"name": s.get("name", s["trajectory"]), "predict_fn": pf, "gts":
gts})
74         return videos
75
76
77     def run(specs: List[Dict], iou: float = 0.5, metric: str = "f1") -> dict:
78         videos = build_videos(specs)
79         grid = local_grid()
80         print(f"=== Local-only calibration vs Gemini silver-GT ({len(videos)} videos, "
81               f"{len(grid)} configs, IoU>={iou}, optimise {metric}) ===")
82
83         # Per-video: today's baseline vs the best-fit local config (optimistic ? fit on that
video).
84         per_video = []
85         for v in videos:
86             best_cfg, best_val = select_best(v["predict_fn"], grid, v["gts"], metric, iou)
87             imp = improvement_report(v["predict_fn"], BASELINE_LOCAL, best_cfg, v["gts"],
iou)
88             b, t = imp["baseline"], imp["tuned"]
89             print(f"\n[{v['name']}] {len(v['gts'])} Gemini rallies")
90             print(f"  baseline {BASELINE_LOCAL}: P={b['precision']:.3f} R={b['recall']:.3f}
F1={b['f1']:.3f}")
91             print(f"  best-fit {best_cfg}: P={t['precision']:.3f} R={t['recall']:.3f}
F1={t['f1']:.3f}")
92             f"(F1 {imp['f1_delta_pct']:+.0f}%, fit-on-self = optimistic)"
93             per_video.append({"name": v["name"], "best_cfg": best_cfg, "improvement": imp})
94
95         # Honest cross-video: pick config on the OTHER videos, score the held-out one.
96         lovo = leave_one_video_out(videos, grid, metric, iou)
97         # Baseline's held-out score (no tuning) for the honest before/after.
98         base_held = [evaluate(v["predict_fn"](BASELINE_LOCAL), v["gts"], iou) for v in
videos]
99         base_mean = sum(getattr(s, metric) for s in base_held) / len(base_held)
100
101         print("\n--- HONEST leave-one-video-out (the number to trust) ---")
102         if "error" in lovo:
103             print(f"  {lovo['error']} (need >=2 videos)")
104         else:
105             print(f"  baseline (untuned) mean held-out {metric}: {base_mean:.3f}")
106             print(f"  calibrated mean held-out {metric}: {lovo['mean_held_out']:.3f}")
107             f"± {lovo['std_held_out']:.3f} (gap {lovo['generalization_gap']:+.3f}")
108             for p in lovo["per_video"]:
109                 print(f"    held-out {p['video']}: {p['held_out']:.3f} (cfg picked on

```

```

others: {p['config']})")
110     return {"per_video": per_video, "lovo": lovo, "baseline_held_out": base_mean,
"metric": metric}
111
112
113     def main() -> None:
114         ap = argparse.ArgumentParser(description="Calibrate the local-only pipeline against
Gemini silver labels.")
115         ap.add_argument("--manifest", required=True,
116                         help="JSON: [{name, trajectory, fps, frame_width, labels}, ...]")
117         ap.add_argument("--iou", type=float, default=0.5)
118         ap.add_argument("--metric", default="f1", choices=["f1", "precision", "recall"])
119         args = ap.parse_args()
120         with open(args.manifest, encoding="utf-8") as f:
121             specs = json.load(f)
122         run(specs, iou=args.iou, metric=args.metric)
123
124
125     if __name__ == "__main__":
126         main()
127

```

14. user (2026-06-10T06:00:15.071Z)

```

1     """Distill Gemini (teacher) into a LEARNED local rally classifier (student).
2
3     Threshold-tuning the local knobs hit a low ceiling (see calibrate_local). This goes
4     further: learn "is this WASB candidate a real rally?" from Gemini silver labels,
5     using a few resolution/fps-INVARIANT local features, so the model transfers across
6     videos and runs 100% local at inference (no Gemini, privacy lever intact).
7
8     Flow (all local at runtime):
9         WASB trajectory -> recall-oriented candidate windows (propose many; classifier
10         filters) -> per-candidate features [duration, peak_velocity, mean_velocity,
11         active_ratio, net_crossings] -> LogisticRegression(prob >= threshold) -> merge
12         -> rally windows.
13
14     Training labels come from Gemini full-context labels via the SAME IoU matching the
15     evaluator uses (training.label_candidates). Honest leave-one-video-out: train on the
16     other videos' candidates, predict the held-out one, score vs its Gemini labels, and
17     compare against the threshold baseline. Reuses classifier.py, training.label_candidates,
18     rally_gate, metrics, calibrate_wasb.load_golden_gts, gemini_refine.merge_overlaps.
19
20     Run:
21         python -m backend.eval.distill_local --manifest videos.json --model-out
output/local_rally_model.json
22     """
23     from __future__ import annotations
24
25     import json
26     import argparse
27     from typing import Dict, List, Tuple
28
29     import numpy as np
30
31     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
32     from backend.pipeline.detectors import rally_gate
33     from backend.eval.calibrate_wasb import load_golden_gts
34     from backend.eval.training import label_candidates
35     from backend.eval.classifier import LogisticRegression
36     from backend.eval.metrics import score
37     from backend.eval.gemini_refine import merge_overlaps
38
39     Interval = Tuple[float, float]
40

```

```

41     FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
42     # Recall-oriented proposal: deliberately permissive so real rallies are among the
43     # candidates (the classifier removes the junk). (velocity_thresh, merge_gap, min_dur)
44     PROPOSAL = (0.005, 1.0, 0.5)
45     BASELINE_LOCAL = (0.02, 2.0, 2.0) # today's production-ish windowing, no learning
46
47
48     def build_candidates(trajecory_csv: str, fps: float, frame_width: float,
49                         proposal: Tuple[float, float, float] = PROPOSAL) ->
Tuple[List[Interval], List[List[float]]]:
50         """WASB trajectory -> (candidate intervals, fps/resolution-invariant feature
rows)."""
51         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
52         vt, mg, mwd = proposal
53         wins = TrackNetRunner.trajectory_to_action_windows(
54             points, fps=fps, frame_width=frame_width,
55             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
56         intervals = [(w["start"], w["end"]) for w in wins]
57         gated = rally_gate.gate_windows(points, intervals, fps, min_crossings=0) # for net-
crossings
58         X: List[List[float]] = []
59         for w, g in zip(wins, gated):
60             dur = max(1e-6, w["end"] - w["start"])
61             win_frames = max(1.0, dur * fps)
62             X.append([
63                 dur, # rally length (sec) ?
64                 float(w["peak_velocity"]), # already normalized by
65                 float(w["mean_velocity"]),
66                 float(w["active_frame_count"]) / win_frames, # active-shuttle ratio ?
67                 float(g.crossings), # net crossings (count) ?
68             ])
69         return intervals, X
70
71
72     def baseline_windows(trajecory_csv: str, fps: float, frame_width: float) ->
List[Interval]:
73         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
74         vt, mg, mwd = BASELINE_LOCAL
75         wins = TrackNetRunner.trajectory_to_action_windows(
76             points, fps=fps, frame_width=frame_width,
77             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
78         return [(w["start"], w["end"]) for w in wins]
79
80
81     def load_video(spec: Dict, iou: float = 0.5) -> Dict:
82         fps, fw = float(spec["fps"]), float(spec["frame_width"])
83         intervals, X = build_candidates(spec["trajectory"], fps, fw)
84         gts = load_golden_gts(spec["labels"])
85         y = label_candidates(intervals, gts, iou)
86         return {"name": spec.get("name", spec["trajectory"]), "intervals": intervals, "X":
X,
87                 "y": y, "gts": gts, "baseline": baseline_windows(spec["trajectory"], fps,
fw)}
88
89
90     def predict_rallies(model: LogisticRegression, X: List[List[float]], intervals:
List[Interval],
91                         threshold: float = 0.5) -> List[Interval]:
92         if not intervals:
93             return []

```

```

94     proba = model.predict_proba(X)
95     pos = [iv for iv, p in zip(intervals, proba) if p >= threshold]
96     return merge_overlaps(pos, gap_tol=1.0)
97
98
99     def run(specs: List[Dict], iou: float = 0.5, threshold: float = 0.5, model_out: str =
None) -> dict:
100         videos = [load_video(s, iou) for s in specs]
101         print(f"=== Distilled local rally classifier vs Gemini silver-GT "
102               f"({len(videos)} videos, IoU>={iou}, prob>={threshold}) ===")
103         for v in videos:
104             ceil = score(v["intervals"], v["gts"], iou).recall
105             print(f"[{v['name']}] {len(v['intervals'])} candidates ({sum(v['y'])} pos) | "
106                   f"proposal recall ceiling {ceil:.3f} | {len(v['gts'])} Gemini rallies")
107
108         result: dict = {"videos": [v["name"] for v in videos]}
109         if len(videos) >= 2:
110             print("\n--- HONEST leave-one-video-out: learned vs threshold baseline (held-
out) ---")
111             clf, base = [], []
112             for i, held in enumerate(videos):
113                 others = [v for j, v in enumerate(videos) if j != i]
114                 X = [row for v in others for row in v["X"]]
115                 y = [t for v in others for t in v["y"]]
116                 model = LogisticRegression().fit(X, y, FEATURE_NAMES)
117                 preds = predict_rallies(model, held["X"], held["intervals"], threshold)
118                 sc = score(preds, held["gts"], iou)
119                 bsc = score(held["baseline"], held["gts"], iou)
120                 clf.append(sc)
121                 base.append(bsc)
122                 print(f"    held-out {held['name']}: LEARNED P={sc.precision:.3f}
R={sc.recall:.3f} F1={sc.f1:.3f}"
123                       f"    | baseline P={bsc.precision:.3f} R={bsc.recall:.3f}
F1={bsc.f1:.3f}")
124             mean = lambda xs, a: sum(getattr(s, a) for s in xs) / len(xs)
125             print(f"    MEAN held-out: LEARNED P={mean(clf, 'precision'):.3f}
R={mean(clf, 'recall'):.3f} "
126                   f"F1={mean(clf, 'f1'):.3f} | baseline P={mean(base, 'precision'):.3f} "
127                   f"F1={mean(base, 'f1'):.3f} R={mean(base, 'recall'):.3f} F1={mean(base, 'f1'):.3f}")
128             result["mean_learned_f1"] = mean(clf, "f1")
129             result["mean_baseline_f1"] = mean(base, "f1")
130
131             # Final model on ALL videos (for production / running on a fresh held-out video).
132             Xall = [row for v in videos for row in v["X"]]
133             yall = [t for v in videos for t in v["y"]]
134             final = LogisticRegression().fit(Xall, yall, FEATURE_NAMES)
135             weights = dict(zip(FEATURE_NAMES, [round(float(w), 3) for w in final.w]))
136             print(f"\nfinal model standardized weights (sign/size = influence): {weights}")
137             if model_out:
138                 final.save(model_out)
139                 print(f"final model (all videos) -> {model_out}")
140                 result["model_out"] = model_out
141             return result
142
143
144     def main() -> None:
145         ap = argparse.ArgumentParser(description="Distill Gemini labels into a local rally
classifier.")
146         ap.add_argument("--manifest", required=True,
147                         help="JSON: [{name, trajectory, fps, frame_width, labels}, ...]
(labels = Gemini CSV)")
148         ap.add_argument("--iou", type=float, default=0.5)
149         ap.add_argument("--threshold", type=float, default=0.5, help="classifier probability
cutoff")
150         ap.add_argument("--model-out", default="output/local_rally_model.json")

```

```

151     args = ap.parse_args()
152     with open(args.manifest, encoding="utf-8") as f:
153         specs = json.load(f)
154         run(specs, iou=args.iou, threshold=args.threshold, model_out=args.model_out)
155
156
157 if __name__ == "__main__":
158     main()
159

```

15. user (2026-06-10T06:00:19.644Z)

```

1     """Export tuned WASB rally segments for manual eyeball verification.
2
3     After calibration picks a tuned threshold config, this turns a cached WASB
4     trajectory into the actual rally [start,end] segments that config produces, so the
5     user can answer "are the rallies actually better?" ? by scrubbing the video to each
6     segment, by diffing tuned-vs-golden quantitatively, or by cutting clips.
7
8     Reuses (no new windowing/metrics/cutting logic):
9     - calibrate_wasb.make_predict_fn -> trajectory -> segments for a given config
10    - eval.metrics.match_intervals/score -> tuned-vs-golden IoU comparison
11    - tools.rally_slicer.slice_rallies -> optional clip cutting (the exported CSV is
12      written in the golden/slicer schema, so it feeds straight back in)
13
14    Run (uses the tuned config from the first calibration by default):
15    python -m backend.eval.export_wasb_segments --trajectory
~/clips/sample_long_traj.csv \
16        --labels "Annotation Setup/golden_labels_badminton.csv" --fps 59.94 --frame-
width 1920
17        # add --video <mp4> --slice to also cut one padded clip per tuned rally
18    """
19    from __future__ import annotations
20
21    import csv
22    import os
23    import os.path as osp
24    import argparse
25    from typing import List, Tuple, Dict, Optional
26
27    from backend.eval.calibrate_wasb import make_predict_fn, load_golden_gts, BASELINE
28    from backend.eval.metrics import match_intervals, score
29    from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
30    from backend.pipeline.detectors import rally_gate
31
32    Interval = Tuple[float, float]
33    Config = Tuple[float, float, float]
34
35    # The tuned config from the first per-video calibration (velocity, merge_gap, min_dur);
36    # F1 0.588 (baseline) -> 0.746. Override on the CLI to export a different operating
point.
37    TUNED: Config = (0.05, 1.0, 1.0)
38
39    SEG_FIELDS = ["rally_number", "start_time", "end_time", "duration", "start_mmss",
"end_mmss"]
40    COMP_FIELDS = ["type", "ref", "start_time", "end_time", "mmss", "status",
41                  "iou", "pred_start", "pred_end", "start_err_s", "end_err_s"]
42
43
44    def seconds_to_mmss(t: float) -> str:
45        """Human-scrubbable timestamp, e.g. 106.031 -> '1:46.031'."""
46        t = max(0.0, float(t))
47        m = int(t // 60)
48        return f"{m}:{t - m * 60:06.3f}"
49

```

```

50
51 def segments_to_rows(segments: List[Interval]) -> List[Dict[str, str]]:
52     """Number segments 1..N in time order; emit the golden/slicer CSV schema."""
53     rows = []
54     for i, (s, e) in enumerate(segments, start=1):
55         rows.append({
56             "rally_number": str(i),
57             "start_time": f"{s:.3f}",
58             "end_time": f"{e:.3f}",
59             "duration": f"{e - s:.3f}",
60             "start_mmss": seconds_to_mmss(s),
61             "end_mmss": seconds_to_mmss(e),
62         })
63     return rows
64
65
66 def _ensure_parent(path: str) -> None:
67     os.makedirs(osp.dirname(osp.abspath(path)) or ".", exist_ok=True)
68
69
70 def write_segments_csv(path: str, segments: List[Interval]) -> str:
71     """Write segments in the golden schema (rally_number/start_time/end_time + mm:ss).
72
73     The schema is what tools.rally_slicer.load_rally_labels consumes, so the file
74     doubles as input to clip cutting.
75     """
76     _ensure_parent(path)
77     with open(path, "w", newline="", encoding="utf-8") as f:
78         w = csv.DictWriter(f, fieldnames=SEG_FIELDS)
79         w.writeheader()
80         w.writerows(segments_to_rows(segments))
81     return path
82
83
84 def build_comparison(preds: List[Interval], gts: List[Interval],
85                     iou_threshold: float = 0.5) -> dict:
86     """Golden-centric comparison: per golden rally, did a tuned segment match (IoU),
87     plus the boundary errors and any extra (false-positive) tuned segments."""
88     mr = match_intervals(preds, gts, iou_threshold)
89     sc = score(preds, gts, iou_threshold, match_result=mr)
90     by_gt = {m.gt_index: m for m in mr.matches}
91
92     rows: List[Dict[str, str]] = []
93     for gi, (gs, ge) in enumerate(gts, start=0):
94         m = by_gt.get(gi)
95         if m is not None:
96             ps, pe = preds[m.pred_index]
97             rows.append({
98                 "type": "golden", "ref": str(gi + 1),
99                 "start_time": f"{gs:.3f}", "end_time": f"{ge:.3f}",
100                 "mmss": f"{seconds_to_mmss(gs)}-{seconds_to_mmss(ge)}",
101                 "status": "matched", "iou": f"{m.iou:.3f}",
102                 "pred_start": f"{ps:.3f}", "pred_end": f"{pe:.3f}",
103                 "start_err_s": f"{ps - gs:+.3f}", "end_err_s": f"{pe - ge:+.3f}",
104             })
105         else:
106             rows.append({
107                 "type": "golden", "ref": str(gi + 1),
108                 "start_time": f"{gs:.3f}", "end_time": f"{ge:.3f}",
109                 "mmss": f"{seconds_to_mmss(gs)}-{seconds_to_mmss(ge)}",
110                 "status": "MISS", "iou": "0.000",
111                 "pred_start": "", "pred_end": "", "start_err_s": "", "end_err_s": "",
112             })
113     for pi in mr.unmatched_pred_indices:
114         # extra tuned segments = false
115         positives

```

```

114         ps, pe = preds[pi]
115         rows.append({
116             "type": "pred_extra", "ref": str(pi + 1),
117             "start_time": f"{ps:.3f}", "end_time": f"{pe:.3f}",
118             "mmss": f"{seconds_to_mmss(ps)}-{seconds_to_mmss(pe)}",
119             "status": "false_positive", "iou": "",
120             "pred_start": "", "pred_end": "", "start_err_s": "", "end_err_s": "",
121         })
122     return {"score": sc, "rows": rows}
123
124
125 def write_comparison_csv(path: str, comparison: dict) -> str:
126     _ensure_parent(path)
127     with open(path, "w", newline="", encoding="utf-8") as f:
128         w = csv.DictWriter(f, fieldnames=COMP_FIELDS)
129         w.writeheader()
130         w.writerows(comparison["rows"])
131     return path
132
133
134 def _with_suffix(path: str, suffix: str) -> str:
135     root, ext = osp.splitext(path)
136     return root + suffix + ext
137
138
139 def run(trajecory_csv: str, fps: float, frame_width: float, cfg: Config = TUNED,
140         out_csv: str = "output/wasb_tuned_segments.csv", labels_csv: Optional[str] =
None,
141         iou: float = 0.5, baseline: bool = False,
142         comparison_out: Optional[str] = None, min_crossings: int = 0) -> dict:
143     predict_fn = make_predict_fn(trajecory_csv, fps, frame_width)
144     # Gate A: drop between-point false fires (single service-feed tosses) by requiring
145     # the shuttle to cross the net >= min_crossings times within the window.
146     points = TrackNetRunner.parse_trajectory_csv(trajecory_csv) if min_crossings > 0
147 else None
148     def _gate(segs: List[Interval], label: str) -> List[Interval]:
149         if not points:
150             return segs
151         kept = rally_gate.apply_gate(points, segs, fps, min_crossings=min_crossings)
152         print(f"[gate] {label}: {len(segs)} -> {len(kept)} segments ")
153         f"(removed {len(segs) - len(kept)} with <{min_crossings} net-crossings)")
154         return kept
155
156     tuned_segs = _gate(predict_fn(cfg), "tuned")
157     write_segments_csv(out_csv, tuned_segs)
158     total = sum(e - s for s, e in tuned_segs)
159     print("=== WASB tuned-segment export ===")
160     print(f"tuned cfg {cfg}: {len(tuned_segs)} rally segments, {total:.1f}s of play ->
{out_csv}")
161     result: dict = {"tuned_cfg": cfg, "num_tuned": len(tuned_segs), "out_csv": out_csv}
162
163     if baseline:
164         base_segs = _gate(predict_fn(BASELINE), "baseline")
165         base_out = _with_suffix(out_csv, "_baseline")
166         write_segments_csv(base_out, base_segs)
167         print(f"baseline cfg {BASELINE}: {len(base_segs)} segments -> {base_out}")
168         result["num_baseline"] = len(base_segs)
169         result["baseline_csv"] = base_out
170
171     if labels_csv:
172         gts = load_golden_gts(labels_csv)
173         if not gts:
174             raise SystemExit(f"no usable golden rallies in {labels_csv}")
175         comp = build_comparison(tuned_segs, gts, iou)

```

```

176         sc = comp["score"]
177         comp_out = comparison_out or _with_suffix(out_csv, "_vs_golden")
178         write_comparison_csv(comp_out, comp)
179         print(f"\n--- tuned vs golden ({len(gts)} rallies, IoU>={iou}) ---")
180         print(f"matched (TP) {sc.true_positives} | misses (FN) {sc.false_negatives} | "
181               f"extra (FP) {sc.false_positives}")
182         print(f"precision {sc.precision:.3f} | recall {sc.recall:.3f} | f1 {sc.f1:.3f} | "
183               f"meanIoU {sc.mean_iou:.3f}")
184         print(f"mean boundary error: start {sc.mean_start_error:.2f}s | end "
185               f"{sc.mean_end_error:.2f}s")
186         print(f"per-rally detail -> {comp_out}")
187         result["comparison"] = sc.as_dict()
188         result["comparison_csv"] = comp_out
189     return result
190
191 def _maybe_slice(video: str, segments_csv: str, clips_dir: str) -> None:
192     """Cut one padded clip per exported rally (reuses the rally slicer)."""
193     from backend.tools.rally_slicer import slice_rallies
194     written = slice_rallies(video, segments_csv, ["all"], clips_dir)
195     print(f"\ncut {len(written)} clip(s) -> {clips_dir}")
196
197 def main() -> None:
198     ap = argparse.ArgumentParser(description="Export tuned WASB rally segments for "
199                                     "eyeballing.")
200     ap.add_argument("--trajectory", required=True, help="WASB trajectory CSV "
201                 "(Frame,Visibility,X,Y)")
202     ap.add_argument("--fps", type=float, default=59.94)
203     ap.add_argument("--frame-width", type=float, default=1920.0)
204     ap.add_argument("--velocity-thresh", type=float, default=TUNED[0])
205     ap.add_argument("--merge-gap", type=float, default=TUNED[1])
206     ap.add_argument("--min-duration", type=float, default=TUNED[2])
207     ap.add_argument("--labels", default=None, help="golden labels CSV -> tuned-vs-golden "
208                 "comparison")
209     ap.add_argument("--iou", type=float, default=0.5)
210     ap.add_argument("--out", default="output/wasb_tuned_segments.csv")
211     ap.add_argument("--comparison-out", default=None)
212     ap.add_argument("--baseline", action="store_true", help="also export baseline-config "
213                 "segments")
214     ap.add_argument("--min-crossings", type=int, default=0,
215                 help="gate A: drop windows whose shuttle crosses the net < N times "
216                 "(kills single-toss service-feed false fires; 0=off, 2 "
217                 "recommended)")
218     ap.add_argument("--video", default=None, help="source video (with --slice) to cut "
219                 "clips from")
220     ap.add_argument("--slice", action="store_true", help="cut one padded clip per tuned "
221                 "rally")
222     ap.add_argument("--clips-dir", default="output/wasb_tuned_clips")
223     args = ap.parse_args()
224
225     cfg = (args.velocity_thresh, args.merge_gap, args.min_duration)
226     res = run(args.trajectory, args.fps, args.frame_width, cfg=cfg, out_csv=args.out,
227               labels_csv=args.labels, iou=args.iou, baseline=args.baseline,
228               comparison_out=args.comparison_out, min_crossings=args.min_crossings)
229     if args.slice:
230         if not args.video:
231             raise SystemExit("--slice requires --video")
232         _maybe_slice(args.video, res["out_csv"], args.clips_dir)
233
234 if __name__ == "__main__":
235     main()
236

```

16. user (2026-06-10T06:00:26.497Z)

```
1      """Refine WASB rally candidates with Gemini (verify boundaries + reject false fires).
2
3      WASB (shuttle-only) over-fires on between-point service feeds. This drives the
4      EXISTING Gemini provider over the SHORT WASB candidate clips (not the whole video):
5      each candidate window is cut to a small padded clip and handed to Gemini, which ?
6      seeing the actual footage ? returns the true rally inside it (tight boundaries) or
7      nothing (it was a toss / dead time). Cheap because uploads are seconds-long clips,
8      not GB-scale chunks; high-precision because Gemini judges each candidate visually.
9
10     Reuses (no reinvented wheels): providers.GeminiProvider (via provider_registry),
11     sports.badminton prompt (sport_registry), utils.video.extract_video_chunk,
12     calibrate_wasb.make_predict_fn, export_wasb_segments.write_segments_csv.
13
14     This mirrors wasb_hybrid's Phase-3 handoff but runs standalone on a cached
15     trajectory (no WASB re-run, no DB) ? an eval/tuning driver, not the live pipeline.
16
17     Run (GEMINI_API_KEY in env):
18         python -m backend.eval.gemini_refine --video Adarsh.MP4 --trajectory adarsh_traj.csv \
19             --fps 59.94 --frame-width 1920 --out output/adarsh_gemini.csv [--limit 4]
20     """
21     from __future__ import annotations
22
23     import os
24     import os.path as osp
25     import time
26     import argparse
27     import concurrent.futures
28     from typing import List, Tuple, Optional
29
30     from backend.eval.calibrate_wasb import make_predict_fn, BASELINE
31     from backend.eval.export_wasb_segments import TUNED, write_segments_csv, seconds_to_mmss
32     from backend.utils.video import get_video_duration, extract_video_chunk
33     from backend.sports import sport_registry
34     from backend.providers import provider_registry
35
36     Interval = Tuple[float, float]
37
38
39     def merge_overlaps(intervals: List[Interval], gap_tol: float = 0.0) -> List[Interval]:
40         """Union of overlapping intervals, optionally stitching across a small gap.
41
42         gap_tol > 0 merges two intervals separated by <= gap_tol seconds ? used to heal
43         shuttle-invisibility splits WITHIN one rally, while keeping genuine between-rally
44         gaps
45         (several seconds) as separate rallies. (Replaces the old envelope-per-candidate
46         which over-merged multiple real rallies WASB had bundled into one candidate.)"""
47         merged: List[Interval] = []
48         for s, e in sorted(intervals):
49             if merged and s <= merged[-1][1] + gap_tol:
50                 merged[-1] = (merged[-1][0], max(merged[-1][1], e))
51             else:
52                 merged.append((s, e))
53         return merged
54
55     def _offset_segments(raw: list, clip_start: float, clip_end: float) -> List[Interval]:
56         """Map a Gemini clip's returned segments back to full-video seconds, clamped to
57         clip."""
58         out: List[Interval] = []
59         for sg in raw or []:
```

```

59         try:
60             st = float(sg["start_time"]) + clip_start
61             en = float(sg["end_time"]) + clip_start
62         except (TypeError, ValueError, KeyError):
63             continue
64         st = max(clip_start, st)
65         en = min(clip_end, en)
66         if en > st:
67             out.append((round(st, 3), round(en, 3)))
68     return out
69
70
71     def refine_window(api_key: str, model: str, sport_profile, video: str, window: Interval,
72                     pad: float, duration: float, tmpdir: str, idx: int,
73                     clip_scale_width: Optional[int] = 1280) -> Tuple[int, Interval,
74 List[Interval]]:
75     # Own provider/client per call: the genai client isn't reliably thread-safe, and a
76     # shared one across workers throws socket errors. A fresh lazy client is cheap.
77     provider = provider_registry.get("gemini", api_key=api_key, model_name=model)
78     s, e = window
79     cs = max(0.0, s - pad)
80     ce = min(duration, e + pad) if duration > 0 else e + pad
81     clip = osp.join(tmpdir, f"cand_{idx:04d}.mp4")
82     # Downscale the uploaded clip (Gemini analyses at low res) -> small, fast, under the
83     # 500MB upload cap even for 4K/5K sources.
84     if not extract_video_chunk(video, cs, ce - cs, clip, reencode=True,
85 scale_width=clip_scale_width):
86         return idx, window, []
87     try:
88         prompt = sport_profile.get_prompt(chunk_duration=ce - cs)
89         raw, _ = provider.analyze_video(clip, prompt, chunk_duration=ce - cs,
90 label=f"cand{idx}")
91         mapped = _offset_segments(raw, cs, ce)
92     finally:
93         try:
94             os.remove(clip)
95         except OSError:
96             pass
97     # Keep Gemini's segments separate ? a WASB candidate can bundle several real rallies
98     # (merge_gap bridged the between-point gaps), and collapsing them to one envelope
99     swept
100     # dead time into a "rally" (precision loss). Tiny invisibility splits are healed
101     later
102     # by the gap-tolerant global merge in run().
103     return idx, window, mapped
104
105
106     def run(video: str, trajectory: str, fps: float, frame_width: float,
107            cfg=BASELINE, pad: float = 3.0, model: str = "gemini-flash-latest",
108            max_workers: int = 4, out_csv: str = "output/gemini_refined.csv",
109            min_dur: float = 2.0, limit: int = 0, clip_scale_width: Optional[int] = 1280) ->
110 dict:
111     api_key = os.environ.get("GEMINI_API_KEY")
112     if not api_key:
113         raise SystemExit("GEMINI_API_KEY not set in env")
114     sport_profile = sport_registry.get("badminton")
115     duration = get_video_duration(video)
116
117     candidates = make_predict_fn(trajectory, fps, frame_width)(cfg)
118     if limit:
119         candidates = candidates[:limit]
120     tmpdir = osp.join("output", "gemini_refine_clips")
121     os.makedirs(tmpdir, exist_ok=True)
122
123     print(f"[gemini_refine] {len(candidates)} WASB candidates (cfg {cfg}) -> Gemini

```

```

{model}, "
118         f"pad {pad}s, {max_workers} workers", flush=True)
119     t0 = time.time()
120     per_candidate = []
121     with concurrent.futures.ThreadPoolExecutor(max_workers=max_workers) as ex:
122         futs = [ex.submit(refine_window, api_key, model, sport_profile, video, w, pad,
duration,
123                         tmpdir, i, clip_scale_width)
124                 for i, w in enumerate(candidates)]
125     for fut in concurrent.futures.as_completed(futs):
126         idx, window, mapped = fut.result()
127         per_candidate.append((idx, window, mapped))
128         tag = "REJECT (no rally)" if not mapped else \
129             " ".join(f"{seconds_to_mmss(a)}-{seconds_to_mmss(b)}" for a, b in
mapped)
130         print(f"  cand {idx:>2}
{seconds_to_mmss(window[0])}-{seconds_to_mmss(window[1])} -> {tag}", flush=True)
131
132     raw_intervals = [iv for _, _, m in per_candidate for iv in m]
133     refined = [(s, e) for s, e in merge_overlaps(raw_intervals, gap_tol=1.0) if e - s >=
min_dur]
134     write_segments_csv(out_csv, refined)
135     try:
136         os.rmdir(tmpdir)
137     except OSError:
138         pass
139
140     kept_cands = sum(1 for _, _, m in per_candidate if m)
141     play = sum(e - s for s, e in refined)
142     print(f"\n[gemini_refine] {kept_cands}/{len(candidates)} candidates had a rally; "
143           f"{len(refined)} merged rallies, {play:.1f}s play -> {out_csv} "
144           f"({time.time() - t0:.0f}s)", flush=True)
145     return {"refined": refined, "num_candidates": len(candidates),
146           "candidates_with_rally": kept_cands, "out_csv": out_csv}
147
148
149     def full_video_rallies(video: str, model: str = "gemini-flash-latest",
150                           out_csv: str = "output/gemini_fullvideo.csv", clip_scale_width:
Optional[int] = 1280,
151                           min_dur: float = 2.0, chunk_sec: float = 120.0, overlap: float =
10.0,
152                           max_workers: int = 3) -> List[Interval]:
153         """Full-context pass: hand Gemini the WHOLE clip (downscaled, chunked to stay under
the
154         upload cap), so it finds every rally itself ? not bounded by WASB's candidate
recall.
155         Independent of the trajectory; complements refine()."""
156         api_key = os.environ.get("GEMINI_API_KEY")
157         if not api_key:
158             raise SystemExit("GEMINI_API_KEY not set in env")
159         sport_profile = sport_registry.get("badminton")
160         duration = get_video_duration(video)
161         starts, t = [], 0.0
162         while t < duration:
163             starts.append(t)
164             t += max(1.0, chunk_sec - overlap)
165         tmpdir = osp.join("output", "gemini_full_clips")
166         os.makedirs(tmpdir, exist_ok=True)
167
168         def do_chunk(ci: int, cstart: float) -> List[Interval]:
169             cend = min(cstart + chunk_sec, duration)
170             clip = osp.join(tmpdir, f"chunk_{ci:03d}.mp4")
171             if not extract_video_chunk(video, cstart, cend - cstart, clip, reencode=True,
scale_width=clip_scale_width):
172                 return []

```

```

173         try:
174             provider = provider_registry.get("gemini", api_key=api_key,
model_name=model)
175             prompt = sport_profile.get_prompt(chunk_duration=cend - cstart)
176             raw, _ = provider.analyze_video(clip, prompt, chunk_duration=cend - cstart,
label=f"chunk{ci}")
177             return _offset_segments(raw, cstart, cend)
178         finally:
179             try:
180                 os.remove(clip)
181             except OSError:
182                 pass
183
184         print(f"[gemini_full] {len(starts)} chunk(s) of {video} (<= {chunk_sec}s, width
{clip_scale_width}) "
185             f"-> Gemini {model}", flush=True)
186         with concurrent.futures.ThreadPoolExecutor(max_workers=max_workers) as ex:
187             chunks = [f.result() for f in [ex.submit(do_chunk, i, s) for i, s in
enumerate(starts)]]
188             rallies = [(s, e) for s, e in merge_overlaps([iv for c in chunks for iv in c],
gap_tol=1.0) if e - s >= min_dur]
189             write_segments_csv(out_csv, rallies)
190             try:
191                 os.rmdir(tmpdir)
192             except OSError:
193                 pass
194             play = sum(e - s for s, e in rallies)
195             print(f"[gemini_full] {len(rallies)} rallies, {play:.1f}s play -> {out_csv}",
flush=True)
196             return rallies
197
198
199     def main() -> None:
200         ap = argparse.ArgumentParser(description="Refine WASB candidates with Gemini.")
201         ap.add_argument("--video", required=True)
202         ap.add_argument("--trajectory", default=None, help="WASB trajectory CSV (required
unless --full-video)")
203         ap.add_argument("--fps", type=float, default=59.94)
204         ap.add_argument("--frame-width", type=float, default=1920.0)
205         ap.add_argument("--config", choices=["baseline", "tuned"], default="baseline",
206             help="WASB candidate config to feed Gemini (baseline = whole
rallies, good recall)")
207         ap.add_argument("--pad", type=float, default=3.0)
208         ap.add_argument("--model", default="gemini-flash-latest")
209         ap.add_argument("--max-workers", type=int, default=4)
210         ap.add_argument("--min-dur", type=float, default=2.0)
211         ap.add_argument("--out", default="output/gemini_refined.csv")
212         ap.add_argument("--limit", type=int, default=0, help="only the first N candidates
(smoke test)")
213         ap.add_argument("--clip-scale-width", type=int, default=1280,
214             help="downscale uploaded clips to this width (0=keep source res)")
215         ap.add_argument("--full-video", action="store_true",
216             help="full-context pass: send the whole clip (chunked/downscaled) to
Gemini "
217                 "instead of refining WASB candidates (needs no --trajectory)")
218         args = ap.parse_args()
219         scale = args.clip_scale_width or None
220         if args.full_video:
221             full_video_rallies(args.video, model=args.model, out_csv=args.out,
222                 clip_scale_width=scale, min_dur=args.min_dur,
max_workers=args.max_workers)
223         return
224         if not args.trajectory:
225             raise SystemExit("--trajectory is required unless --full-video")
226         cfg = BASELINE if args.config == "baseline" else TUNED

```

```

227         run(args.video, args.trajectory, args.fps, args.frame_width, cfg=cfg, pad=args.pad,
228             model=args.model, max_workers=args.max_workers, out_csv=args.out,
229             min_dur=args.min_dur, limit=args.limit, clip_scale_width=scale)
230
231
232 if __name__ == "__main__":
233     main()
234

```

17. user (2026-06-10T06:00:30.980Z)

```

1     """Evaluate the rally pipeline against PARTIAL human golden labels.
2
3     The VLC rally-annotator produces a human-rated CSV (rally_number,start_time,end_time,
4     ending_reason,sport). While a video is only partially tagged, the labels cover a
5     PREFIX of the video ? every rally up to some point is marked, and everything after is
6     simply un-watched (not "no rally there"). Scoring naively against such labels is wrong:
7     the detector's *correct* rallies in the un-labeled tail would be counted as false
8     positives and crush precision.
9
10    This harness fixes that by restricting the evaluation to the LABELED WINDOW
11    [window_start, window_end] (window_end defaults to the last labeled rally's end) and
12    dropping predicted segments that fall outside it. Within that window the human labels
13    ARE complete, so precision/recall/F1 are honest.
14
15    It reuses everything from the existing eval stack (no new windowing/metrics/gate logic):
16    - calibrate_wasb.make_predict_fn / load_golden_gts / BASELINE
17    - pipeline.detectors.rally_gate.apply_gate (net-crossing "gate A")
18    - eval.metrics.match_intervals / score
19    - export_wasb_segments.build_comparison / write_comparison_csv (per-rally detail CSV)
20
21    Run:
22    python -m backend.eval.eval_partial_labels \
23        --trajectory C:/Users/avidu/AppData/Local/Temp/adarsh_avi_traj.csv \
24        --labels "C:/Users/avidu/Projects/Annotation Setup/Collect/Adarsh and
25    Avi.rallies.csv" \
26        --fps 59.94 --frame-width 1920
27
28    """
29    from __future__ import annotations
30
31    import argparse
32    import os.path as osp
33    from typing import List, Optional, Tuple
34
35    from backend.eval.calibrate_wasb import make_predict_fn, load_golden_gts, BASELINE
36    from backend.eval.metrics import score
37    from backend.eval.export_wasb_segments import build_comparison, write_comparison_csv
38    from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
39    from backend.pipeline.detectors import rally_gate
40
41    Interval = Tuple[float, float]
42    Config = Tuple[float, float, float]
43
44    # Operating points to compare. The tuned config wins on its fit-video but over-segments
45    # and does not transfer; baseline (+ net-crossing gate) is the stronger general point.
46    TUNED: Config = (0.05, 1.0, 1.0)
47
48    def restrict_to_window(intervals: List[Interval], w0: float, w1: float) ->
49    List[Interval]:
50        """Keep intervals that overlap the labeled window [w0, w1].
51
52        A predicted segment is evaluated iff it overlaps the labeled region; segments wholly
53        in the un-labeled tail (start >= w1) or before the window (end <= w0) are dropped so
54        they are neither rewarded nor penalized. GTs are assumed already inside the window.

```

```

53     """
54     return [(s, e) for (s, e) in intervals if e > w0 and s < w1]
55
56
57     def _eval_point(predict_fn, cfg: Config, points, fps: float, gts: List[Interval],
58                     window: Tuple[float, float], gate_crossings: int, iou: float) -> dict:
59         """Score one (config, gate) operating point inside the labeled window."""
60         preds = predict_fn(cfg)
61         if gate_crossings > 0 and points:
62             preds = rally_gate.apply_gate(points, preds, fps, min_crossings=gate_crossings)
63             preds = restrict_to_window(preds, *window)
64             sc = score(preds, gts, iou)
65             return {"cfg": cfg, "gate": gate_crossings, "n_pred": len(preds), "score": sc,
66 "preds": preds}
67
68     def run(trajjectory_csv: str, labels_csv: str, fps: float, frame_width: float,
69            window_start: float = 0.0, window_end: Optional[float] = None,
70            gate_crossings: int = 2, iou: float = 0.5,
71            comparison_out: Optional[str] = None) -> dict:
72         gts_all = load_golden_gts(labels_csv)
73         if not gts_all:
74             raise SystemExit(f"no usable human rallies in {labels_csv}")
75         if window_end is None:
76             window_end = max(e for _, e in gts_all)
77         window = (window_start, window_end)
78         gts = restrict_to_window(gts_all, *window)
79
80         predict_fn = make_predict_fn(trajjectory_csv, fps, frame_width)
81         points = TrackNetRunner.parse_trajectory_csv(trajjectory_csv)
82         traj_end = (max(p.frame for p in points) / fps) if points else 0.0
83
84         print("=== Pipeline vs PARTIAL human labels ===")
85         print(f"labels: {labels_csv}")
86         print(f" {len(gts_all)} human rallies; labeled window [{window_start:.1f}s,
87 {window_end:.1f}s] "
88             f"-> {len(gts)} rallies in-window")
89         print(f"trajectory covers ~0-{traj_end:.1f}s "
90             f"({'covers the window' if traj_end >= window_end - 1 else 'WARNING: shorter
91 than window'})")
92         print(f"IoU>={iou}; net-crossing gate K={gate_crossings}\n")
93
94         # Compare the standard operating points: baseline / baseline+gate / tuned /
95         tuned+gate.
96         points_for_gate = points
97         candidates = [
98             ("baseline", BASELINE, 0),
99             (f"baseline+gate{gate_crossings}", BASELINE, gate_crossings),
100             ("tuned", TUNED, 0),
101             (f"tuned+gate{gate_crossings}", TUNED, gate_crossings),
102         ]
103         results = []
104         header = f"{'operating point':<18} {'pred':>4} {'TP':>3} {'FN':>3} {'FP':>3} " \
105             f"{'prec':>6} {'recall':>6} {'F1':>6} {'mIoU':>6} {'sErr':>6} {'eErr':>6}"
106         print(header)
107         print("-" * len(header))
108         for name, cfg, gate in candidates:
109             r = _eval_point(predict_fn, cfg, points_for_gate, fps, gts, window, gate, iou)
110             sc = r["score"]
111             r["name"] = name
112             results.append(r)
113             print(f"{name:<18} {r['n_pred']:>4} {sc.true_positives:>3}
114 {sc.false_negatives:>3} "
115                 f"{sc.false_positives:>3} {sc.precision:>6.3f} {sc.recall:>6.3f}
116 {sc.f1:>6.3f} ")

```

```

112         f"{sc.mean_iou:>6.3f} {sc.mean_start_error:>6.2f}
{sc.mean_end_error:>6.2f}")
113
114     best = max(results, key=lambda r: r["score"].f1)
115     print(f"\nbest operating point by F1: {best['name']} (F1 {best['score'].f1:.3f})")
116
117     # Per-rally detail for the best point: which human rallies matched/missed + extra
fires.
118     comp = build_comparison(best["preds"], gts, iou)
119     comp_out = comparison_out or osp.join("output", "adarsh_avi_vs_human.csv")
120     write_comparison_csv(comp_out, comp)
121     print(f"per-rally detail ({best['name']}) -> {comp_out}")
122
123     return {
124         "window": window, "n_human_in_window": len(gts),
125         "results": [{"name": r["name"], "cfg": r["cfg"], "gate": r["gate"],
126                     "n_pred": r["n_pred"], **r["score"].as_dict()} for r in results],
127         "best": best["name"], "comparison_csv": comp_out,
128     }
129
130
131     # Grid centered on the two levers the human-GT failure analysis flagged:
132     # min_window_duration DOWN (recover short 3-5s rallies) and merge_gap UP (heal
133     # fragmented long rallies). Gate is swept on/off because it can over-prune.
134     DEFAULT_GRID = [
135         (vt, mg, mwd)
136         for vt in (0.01, 0.02, 0.03, 0.05)
137         for mg in (1.0, 2.0, 3.0, 4.0)
138         for mwd in (0.5, 1.0, 1.5, 2.0)
139     ]
140
141
142     def sweep(trajjectory_csv: str, labels_csv: str, fps: float, frame_width: float,
143             window_start: float = 0.0, window_end: Optional[float] = None,
144             gate_options: Tuple[int, ...] = (0, 2), iou: float = 0.5,
145             grid: Optional[List[Config]] = None, top: int = 12) -> dict:
146         """Grid-search (velocity, merge_gap, min_dur) x gate vs human labels, in-window.
147
148         WARNING: this is fit-on-self on ONE video -> optimistic. It shows the achievable
149         ceiling + which levers matter, NOT a transferable config. The honest operating
150         point comes from leave-one-video-out once >=3 videos are labeled.
151         """
152         grid = grid or DEFAULT_GRID
153         gts_all = load_golden_gts(labels_csv)
154         if not gts_all:
155             raise SystemExit(f"no usable human rallies in {labels_csv}")
156         if window_end is None:
157             window_end = max(e for _, e in gts_all)
158         window = (window_start, window_end)
159         gts = restrict_to_window(gts_all, *window)
160         predict_fn = make_predict_fn(trajjectory_csv, fps, frame_width)
161         points = TrackNetRunner.parse_trajectory_csv(trajjectory_csv)
162
163         rows: List[dict] = []
164         for cfg in grid:
165             for gate in gate_options:
166                 r = _eval_point(predict_fn, cfg, points, fps, gts, window, gate, iou)
167                 sc = r["score"]
168                 rows.append({"cfg": cfg, "gate": gate, "n_pred": r["n_pred"], "f1": sc.f1,
169                             "precision": sc.precision, "recall": sc.recall, "mean_iou":
sc.mean_iou,
170                             "tp": sc.true_positives, "fp": sc.false_positives, "fn":
sc.false_negatives})
171         rows.sort(key=lambda x: (x["f1"], x["mean_iou"]), reverse=True)
172

```

```

173         print(f"=== Config sweep vs human labels ({len(gts)} rallies in "
174               f"[{window_start:.0f},{window_end:.0f}]s; {len(grid)}x{len(gate_options)}
configs) ===")
175         print("NOTE: fit-on-self on ONE video = optimistic; real test is leave-one-video-
out.\n")
176         hdr = f"{'#':>3} {'veloc':>6} {'merge':>6} {'minDur':>6} {'gate':>4} " \
177               f"{'F1':>6} {'prec':>6} {'recall':>6} {'mIoU':>6} {'TP':>3} {'FP':>3}
{'FN':>3}"
178         print(hdr); print("-" * len(hdr))
179         for i, r in enumerate(rows[:top], 1):
180             vt, mg, mwd = r["cfg"]
181             print(f"{i:>3} {vt:>6.3f} {mg:>6.1f} {mwd:>6.1f} {r['gate']:>4} "
182                   f"{r['f1']:>6.3f} {r['precision']:>6.3f} {r['recall']:>6.3f}
{r['mean_iou']:>6.3f} "
183                   f"{r['tp']:>3} {r['fp']:>3} {r['fn']:>3}")
184         b = rows[0]
185         print(f"\nbest: velocity={b['cfg'][0]} merge_gap={b['cfg'][1]} min_dur={b['cfg'][2]}
"
186               f"gate={b['gate']} -> F1 {b['f1']:.3f} (P {b['precision']:.3f} / R
{b['recall']:.3f})")
187         return {"window": window, "n_gt": len(gts), "ranked": rows, "best": b}
188
189
190     def main() -> None:
191         ap = argparse.ArgumentParser(description=__doc__)
192         ap.add_argument("--trajectory", required=True, help="WASB trajectory CSV
(Frame,Visibility,X,Y)")
193         ap.add_argument("--labels", required=True, help="human golden CSV
(rally_number,start_time,end_time,...)")
194         ap.add_argument("--fps", type=float, default=59.94)
195         ap.add_argument("--frame-width", type=float, default=1920.0)
196         ap.add_argument("--window-start", type=float, default=0.0)
197         ap.add_argument("--window-end", type=float, default=None,
198                           help="end of the labeled region (default: last labeled rally end)")
199         ap.add_argument("--gate", type=int, default=2, help="net-crossing gate K (0=off, 2
recommended)")
200         ap.add_argument("--iou", type=float, default=0.5)
201         ap.add_argument("--comparison-out", default=None)
202         ap.add_argument("--sweep", action="store_true",
203                           help="grid-search configs vs labels (fit-on-self; finds the ceiling
+ best levers)")
204         ap.add_argument("--top", type=int, default=12, help="how many sweep rows to print")
205         args = ap.parse_args()
206         if args.sweep:
207             sweep(args.trajectory, args.labels, args.fps, args.frame_width,
208                   window_start=args.window_start, window_end=args.window_end,
209                   iou=args.iou, top=args.top)
210         else:
211             run(args.trajectory, args.labels, args.fps, args.frame_width,
212                 window_start=args.window_start, window_end=args.window_end,
213                 gate_crossings=args.gate, iou=args.iou, comparison_out=args.comparison_out)
214
215
216     if __name__ == "__main__":
217         main()
218

```

18. user (2026-06-10T06:00:35.605Z)

```

1     """PROTOTYPE: a learned per-FRAME rally segmenter (vs the unsupervised windowing
heuristic).
2
3     Motivation (2026-06-08 finding): WASB tracks the shuttle well on Kushagra (97% of GT
4     rallies contain >=2 net-crossings) yet NO velocity-windowing config segments it
5     (per-video F1 ceiling 0.163), while it works on Adarsh (0.73). The rally SIGNAL is in

```

```

6     the trajectory; the hand-tuned heuristic just can't extract it across footage. This
7     prototype tests the hypothesis that a LEARNED model on per-frame trajectory features
8     recovers the rally segments ? including on the footage the heuristic fails on.
9
10    Approach (deliberately small, dependency-light: numpy + scipy + the repo's pure-python
11    LogisticRegression ? no torch/sklearn):
12        1. Per-frame features over a local window from the WASB trajectory: visibility rate,
13           mean/max shuttle speed (frame-width-normalized), net-crossing RATE, x/y spread.
14           All are scale/camera-robust RATES (unlike a raw velocity threshold).
15        2. Per-frame label = inside a human GT rally or not.
16        3. LEAVE-ONE-VIDEO-OUT: train the classifier on the other match(es), test on the
17           held-out one (the honest cross-footage number; no within-video leakage).
18        4. Smooth probabilities -> threshold (tuned on TRAIN) -> contiguous in-rally segments
19           -> merge tiny gaps / drop short -> score vs GT at IoU>=0.5 (same metric as the
20    heuristic).
21
22    This is a directional PROOF-OF-SIGNAL, not a production model (n=2 videos, linear model,
23    hand features). It exists to answer one question: is the rally boundary learnable from
24    the trajectory where the heuristic fails? Compare its held-out F1 to the heuristic LOVO.
25
26    Run:
27        python -m backend.eval.rally_seq_proto --manifest output/human_lovo_manifest.json
28
29    from __future__ import annotations
30
31    import json
32    import argparse
33    from typing import Dict, List, Tuple
34
35    import numpy as np
36    from scipy.ndimage import uniform_filter1d, maximum_filter1d
37
38    from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
39    from backend.eval.classifier import LogisticRegression
40    from backend.eval.calibrate_wasb import load_golden_gts
41    from backend.eval.metrics import score
42
43    FEAT_NAMES = ["vis_rate", "spd_mean", "spd_max", "cross_rate", "x_std", "y_std"]
44    Interval = Tuple[float, float]
45
46    def _spread(a: np.ndarray) -> float:
47        return float(np.percentile(a, 95) - np.percentile(a, 5)) if a.size > 1 else 0.0
48
49
50    def build_frame_arrays(points, frame_width: float) -> Dict:
51        """Per-frame trajectory arrays + per-visible-frame speed and net-crossing events."""
52        pts = sorted(points, key=lambda p: p.frame)
53        N = (pts[-1].frame + 1) if pts else 1
54        vis = np.zeros(N); x = np.zeros(N); y = np.zeros(N)
55        for p in pts:
56            if 0 <= p.frame < N and p.visible and p.x >= 0:
57                vis[p.frame] = 1.0; x[p.frame] = float(p.x); y[p.frame] = float(p.y)
58
59        mask = vis > 0
60        play_is_x = _spread(x[mask]) >= _spread(y[mask])
61        coord = x if play_is_x else y
62        net = float(np.median(coord[mask])) if mask.any() else 0.0
63        deadband = 0.06 * (_spread(coord[mask]) if mask.sum() > 1 else 1.0)
64
65        speed = np.zeros(N); spd_mask = np.zeros(N); cross = np.zeros(N)
66        prev = None
67        for p in pts:
68            f = p.frame
69            if not (0 <= f < N and vis[f]):

```

```

70         continue
71     if prev is not None:
72         df = f - prev
73         if df > 0:
74             speed[f] = float(np.hypot(x[f] - x[prev], y[f] - y[prev]) / frame_width
/ df)
75             spd_mask[f] = 1.0
76             c, pc = coord[f] - net, coord[prev] - net
77             if abs(c) > deadband and abs(pc) > deadband and (c > 0) != (pc > 0):
78                 cross[f] = 1.0
79         prev = f
80     return dict(N=N, vis=vis, x=x, y=y, speed=speed, spd_mask=spd_mask, cross=cross,
fw=frame_width)
81
82
83     def features(arr: Dict, fps: float, win_sec: float = 0.75, stride_sec: float = 0.1):
84         """Windowed per-frame features sampled at a stride -> (X[n_samples,6],
times[n_samples])."""
85         N = arr["N"]; fw = arr["fw"]
86         size = max(3, int(win_sec * fps) * 2 + 1)
87         vis, speed, spd_mask, cross, x, y = (arr[k] for k in ("vis", "speed", "spd_mask",
"cross", "x", "y"))
88
89         n = uniform_filter1d(vis, size, mode="constant") # mean
visibility (== vis_rate)
90         sm = uniform_filter1d(spd_mask, size, mode="constant")
91         spd_mean = uniform_filter1d(speed, size, mode="constant") / np.maximum(sm, 1e-9)
92         spd_max = maximum_filter1d(speed, size, mode="constant")
93         cross_rate = uniform_filter1d(cross, size, mode="constant") * fps # crossings
/ second
94         xm = uniform_filter1d(x * vis, size, mode="constant") / np.maximum(n, 1e-9)
95         x2 = uniform_filter1d((x * vis) ** 2, size, mode="constant") / np.maximum(n, 1e-9)
96         x_std = np.sqrt(np.maximum(x2 - xm ** 2, 0.0)) / fw
97         ym = uniform_filter1d(y * vis, size, mode="constant") / np.maximum(n, 1e-9)
98         y2 = uniform_filter1d((y * vis) ** 2, size, mode="constant") / np.maximum(n, 1e-9)
99         y_std = np.sqrt(np.maximum(y2 - ym ** 2, 0.0)) / fw
100
101         idx = np.arange(0, N, max(1, int(stride_sec * fps)))
102         X = np.stack([n[idx], spd_mean[idx], spd_max[idx], cross_rate[idx], x_std[idx],
y_std[idx]], axis=1)
103         return np.nan_to_num(X), idx / fps
104
105
106     def labels_for(times: np.ndarray, gts: List[Interval]) -> np.ndarray:
107         y = np.zeros(len(times))
108         for i, t in enumerate(times):
109             for s, e in gts:
110                 if s <= t <= e:
111                     y[i] = 1.0
112                     break
113         return y
114
115
116     def roc_auc(y: np.ndarray, p: np.ndarray) -> float:
117         """Rank-based AUC (Mann-Whitney); pure numpy."""
118         pos, neg = p[y > 0.5], p[y <= 0.5]
119         if pos.size == 0 or neg.size == 0:
120             return float("nan")
121         order = np.argsort(p, kind="mergesort")
122         ranks = np.empty(len(p)); ranks[order] = np.arange(1, len(p) + 1)
123         return float((ranks[y > 0.5].sum() - pos.size * (pos.size + 1) / 2) / (pos.size *
neg.size))
124
125
126     def probs_to_segments(probs: np.ndarray, times: np.ndarray, threshold: float,

```

```

127             merge_gap: float = 1.0, min_dur: float = 1.0, smooth: int = 7) ->
List[Interval]:
128     p = uniform_filter1d(probs, max(1, smooth), mode="nearest")
129     on = p >= threshold
130     runs: List[Interval] = []
131     i, n = 0, len(on)
132     while i < n:
133         if on[i]:
134             j = i
135             while j + 1 < n and on[j + 1]:
136                 j += 1
137             runs.append((float(times[i]), float(times[j])))
138             i = j + 1
139         else:
140             i += 1
141     merged: List[Interval] = []
142     for s, e in runs:
143         if merged and s - merged[-1][1] <= merge_gap:
144             merged[-1] = (merged[-1][0], e)
145         else:
146             merged.append((s, e))
147     return [(s, e) for s, e in merged if e - s >= min_dur]
148
149
150 def _seg_f1(clf, vids: List[Dict], thr: float) -> float:
151     f1s = []
152     for v in vids:
153         segs = probs_to_segments(clf.predict_proba(v["X"]), v["times"], thr)
154         f1s.append(score(segs, v["gts"], 0.5).f1)
155     return sum(f1s) / max(len(f1s), 1)
156
157
158 def run(specs: List[Dict]) -> dict:
159     data = {}
160     for s in specs:
161         pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
162         arr = build_frame_arrays(pts, float(s["frame_width"]))
163         X, times = features(arr, float(s["fps"]))
164         gts = load_golden_gts(s["labels"])
165         data[s["name"]] = {"X": X, "times": times, "y": labels_for(times, gts), "gts":
gts}
166     names = list(data)
167     print(f"=== Learned per-frame rally segmenter ? LEAVE-ONE-VIDEO-OUT ({len(names)}
videos) ===")
168     print(f"features: {FEAT_NAMES}\n")
169
170     results = []
171     for test in names:
172         train = [data[n] for n in names if n != test]
173         Xtr = np.vstack([v["X"] for v in train]); ytr = np.concatenate([v["y"] for v in
train])
174         clf = LogisticRegression(lr=0.2, epochs=2000, l2=1e-2)
175         clf.fit(Xtr, ytr, feature_names=FEAT_NAMES)
176         # tune the decision threshold on TRAIN segment-F1 (no test leakage)
177         thr = max(np.arange(0.2, 0.81, 0.05), key=lambda t: _seg_f1(clf, train,
float(t)))
178         v = data[test]
179         probs = clf.predict_proba(v["X"])
180         auc = roc_auc(v["y"], probs)
181         segs = probs_to_segments(probs, v["times"], float(thr))
182         sc = score(segs, v["gts"], 0.5)
183         # Oracle threshold (fit-on-TEST, upper bound): isolates the representation
ceiling
184         # from the threshold-transfer gap. Optimistic ? for diagnosis only.
185         othr = max(np.arange(0.2, 0.81, 0.05),

```

```

186             key=lambda t: score(probs_to_segments(probs, v["times"], float(t)),
v["gts"], 0.5).f1)
187         osc = score(probs_to_segments(probs, v["times"], float(othr)), v["gts"], 0.5)
188         results.append({"video": test, "thr": float(thr), "auc": auc,
189                        "f1": sc.f1, "precision": sc.precision, "recall": sc.recall,
190                        "n_pred": len(segs), "n_gt": len(v["gts"]),
191                        "oracle_thr": float(othr), "oracle_f1": osc.f1})
192         print(f"[held-out {test}] per-frame AUC {auc:.3f} | LOVO-threshold F1
{sc.f1:.3f} "
193               f"(P {sc.precision:.3f} R {sc.recall:.3f}, thr {thr:.2f} from train) | "
194               f"oracle-threshold F1 {osc.f1:.3f} (thr {othr:.2f}, fit-on-test upper
bound)")
195         mean_f1 = sum(r["f1"] for r in results) / len(results)
196         print(f"\nlearned LOVO mean held-out F1: {mean_f1:.3f}")
197         return {"per_video": results, "mean_f1": mean_f1}
198
199
200     def main() -> None:
201         ap = argparse.ArgumentParser(description=__doc__)
202         ap.add_argument("--manifest", required=True, help="JSON:
[{name,trajectory,fps,frame_width,labels}, ...]")
203         args = ap.parse_args()
204         with open(args.manifest, encoding="utf-8") as f:
205             run(json.load(f))
206
207
208     if __name__ == "__main__":
209         main()
210

```

19. user (2026-06-10T06:00:40.191Z)

```

1     """E1 ? audio GO/NO-GO probe (docs/AUDIO_RALLY_DETECTION_PLAN.md).
2
3     Answers, on REAL audio: does the shuttle "thwack" survive well enough that hit
4     peaks (a) cluster inside rallies far more than dead time, (b) recover rallies the
5     WASB windows miss (our recall bottleneck), and (c) form usable audio-only rally
6     proposals? No training, no fusion ? purely diagnostic, so we know whether to build
7     the rest of the audio path.
8
9     Reuses: audio.hit_detector, calibrate_wasb.load_golden_gts/make_predict_fn,
10    eval.metrics. Labels may be human-gold OR Gemini-silver ? silver is fine for the
11    SNR/efficacy question E1 asks.
12    """
13    from __future__ import annotations
14
15    import argparse
16    from typing import List, Tuple
17
18    from backend.pipeline.audio.hit_detector import detect_hits_in_video, HitEvent
19    from backend.eval.calibrate_wasb import load_golden_gts, make_predict_fn, BASELINE
20    from backend.eval.metrics import match_intervals, score
21    from backend.utils.video import get_video_duration
22
23    Interval = Tuple[float, float]
24
25
26    def cluster_hits(hits: List[HitEvent], gap_sec: float = 2.0, min_hits: int = 2,
27                    pad: float = 0.3) -> List[Interval]:
28        """Group hits whose inter-hit gap <= gap_sec into rally proposals; keep clusters
29        with >= min_hits (a lone hit = a service feed, not a rally)."""
30        ts = sorted(h.t for h in hits)
31        if not ts:
32            return []
33        out: List[Interval] = []

```

```

34     start = last = ts[0]
35     cnt = 1
36     for t in ts[1:]:
37         if t - last <= gap_sec:
38             last, cnt = t, cnt + 1
39         else:
40             if cnt >= min_hits:
41                 out.append((max(0.0, start - pad), last + pad))
42                 start = last = t
43                 cnt = 1
44             if cnt >= min_hits:
45                 out.append((max(0.0, start - pad), last + pad))
46     return out
47
48
49     def _hits_in(hits: List[HitEvent], s: float, e: float) -> int:
50         return sum(1 for h in hits if s <= h.t <= e)
51
52
53     def run(video: str, labels_csv: str, trajectory: str, fps: float, frame_width: float,
54            k: float = 2.5, cluster_gap: float = 2.0, min_hits: int = 2, iou: float = 0.5)
-> dict:
55         hits = detect_hits_in_video(video, k=k)
56         gts = load_golden_gts(labels_csv)
57         wasb = make_predict_fn(trajectory, fps, frame_width)(BASELINE)
58         dur = get_video_duration(video) or (max((e for _, e in gts + wasb), default=0.0))
59
60         print(f"=== E1 audio probe: {video} ===")
61         print(f" {len(hits)} hits detected (k={k}) over {dur:.0f}s =
{len(hits)/max(1,dur)*60:.1f} hits/min | "
62               f"{len(gts)} labeled rallies, {len(wasb)} WASB baseline windows")
63
64         # (1) Discrimination: hit density inside rallies vs dead time.
65         rally_dur = sum(e - s for s, e in gts)
66         dead_dur = max(1e-6, dur - rally_dur)
67         hits_in = sum(_hits_in(hits, s, e) for s, e in gts)
68         hits_out = len(hits) - hits_in
69         dens_in = hits_in / max(1e-6, rally_dur)
70         dens_out = hits_out / dead_dur
71         ratio = dens_in / dens_out if dens_out > 0 else float("inf")
72         rallies_with_hits = sum(1 for s, e in gts if _hits_in(hits, s, e) >= 2)
73         print(f" [discrimination] in-rally {dens_in:.2f} hits/s vs dead-time {dens_out:.2f}
hits/s "
74               f"? ratio {ratio:.1f}x | rallies with >=2 hits: {rallies_with_hits}/{len(gts)}
"
75               f"({100*rallies_with_hits/max(1,len(gts)):.0f}%)")
76
77         # (2) Recall recovery: of rallies WASB MISSES, how many have an audio hit-cluster?
78         mr = match_intervals(wasb, gts, iou)
79         missed = [gts[i] for i in mr.unmatched_gt_indices]
80         recoverable = [r for r in missed if _hits_in(hits, *r) >= min_hits]
81         rec_rate = len(recoverable) / len(missed) if missed else 0.0
82         print(f" [recall recovery] WASB misses {len(missed)}/{len(gts)} rallies; "
83               f"audio hit-clusters sit in {len(recoverable)} of them ? {100*rec_rate:.0f}%
recoverable")
84
85         # (3) Audio-only rally proposals vs labels.
86         props = cluster_hits(hits, cluster_gap, min_hits)
87         sc = score(props, gts, iou)
88         print(f" [audio-only proposals] {len(props)} rallies | P={sc.precision:.3f}
R={sc.recall:.3f} "
89               f"F1={sc.f1:.3f} (vs WASB-only F1 {score(wasb, gts, iou).f1:.3f})")
90     return {
91         "video": video, "k": k, "n_hits": len(hits), "duration": dur,
92         "discrimination_ratio": ratio, "rallies_with_hits_pct": 100 * rallies_with_hits

```

```

/ max(1, len(gts)),
93         "wasb_missed": len(missed), "recall_recoverable_pct": 100 * rec_rate,
94         "audio_only": sc.as_dict(), "wasb_only_f1": score(wasb, gts, iou).f1,
95     }
96
97
98     def main() -> None:
99         ap = argparse.ArgumentParser(description="E1 audio go/no-go probe.")
100         ap.add_argument("--video", required=True)
101         ap.add_argument("--labels", required=True, help="golden or Gemini-silver rally CSV")
102         ap.add_argument("--trajectory", required=True, help="WASB trajectory CSV")
103         ap.add_argument("--fps", type=float, default=59.94)
104         ap.add_argument("--frame-width", type=float, default=1920.0)
105         ap.add_argument("--k", type=float, default=2.5, help="onset sensitivity (lower =
more hits)")
106         ap.add_argument("--cluster-gap", type=float, default=2.0)
107         ap.add_argument("--min-hits", type=int, default=2)
108         ap.add_argument("--iou", type=float, default=0.5)
109         args = ap.parse_args()
110         run(args.video, args.labels, args.trajectory, args.fps, args.frame_width,
111             k=args.k, cluster_gap=args.cluster_gap, min_hits=args.min_hits, iou=args.iou)
112
113
114     if __name__ == "__main__":
115         main()
116

```

20. user (2026-06-10T06:00:50.010Z)

```

1     """Slice individual rally clips out of a (long) annotated video.
2
3     Use cases:
4     - **Offline verification** ? pull rally N out of a 30-min match to eyeball its
5       boundaries against the labels.
6     - **Per-video calibration/training** ? grab a handful of labeled rallies as the
7       supervision the segmenter is tuned/adapted on for that specific video.
8
9     Given a labels CSV (the golden `rally_number,start_time,end_time[,start_mmss,
10    end_mmss]` shape) and a set of rally numbers, it writes one padded clip per rally:
11        <out_dir>/<video_stem>__rally_<n>.mp4
12
13    Padding matches the highlight compiler (begin 0.5 s / end 1.0 s) so a sliced clip
14    shows the same lead-in/out the model is trained and evaluated against. The window
15    math (`compute_clip_windows`) is pure and unit-tested; ffmpeg is only touched in
16    `slice_rallies`.
17    """
18    from __future__ import annotations
19
20    import argparse
21    import csv
22    import os
23    import subprocess
24    from dataclasses import dataclass
25    from typing import Dict, List, Optional, Tuple
26
27    from backend.eval.annotations import parse_timestamp
28
29    # Match VideoCompiler defaults so sliced clips frame a rally the same way.
30    BEGIN_PADDING = 0.5
31    END_PADDING = 1.0
32
33
34    @dataclass
35    class ClipWindow:
36        rally: str          # kept as a string to allow inserted ids like "7.5"

```

```

37         start: float          # padded, clamped
38         end: float            # padded, clamped
39
40
41     def _read_time(row: Dict[str, str], decimal_key: str, mmss_key: str) -> Optional[float]:
42         """Prefer the decimal-seconds column; fall back to mm:ss. None if unusable."""
43         for key in (decimal_key, mmss_key):
44             val = (row.get(key) or "").strip()
45             if val:
46                 try:
47                     return parse_timestamp(val)
48                 except ValueError:
49                     continue
50         return None
51
52
53     def load_rally_labels(csv_path: str) -> Dict[str, Tuple[float, float]]:
54         """Map rally_number -> (start, end) seconds, skipping rows with no usable times.
55
56         Accepts the golden schema (start_time/end_time decimal, optional start_mmss/
57         end_mmss). Rally numbers are kept as strings so inserted ids ("7.5") survive.
58         """
59         labels: Dict[str, Tuple[float, float]] = {}
60         with open(csv_path, newline="", encoding="utf-8") as f:
61             reader = csv.DictReader(f)
62             reader.fieldnames = [(n or "").strip().lower() for n in (reader.fieldnames or
63 [])]
64             for row in reader:
65                 row = {(k or "").strip().lower(): v for k, v in row.items()}
66                 rally = (row.get("rally_number") or "").strip()
67                 if not rally:
68                     continue
69                 start = _read_time(row, "start_time", "start_mmss")
70                 end = _read_time(row, "end_time", "end_mmss")
71                 if start is None or end is None or end <= start:
72                     continue # missing/degenerate (e.g. a not-yet-filled MISSED row)
73                 labels[rally] = (start, end)
74         return labels
75
76     def compute_clip_windows(labels: Dict[str, Tuple[float, float]],
77                               rally_numbers: List[str],
78                               begin_pad: float = BEGIN_PADDING,
79                               end_pad: float = END_PADDING,
80                               video_duration: Optional[float] = None) -> List[ClipWindow]:
81         """Apply padding and clamp to the video, for the requested rallies.
82
83         rally_numbers may be ["all"] to take every labeled rally (in time order).
84         Unknown rally ids are skipped (caller can diff to warn).
85         """
86         if len(rally_numbers) == 1 and rally_numbers[0].lower() == "all":
87             wanted = sorted(labels, key=lambda r: labels[r][0])
88         else:
89             wanted = [r for r in rally_numbers if r in labels]
90
91         windows: List[ClipWindow] = []
92         for rally in wanted:
93             start, end = labels[rally]
94             ps = max(0.0, start - begin_pad)
95             pe = end + end_pad
96             if video_duration is not None and video_duration > 0:
97                 if ps >= video_duration:
98                     continue # entirely past the end
99                 pe = min(pe, video_duration)
100             if pe <= ps:

```

```

101         continue
102         windows.append(ClipWindow(rally=rally, start=round(ps, 3), end=round(pe, 3)))
103     return windows
104
105
106 def _probe_duration(video_path: str) -> Optional[float]:
107     try:
108         out = subprocess.run(
109             ["ffprobe", "-v", "error", "-show_entries", "format=duration",
110              "-of", "default=noprint_wrappers=1:nokey=1", video_path],
111             capture_output=True, text=True, check=True,
112             ).stdout.strip()
113         return float(out)
114     except (subprocess.SubprocessError, OSError, ValueError):
115         return None
116
117
118 def slice_rallies(video_path: str, labels_csv: str, rally_numbers: List[str],
119                  out_dir: str, begin_pad: float = BEGIN_PADDING,
120                  end_pad: float = END_PADDING) -> List[str]:
121     """Write one padded clip per requested rally; returns the clip paths written."""
122     if not os.path.exists(video_path):
123         raise FileNotFoundError(f"video not found: {video_path}")
124     labels = load_rally_labels(labels_csv)
125     if not labels:
126         raise ValueError(f"no usable rally rows in {labels_csv}")
127     os.makedirs(out_dir, exist_ok=True)
128
129     duration = _probe_duration(video_path)
130     windows = compute_clip_windows(labels, rally_numbers, begin_pad, end_pad, duration)
131     if not windows:
132         raise ValueError("no rallies matched the request (check --rallies vs the CSV)")
133
134     stem = os.path.splitext(os.path.basename(video_path))[0]
135     written: List[str] = []
136     for w in windows:
137         safe_rally = w.rally.replace(".", "_")
138         out_path = os.path.join(out_dir, f"{stem}__rally_{safe_rally}.mp4")
139         cmd = [
140             "ffmpeg", "-y",
141             "-ss", str(w.start), "-to", str(w.end), "-i", video_path,
142             "-c:v", "libx264", "-preset", "superfast", "-crf", "22",
143             "-c:a", "aac", "-b:a", "128k",
144             out_path,
145         ]
146         try:
147             subprocess.run(cmd, capture_output=True, check=True)
148         except subprocess.CalledProcessError as e:
149             err = e.stderr.decode(errors="replace").strip()[-300:]
150             print(f"[slicer] rally {w.rally}: ffmpeg failed: {err}")
151             continue
152         if os.path.exists(out_path) and os.path.getsize(out_path) > 0:
153             written.append(out_path)
154             print(f"[slicer] rally {w.rally} -> {out_path}")
155     return written
156
157
158 def main() -> None:
159     p = argparse.ArgumentParser(description="Slice rally clips from an annotated
160 video.")
161     p.add_argument("--video", required=True, help="Source video file")
162     p.add_argument("--labels", required=True, help="Labels CSV (golden rally format)")
163     p.add_argument("--rallies", required=True,
164                     help="Comma-separated rally numbers (e.g. 7,7.5,16) or 'all'")

```

```

164     p.add_argument("--out-dir", default="output/rally_clips", help="Output directory")
165     p.add_argument("--begin-pad", type=float, default=BEGIN_PADDING)
166     p.add_argument("--end-pad", type=float, default=END_PADDING)
167     args = p.parse_args()
168
169     rally_numbers = [r.strip() for r in args.rallies.split(",") if r.strip()]
170     written = slice_rallies(args.video, args.labels, rally_numbers,
171                             args.out_dir, args.begin_pad, args.end_pad)
172     print(f"\nWrote {len(written)} clip(s) to {args.out_dir}")
173
174
175     if __name__ == "__main__":
176         main()
177

```

21. user (2026-06-10T06:00:55.105Z)

```

1     """CLI to evaluate detected rallies against hand-annotated ground truth.
2
3     Scores predictions already stored in the SQLite DB (run the pipeline first) ? no
4     API calls, fully deterministic. See docs/EVALUATION.md for the workflow.
5
6     Examples
7     -----
8     Create an empty annotation file to fill in:
9         python run_eval.py --scaffold --annotations rallies.csv
10
11     Score a video's detected rallies (identify it by file or by stored video_id):
12         python run_eval.py --video "C:/path/match.mp4" --annotations rallies.csv
13         python run_eval.py --video-id <md5> --annotations rallies.csv --stage both --iou 0.5
14
15     Show which rallies were missed / spurious, and dump a JSON report:
16         python run_eval.py --video-id <md5> --annotations rallies.csv --show-failures --json
report.json
17     """
18
19     import os
20     import sys
21     import json
22     import argparse
23     import logging
24
25     from backend.infrastructure.database import Database
26     from backend.config import load_config
27     from backend.utils.hashing import compute_video_id as get_file_md5 # canonical video_id
28     from backend.eval.annotations import load_annotations, write_scaffold
29     from backend.eval.harness import evaluate, format_report_text, report_to_dict, STAGES
30
31
32     def _default_db_path() -> str:
33         """Resolve the DB path from the typed config (output.output_dir /
output.db_name)."""
34         cfg = load_config()
35         return os.path.join(cfg.output.output_dir, cfg.output.db_name)
36
37
38     _EPILOG = """\
39     Prerequisite:
40         The harness scores predictions read from the SQLite DB ? it does NOT run the
41         pipeline. So you must first PROCESS the video (e.g. via run_process_and_stitch.py)
42         so its detections land in output/sports_indexer.db. Then point this tool at the
43         same video file (--video) or its stored MD5 (--video-id). If you get
44         "database not found" or zero predictions, the video hasn't been processed yet.
45
46     See docs/EVALUATION.md for the annotation format and how to read the output.

```

```

47     """
48
49
50     def build_parser() -> argparse.ArgumentParser:
51         p = argparse.ArgumentParser(
52             description="Evaluate detected rallies vs ground-truth annotations.",
53             epilog=_EPILOG,
54             formatter_class=argparse.RawDescriptionHelpFormatter,
55         )
56         p.add_argument("--annotations", help="Path to the ground-truth rally CSV (start,end
57 per row).")
58         p.add_argument("--video", help="Path to the video file (its MD5 becomes the
59 video_id).")
60         p.add_argument("--video-id", help="Stored video_id (MD5) to evaluate, instead of
61 --video.")
62         p.add_argument("--db", default=None, help="SQLite DB path (default: from
63 config.json).")
64         p.add_argument("--stage", choices=("candidates", "final", "both"), default="both",
65             help="Which prediction stage(s) to score (default: both).")
66         p.add_argument("--detector", default=None,
67             help="Only score candidates from this detector (e.g. yolo_hybrid,
68 tracknet_hybrid).")
69         p.add_argument("--iou", type=float, default=0.5, help="IoU match threshold (default:
70 0.5).")
71         p.add_argument("--pr-curve", action="store_true", help="Include an IoU sweep
72 (0.1..0.9) in the report.")
73         p.add_argument("--show-failures", action="store_true",
74             help="List missed (FN) and spurious (FP) rallies with timestamps.")
75         p.add_argument("--json", dest="json_out", default=None, help="Write a JSON report to
76 this path.")
77         p.add_argument("--scaffold", action="store_true",
78             help="Write an empty annotation CSV to --annotations and exit.")
79         return p
80
81
82     def main(argv=None) -> int:
83         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
84         args = build_parser().parse_args(argv)
85
86         # Scaffold mode: just write the template and exit.
87         if args.scaffold:
88             if not args.annotations:
89                 print("ERROR: --scaffold requires --annotations <path>", file=sys.stderr)
90                 return 2
91             write_scaffold(args.annotations)
92             print(f"Wrote empty annotation file: {args.annotations}")
93             return 0
94
95         if not args.annotations:
96             print("ERROR: --annotations is required (or use --scaffold).", file=sys.stderr)
97             return 2
98
99         # Resolve the video_id.
100         if args.video_id:
101             video_id = args.video_id
102         elif args.video:
103             if not os.path.exists(args.video):
104                 print(f"ERROR: video not found: {args.video}", file=sys.stderr)
105                 return 2
106             print("Computing video MD5...")
107             video_id = get_file_md5(args.video)
108             print(f"video_id = {video_id}")
109         else:
110             print("ERROR: provide --video <path> or --video-id <md5>.", file=sys.stderr)
111             return 2

```

```

104
105     db_path = args.db or _default_db_path()
106     if not os.path.exists(db_path):
107         print(f"ERROR: database not found at {db_path}. Run the pipeline first.",
file=sys.stderr)
108         return 2
109
110     gt = load_annotations(args.annotations)
111     if not gt:
112         print(f"WARNING: no ground-truth rallies in {args.annotations}.",
file=sys.stderr)
113
114     stages = list(STAGES) if args.stage == "both" else [args.stage]
115     db = Database(db_path)
116     report = evaluate(db, video_id, gt, stages, iou_threshold=args.iou,
117                     detector=args.detector, with_pr_curve=args.pr_curve)
118
119     print(format_report_text(report, show_failures=args.show_failures))
120
121     if args.json_out:
122         with open(args.json_out, "w") as f:
123             json.dump(report_to_dict(report), f, indent=2)
124             print(f"\nJSON report written to {args.json_out}")
125
126     return 0
127
128
129 if __name__ == "__main__":
130     raise SystemExit(main())
131

```

22. user (2026-06-10T06:01:00.708Z)

```

1     """CLI: regression-test the production tool against golden-set ground truth.
2
3     Obtains ground truth from an annotation provider (a tier), scores the video's
4     stored predictions, and compares to a snapshotted baseline. Exit code 1 on a
5     detected regression so it can gate CI. See docs/GOLDEN_SET_IMPLEMENTATION.md.
6
7     Examples
8     -----
9     Snapshot the current numbers as the baseline (first run / after a real improvement):
10         python run_regression.py --video-id <md5> --tier file --annotations rallies.csv
--update-baseline
11
12     Check for a regression (exit 1 if precision/recall dropped beyond tolerance):
13         python run_regression.py --video-id <md5> --tier file --annotations rallies.csv
14     """
15     import os
16     import sys
17     import json
18     import argparse
19     import logging
20
21     from backend.infrastructure.database import Database
22     from backend.config import load_config
23     from backend.annotations import annotation_registry
24     from backend.eval import regression
25
26
27     def _default_db_path() -> str:
28         cfg = load_config()
29         return os.path.join(cfg.output.output_dir, cfg.output.db_name)
30
31

```

```

32     def build_parser() -> argparse.ArgumentParser:
33         p = argparse.ArgumentParser(description="Regression-test the tool vs golden-set
ground truth.")
34         p.add_argument("--video-id", required=True, help="Stored video_id (MD5) whose
predictions to score.")
35         p.add_argument("--tier", default="file",
36                         help=f"Annotation provider tier. Available:
{annotation_registry.list_available()}")
37         p.add_argument("--annotations", help="video_ref for the provider (e.g. CSV path for
the 'file' tier).")
38         p.add_argument("--stage", choices=("candidates", "final"), default="final")
39         p.add_argument("--detector", default=None, help="Restrict candidate scoring to one
detector.")
40         p.add_argument("--iou", type=float, default=0.5)
41         p.add_argument("--tolerance", type=float, default=regression.DEFAULT_TOLERANCE)
42         p.add_argument("--db", default=None)
43         p.add_argument("--baseline", default=regression.DEFAULT_BASELINE_PATH,
help="Baseline JSON path.")
44         p.add_argument("--update-baseline", action="store_true",
45                         help="Snapshot the current numbers as the new baseline instead of
checking.")
46         p.add_argument("--json", dest="json_out", default=None)
47         return p
48
49
50     def main(argv=None) -> int:
51         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
52         args = build_parser().parse_args(argv)
53
54         db_path = args.db or _default_db_path()
55         if not os.path.exists(db_path):
56             print(f"ERROR: database not found at {db_path}. Run the pipeline first.",
file=sys.stderr)
57             return 2
58         db = Database(db_path)
59
60         result = regression.regress_with_provider(
61             db, video_id=args.video_id, video_ref=args.annotations, tier=args.tier,
62             stage=args.stage, iou_threshold=args.iou, detector=args.detector,
63             tolerance=args.tolerance, baseline_path=args.baseline,
64         )
65
66         print(regression.format_result(result))
67
68         if args.update_baseline:
69             regression.save_baseline(result.video_id, result.stage, result.precision,
70                                     result.recall, result.f1, path=args.baseline)
71             print(f"Baseline updated in {args.baseline}.")
72
73         if args.json_out:
74             with open(args.json_out, "w") as f:
75                 json.dump(result.as_dict(), f, indent=2)
76                 print(f"JSON written to {args.json_out}")
77
78         # Non-zero exit on regression so CI can gate on it.
79         return 1 if result.regressed else 0
80
81
82     if __name__ == "__main__":
83         raise SystemExit(main())
84

```

23. user (2026-06-10T06:01:06.429Z)

```

1     """Phase 3 driver: batch process + evaluate a collector export against ground truth.

```

```

2
3     Walks the `manifest.json` produced by sports-data-collector's `curate export`,
4     processes each video through a segmenter (so detections land in the indexer DB),
5     scores each against its ground-truth rally CSV, and prints per-video plus
6     corpus-aggregate precision/recall/F1, mean IoU, and boundary error.
7
8     Examples
9     -----
10    Free CPU baseline on a single video (validate the loop):
11        python run_phase3_eval.py --manifest ../sports-data-
collector/data/eval_export/manifest.json \
12        --segmenter motion --only shuttlest_1
13
14    Full corpus with the two-stage detector (needs GEMINI_API_KEY; costs API calls):
15        python run_phase3_eval.py --manifest ../manifest.json --segmenter yolo_hybrid \
16        --json output/phase3_report.json
17
18    Re-score only (videos already processed), no re-segmentation:
19        python run_phase3_eval.py --manifest ../manifest.json --score-only
20    """
21
22    import os
23    import sys
24    import json
25    import time
26    import argparse
27    import logging
28
29    from dotenv import load_dotenv
30    load_dotenv() # make GEMINI_API_KEY (etc.) available to the AI-backed segmenters
31
32    from backend.infrastructure.database import Database
33    from backend.config import load_config
34    from backend.utils.hashing import compute_video_id as get_file_md5 # canonical video_id
35    # Importing the package registers all segmenters (motion/gemini/yolo_hybrid/...).
36    from backend.pipeline.segmenters import segmenter_registry
37    from backend.eval.annotations import load_annotations
38    from backend.eval.harness import evaluate, format_report_text, report_to_dict
39    from backend.eval import batch
40
41    logger = logging.getLogger(__name__)
42
43
44    def build_parser() -> argparse.ArgumentParser:
45        p = argparse.ArgumentParser(
46            description="Batch process + evaluate a collector export manifest (Phase 3).",
47            formatter_class=argparse.RawDescriptionHelpFormatter,
48        )
49        p.add_argument("--manifest", required=True, help="Path to the collector's
manifest.json.")
50        p.add_argument("--videos-root", default=None,
51            help="Root for resolving manifest relative local_paths "
52            "(default: the collector repo root, two levels above the
manifest).")
53        p.add_argument("--segmenter", default="motion",
54            help="Segmenter to run (default: motion ? free/CPU). "
55            "yolo_hybrid/gemini need GEMINI_API_KEY and cost API calls.")
56        p.add_argument("--stage", choices=("candidates", "final", "both", "auto"),
57            default="auto",
58            help="Which prediction stage(s) to score (default: auto by
segmenter).")
59        p.add_argument("--detector", default=None, help="Restrict candidate scoring to this
detector.")
60        p.add_argument("--iou", type=float, default=0.5, help="IoU match threshold (default:
0.5).")

```

```

60     p.add_argument("--only", action="append", default=None,
61                    help="Only this video_id (repeatable).")
62     p.add_argument("--limit", type=int, default=None, help="Process at most N videos.")
63     p.add_argument("--max-frames", type=int, default=None,
64                    help="Cap segmenter frames (fast validation).")
65     p.add_argument("--score-only", action="store_true",
66                    help="Skip segmentation; only score detections already in the DB.")
67     p.add_argument("--reprocess", action="store_true",
68                    help="Re-run segmentation even if the video already has segments.")
69     p.add_argument("--db", default=None, help="Indexer DB path (default:
output/<db_name>).")
70     p.add_argument("--json", dest="json_out", default=None, help="Write a JSON report
here.")
71     p.add_argument("--show-failures", action="store_true", help="List missed/spurious
rallies per video.")
72     return p
73
74
75     def main(argv=None) -> int:
76         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
77         args = build_parser().parse_args(argv)
78
79         manifest_path = os.path.abspath(args.manifest)
80         if not os.path.exists(manifest_path):
81             print(f"ERROR: manifest not found: {manifest_path}", file=sys.stderr)
82             return 2
83         manifest_dir = os.path.dirname(manifest_path)
84         videos_root = args.videos_root or os.path.dirname(os.path.dirname(manifest_dir))
85
86         with open(manifest_path) as f:
87             manifest = json.load(f)
88         entries = manifest.get("eval_sets", [])
89
90         cfg = load_config() # typed AppConfig (same object the live pipeline feeds
segmenters)
91         db_path = args.db or os.path.join(cfg.output.output_dir, cfg.output.db_name)
92         os.makedirs(os.path.dirname(db_path) or ".", exist_ok=True)
93         db = Database(db_path)
94
95         stages = batch.default_stages_for(args.segmenter, args.stage)
96
97         # Resolve the segmenter class up front (unless purely scoring).
98         segmenter = None
99         if not args.score_only:
100             seg_cls = segmenter_registry.get(args.segmenter)
101             if not seg_cls:
102                 print(f"ERROR: segmenter '{args.segmenter}' not found. "
103                       f"Available: {list(segmenter_registry.list_available().keys())}",
file=sys.stderr)
104                 return 2
105             segmenter = seg_cls(db, cfg)
106
107         # Filter entries.
108         targets = []
109         for e in entries:
110             if args.only and e["video_id"] not in args.only:
111                 continue
112             targets.append(e)
113         if args.limit:
114             targets = targets[: args.limit]
115
116         per_video = []
117         skipped = []
118         print(f"Phase 3: {len(targets)} target(s) | segmenter={args.segmenter} |
stages={stages} | db={db_path}\n")

```

```

119
120     for e in targets:
121         vid = e["video_id"]
122         video_path = batch.resolve_video_path(e.get("local_path"), videos_root)
123         csv_path = os.path.join(manifest_dir, e["csv"])
124
125         if not video_path or not os.path.exists(video_path):
126             skipped.append((vid, "no video file"))
127             print(f"[skip] {vid}: no video file ({e.get('local_path')})")
128             continue
129         if not os.path.exists(csv_path):
130             skipped.append((vid, "no GT csv"))
131             print(f"[skip] {vid}: GT csv missing ({csv_path})")
132             continue
133
134         video_id = get_file_md5(video_path)
135
136         # Segment unless score-only or already present (and not --reprocess).
137         if not args.score_only:
138             already = db.query_play_segments(video_id, {}) or
db.query_candidate_segments(video_id)
139             if already and not args.reprocess:
140                 print(f"[have] {vid}: detections already in DB (use --reprocess to
redo)")
141             else:
142                 print(f"[proc] {vid}: segmenting with {args.segmenter}...")
143                 db.add_video(video_id, video_path, "processed", [])
144                 t0 = time.time()
145                 segs = segmenter.process_video(video_id, video_path,
max_frames=args.max_frames)
146                 print(f"          -> {len(segs)} segments in {time.time() - t0:.0f}s")
147
148                 gt = load_annotations(csv_path)
149                 report = evaluate(db, video_id, gt, stages, iou_threshold=args.iou,
detector=args.detector)
150                 print(format_report_text(report, show_failures=args.show_failures))
151                 print()
152                 d = report_to_dict(report)
153                 d["collector_video_id"] = vid
154                 per_video.append(d)
155
156         agg = batch.aggregate(per_video, stages)
157         print(batch.format_aggregate(agg))
158         if skipped:
159             print(f"\nSkipped {len(skipped)}: " + ", ".join(f"{v} ({why})" for v, why in
skipped))
160
161         if args.json_out:
162             with open(args.json_out, "w") as f:
163                 json.dump({"segmenter": args.segmenter, "iou": args.iou, "stages": stages,
164                     "aggregate": agg, "per_video": per_video,
165                     "skipped": skipped}, f, indent=2)
166             print(f"\nJSON report written to {args.json_out}")
167
168         return 0
169
170
171 if __name__ == "__main__":
172     raise SystemExit(main())
173

```

24. user (2026-06-10T06:01:10.902Z)

```

1     import os
2     import sys

```

```

3     import json
4     import logging
5     from dotenv import load_dotenv
6
7     load_dotenv()
8
9     # Configure root logger ? all backend modules use logging.getLogger(__name__)
10    logging.basicConfig(
11        level=logging.INFO,
12        format="%(asctime)s [%(name)s] %(levelname)s: %(message)s",
13        datefmt="%H:%M:%S",
14    )
15
16    from backend.infrastructure.database import Database
17    from backend.config import load_config
18    from backend.utils.hashing import compute_video_id
19    from backend.pipeline.validators.base import registry
20    from backend.pipeline.segmenters import segmenter_registry
21    from backend.pipeline.stitcher.compiler import VideoCompiler
22
23    def main():
24        video_path = r"C:\Users\avidu\Videos\Testing\KushagraYashAviFirstVideo.mp4"
25        output_dir = r"C:\Users\avidu\Videos\Testing"
26        output_filename = "Final_2_OutputOfKushagraYashAviFirstVideo.mp4"
27
28        print(f"Loading config...")
29        cfg = load_config() # typed AppConfig ? single source of defaults
30
31        print(f"Verifying files exist...")
32        if not os.path.exists(video_path):
33            print(f"Error: Video file not found at {video_path}")
34            return
35
36        os.makedirs(cfg.output.output_dir, exist_ok=True)
37        db_path = os.path.join(cfg.output.output_dir, cfg.output.db_name)
38        db = Database(db_path)
39
40        # Calculate MD5 of the video to uniquely identify it
41        import time
42        print("Calculating video MD5 (this might take a few seconds on large files)...",
flush=True)
43        t0 = time.time()
44        video_id = compute_video_id(video_path)
45        print(f"Calculated MD5: {video_id} (took {time.time()-t0:.1f}s)", flush=True)
46
47        segments = []
48        cached_video = db.get_video(video_id)
49        cache_hit = False
50
51        if cached_video and cached_video.get("status") == "processed":
52            segments = db.query_play_segments(video_id, {})
53            if len(segments) > 0:
54                print(f"\n[CACHE] Found cached indexed video in database (MD5: {video_id})
with {len(segments)} parsed play segments.", flush=True)
55                choice = input("Would you like to (1) Run stitching on the cached segments,
or (2) Reprocess the video with Gemini from scratch? [1/2]: ").strip()
56                if choice == '1':
57                    print("[CACHE] Using cached play segments. Skipping segmenter API
call.", flush=True)
58                    cache_hit = True
59                else:
60                    print("[CACHE] Ignoring cache...", flush=True)
61            else:
62                print(f"[CACHE] Video exists in DB but contains 0 play segments. Treating as
cache miss.", flush=True)

```

```

63
64     if not cache_hit:
65         print("[CACHE] Initializing clean indexing...", flush=True)
66         db.clear_video_data(video_id)
67
68         print("Bypassing OpenCV validations for performance on massive files...",
flush=True)
69         db.add_video(video_id, video_path, "processed", [])
70
71         print("Running segmenter & indexing...", flush=True)
72         seg_name = cfg.indexing.default_segmenter
73         segmenter_cls = segmenter_registry.get(seg_name)
74         if not segmenter_cls:
75             print(f"[FAIL] Segmenter '{seg_name}' not found.")
76             return
77         print(f"Using segmenter: {seg_name}", flush=True)
78         segmenter = segmenter_cls(db, cfg)
79         segments = segmenter.process_video(video_id, video_path)
80         print(f"Successfully indexed {len(segments)} play segments.", flush=True)
81
82     if not segments:
83         print("No play segments detected. Cannot compile highlights.")
84         return
85
86
87     end_padding = cfg.stitching.end_padding
88     begin_padding = cfg.stitching.begin_padding
89     min_shot_count = cfg.stitching.min_shot_count
90
91     if cache_hit:
92         print("\n--- Stitching Customizations ---")
93         try:
94             bpad_input = input(f"Enter begin_padding in seconds (default
{begin_padding}): ").strip()
95             if bpad_input:
96                 begin_padding = float(bpad_input)
97
98             pad_input = input(f"Enter end_padding in seconds (default {end_padding}):
").strip()
99             if pad_input:
100                 end_padding = float(pad_input)
101
102             shot_input = input(f"Enter min_shot_count to filter by (default
{min_shot_count}): ").strip()
103             if shot_input:
104                 min_shot_count = int(shot_input)
105         except ValueError:
106             print("[WARNING] Invalid input detected, falling back to defaults from
config.json.")
107
108     print(f"\nCompiling highlights (with begin_padding={begin_padding}s,
end_padding={end_padding}s, min_shot_count={min_shot_count})...", flush=True)
109
110     # Filter segments where shot_count >= min_shot_count
111     time_pairs = []
112     skipped_count = 0
113     for s in segments:
114         meta = s.get("metadata", {})
115         if isinstance(meta, str):
116             try:
117                 meta = json.loads(meta)
118             except Exception:
119                 meta = {}
120         shot_count = meta.get("shot_count", 0)
121         if shot_count >= min_shot_count:

```

```

122         time_pairs.append((s["start_time"], s["end_time"]))
123     else:
124         skipped_count += 1
125
126     if skipped_count > 0:
127         print(f"[FILTER] Excluded {skipped_count} play segments with fewer than
{min_shot_count} shots.", flush=True)
128     print(f"[FILTER] Retaining {len(time_pairs)} play segments for stitching.",
flush=True)
129
130     if not time_pairs:
131         print("No play segments passed the filtering criteria. Cannot compile
highlights.")
132         return
133
134     compiler = VideoCompiler(output_dir, end_padding=end_padding,
begin_padding=begin_padding)
135     try:
136         out_path = compiler.compile_highlights(video_path, time_pairs, output_filename)
137         print(f"Stitched highlight video saved to: {out_path}", flush=True)
138     except Exception as e:
139         print(f"Error during stitching: {e}", flush=True)
140
141 if __name__ == "__main__":
142     main()
143

```

25. user (2026-06-10T06:01:17.538Z)

```

1     """TrackNet shuttle-trajectory runner (WSL2 subprocess adapter).
2
3     This bridges the Windows-side indexer to a TensorFlow TrackNetV4 install living
4     inside WSL2 (ADR-001 in docs/DECISIONS.md). The flow is:
5
6         1. Translate the Windows video path to its WSL-visible form (and optionally
7            stage it into the WSL filesystem to avoid slow /mnt/c per-frame reads).
8         2. Invoke ``src/predict.py`` inside the TrackNetV4 conda env via ``wsl``.
9         3. Read back the per-frame ``Frame,Visibility,X,Y`` CSV it writes.
10        4. Convert that trajectory into high-motion "action windows" ? the same
11           candidate-rally shape HybridYoloSegmenter produces ? so the rest of the
12           pipeline (AI handoff, DB population) is unchanged.
13
14     Nothing here imports TensorFlow; all model code stays in WSL.
15     """
16
17     import os
18     import csv
19     import shlex
20     import logging
21     import subprocess
22     from dataclasses import dataclass
23     from typing import List, Dict, Any, Optional, Tuple
24
25     from backend.pipeline.detectors.base import DetectorRunner
26
27     logger = logging.getLogger(__name__)
28
29
30     @dataclass
31     class TrackNetConfig:
32         """Resolved settings for a TrackNet WSL run (built from config.json ->
indexing.tracknet)."""
33         distro: str = "Ubuntu"
34         repo_dir: str = "~/models/TrackNetV4" # WSL path to the cloned repo
35         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"

```

```

36         conda_env: str = "TrackNetV4"
37         weights_path: str = ""                                     # WSL path to the .h5/.keras weights
(REQUIRED)
38         queue_length: int = 5
39         stage_in_wsl: bool = True                                 # copy video into WSL fs first
(perf)
40         wsl_stage_dir: str = "~/clips"                           # where staged videos/outputs live
in WSL
41         timeout_sec: int = 0                                     # 0 = no timeout
42
43     @classmethod
44     def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "TrackNetConfig":
45         tn = (idx_cfg or {}).get("tracknet", {}) or {}
46         return cls(
47             distro=tn.get("wsl_distro", "Ubuntu"),
48             repo_dir=tn.get("repo_dir", "~/models/TrackNetV4"),
49             conda_sh=tn.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
50             conda_env=tn.get("conda_env", "TrackNetV4"),
51             weights_path=tn.get("weights_path", ""),
52             queue_length=int(tn.get("queue_length", 5)),
53             stage_in_wsl=bool(tn.get("stage_in_wsl", True)),
54             wsl_stage_dir=tn.get("wsl_stage_dir", "~/clips"),
55             timeout_sec=int(tn.get("timeout_sec", 0)),
56         )
57
58
59     def to_wsl_mnt_path(win_path: str) -> str:
60         """Translate a Windows path (C:\\Users\\x) to its WSL /mnt form (/mnt/c/Users/x).
61
62         Already-POSIX paths (starting with / or ~) are returned unchanged so the
63         same helper is safe to call on either kind of path.
64         """
65         p = str(win_path)
66         if p.startswith("/") or p.startswith("~"):
67             return p
68         p = p.replace("\\", "/")
69         if len(p) >= 2 and p[1] == ":":
70             drive = p[0].lower()
71             return f"/mnt/{drive}{p[2:]}"
72         return p
73
74
75     @dataclass
76     class TrajectoryPoint:
77         frame: int
78         visible: bool
79         x: float
80         y: float
81
82
83     class TrackNetRunner(DetectorRunner): # DetectorRunner ABC (finish item 6 / new item 1)
84         """Runs TrackNet predict.py in WSL and turns the output CSV into action windows.
85
86         Implements the common DetectorRunner contract so a future Linux-native or
87         alternative trajectory runner can be substituted without touching the hybrids.
88         """
89
90         def __init__(self, cfg: TrackNetConfig):
91             self.cfg = cfg
92
93         # ----- WSL exec
94
95         def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
96             """Run a bash command string inside the configured WSL distro."""
97             full = f"source {self.cfg.conda_sh} && {inner_cmd}"

```

```

98         argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
99         logger.debug("WSL exec: %s", full)
100         return subprocess.run(
101             argv, capture_output=True, text=True,
102             timeout=(self.cfg.timeout_sec or None),
103         )
104
105     def healthcheck(self) -> Tuple[bool, str]:
106         """Verify the WSL env is usable: conda env exists, TF imports, GPU visible.
107
108         Returns (ok, message). Cheap to call before a long run.
109         """
110         cmd = (
111             f"conda activate {self.cfg.conda_env} && "
112             f"python -c \"import tensorflow as tf; "
113             f"print('TF', tf.__version__, 'GPUs', "
114             f"len(tf.config.list_physical_devices('GPU')))\\""
115         )
116         try:
117             res = self._wsl_bash(cmd)
118         except FileNotFoundError:
119             return False, "`wsl` not found ? is this running on Windows with WSL2
120             installed?"
121         except subprocess.TimeoutExpired:
122             return False, "WSL healthcheck timed out."
123         out = (res.stdout or "").strip()
124         if res.returncode != 0:
125             return False, f"WSL env check failed: {(res.stderr or out).strip()}"
126         return True, out
127
128     # ----- inference
129
130     def run_predict(self, video_win_path: str, output_win_dir: str) -> Optional[str]:
131         """Run predict.py on a video. Returns the Windows path to the produced CSV, or
132         None.
133
134         ``output_win_dir`` is a Windows directory (under the repo's output/) that
135         WSL writes into via its /mnt view, so the Windows process can read the CSV
136         back with no copy.
137         """
138         if not self.cfg.weights_path:
139             logger.error(
140                 "TrackNet weights_path is not set. Set indexing.tracknet.weights_path "
141                 "in config.json to the WSL path of your trained TrackNetV4 weights."
142             )
143             return None
144
145         os.makedirs(output_win_dir, exist_ok=True)
146         # Normalize separators so basename/splitext work when tests (or collector)
147         # pass Windows-style paths on a Linux host.
148         video_win_path = str(video_win_path).replace("\\", "/")
149         base = os.path.basename(video_win_path)
150         stem, ext = os.path.splitext(base)
151
152         # predict.py only accepts .avi/.mp4 and names the CSV "<stem>-predict.csv".
153         if ext.lower() not in (".mp4", ".avi"):
154             logger.error("TrackNet predict.py only supports .mp4/.avi (got %s)", ext)
155             return None
156
157         wsl_out = to_wsl_mnt_path(output_win_dir)
158         expected_csv_win = os.path.join(output_win_dir, f"{stem}_predict.csv")
159
160         # Resolve the video path WSL will read.
161         if self.cfg.stage_in_wsl:
162             staged = self._stage_video(video_win_path, base)

```

```

160         if staged is None:
161             return None
162         wsl_video = staged
163     else:
164         wsl_video = to_wsl_mnt_path(video_win_path)
165
166     cmd = (
167         f"conda activate {self.cfg.conda_env} && "
168         f"cd {self.cfg.repo_dir}/src && "
169         f"python predict.py "
170         f"--video_path {shlex.quote(wsl_video)} "
171         f"--model_weights {shlex.quote(self.cfg.weights_path)} "
172         f"--output_dir {shlex.quote(wsl_out)} "
173         f"--queue_length {self.cfg.queue_length}"
174     )
175     logger.info("Running TrackNet inference on %s ...", base)
176     try:
177         res = self._wsl_bash(cmd)
178     except subprocess.TimeoutExpired:
179         logger.error("TrackNet inference timed out after %ss.",
180 self.cfg.timeout_sec)
181         return None
182
183     if res.returncode != 0:
184         logger.error("TrackNet predict.py failed:\n%s", (res.stderr or
185 res.stdout)[-2000:])
186         return None
187
188     if not os.path.exists(expected_csv_win):
189         logger.error("TrackNet finished but expected CSV not found at %s",
190 expected_csv_win)
191         return None
192     logger.info("TrackNet trajectory CSV: %s", expected_csv_win)
193     return expected_csv_win
194
195 def _stage_video(self, video_win_path: str, base: str) -> Optional[str]:
196     """Copy the input video into the WSL filesystem (avoids slow /mnt/c reads).
197
198     Returns the WSL path to the staged copy, or None on failure.
199     """
200     src_mnt = to_wsl_mnt_path(video_win_path)
201     dest = f"{self.cfg.wsl_stage_dir}/{base}"
202     cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)}
203 {shlex.quote(dest)} && echo OK"
204     try:
205         res = self._wsl_bash(cmd)
206     except subprocess.TimeoutExpired:
207         logger.error("Staging video into WSL timed out.")
208         return None
209     if res.returncode != 0 or "OK" not in (res.stdout or ""):
210         logger.error("Failed to stage video into WSL: %s", (res.stderr or
211 res.stdout).strip())
212         return None
213     return dest
214
215 # ----- CSV -> windows
216
217 @staticmethod
218 def parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]:
219     """Parse a TrackNet predict CSV (columns: Frame,Visibility,X,Y)."""
220     points: List[TrajectoryPoint] = []
221     with open(csv_win_path, newline="") as f:
222         reader = csv.DictReader(f)
223         for row in reader:
224             try:

```

```

220         points.append(TrajectoryPoint(
221             frame=int(row["Frame"]),
222             visible=int(row["Visibility"]) == 1,
223             x=float(row["X"]),
224             y=float(row["Y"]),
225         ))
226     except (KeyError, ValueError):
227         continue
228     points.sort(key=lambda p: p.frame)
229     return points
230
231 @staticmethod
232 def trajectory_to_action_windows(
233     points: List[TrajectoryPoint],
234     fps: float,
235     frame_width: float,
236     chunk_start: float = 0.0,
237     velocity_thresh: float = 0.02,
238     merge_gap: float = 2.0,
239     min_window_duration: float = 2.0,
240     chunk_index: Optional[int] = None,
241 ) -> List[Dict[str, Any]]:
242     """Convert a shuttle trajectory into candidate rally windows.
243
244     A frame is "active" when the shuttle is visible AND its inter-frame
245     displacement (normalised by frame width, per second) exceeds
246     ``velocity_thresh`` ? i.e. the shuttle is in flight. Active frames within
247     ``merge_gap`` seconds of each other are grouped into one window; short
248     windows are dropped. Mirrors HybridYoloSegmenter's windowing so the dict
249     shape feeding the AI-handoff phase is identical.
250     """
251     if fps <= 0:
252         fps = 30.0
253     if frame_width <= 0:
254         frame_width = 1280.0
255
256     # Compute per-frame normalised velocity from the last *visible* point.
257     active = [] # (frame_idx, velocity)
258     prev: Optional[TrajectoryPoint] = None
259     for p in points:
260         if not p.visible:
261             prev = None # break the trajectory; absent shuttle = not in flight
262             continue
263         if prev is not None:
264             dframes = p.frame - prev.frame
265             if dframes > 0:
266                 dist = ((p.x - prev.x) ** 2 + (p.y - prev.y) ** 2) ** 0.5
267                 velocity = (dist / frame_width) * (fps / dframes)
268                 if velocity > velocity_thresh:
269                     active.append((p.frame, velocity))
270             prev = p
271
272     if not active:
273         return []
274
275     max_gap_frames = int(merge_gap * fps)
276
277     def _finalize(group: List[Tuple[int, float]]) -> Dict[str, Any]:
278         vels = [v for _, v in group]
279         return {
280             "start": group[0][0] / fps + chunk_start,
281             "end": group[-1][0] / fps + chunk_start,
282             "active_frame_count": len(group),
283             "peak_velocity": max(vels),
284             "mean_velocity": sum(vels) / len(vels),

```

```

285         "track_id_count": 1, # single shuttle; kept for window-shape parity
286         "chunk_index": chunk_index,
287     }
288
289     windows = []
290     group = [active[0]]
291     for af in active[1:]:
292         if af[0] - group[-1][0] <= max_gap_frames:
293             group.append(af)
294         else:
295             windows.append(_finalize(group))
296             group = [af]
297     windows.append(_finalize(group))
298
299     return [w for w in windows if (w["end"] - w["start"]) >= min_window_duration]
300

```

26. user (2026-06-10T06:01:21.793Z)

```

1     """Rally-start gate: suppress between-point false fires using shuttle net-crossings.
2
3     WASB (shuttle-only) over-detects: when a player flicks/tosses the shuttle back to
4     the opponent for service after losing a point, that single trajectory looks like a
5     rally. The discriminator is structural, not duration-based: a real rally has the
6     shuttle crossing the net MULTIPLE times (each shot lands on the other side); a
7     single service feed crosses ~once.
8
9     This module is camera-agnostic and needs NO new model ? it works on the trajectory
10    we already produce (Frame,Visibility,X,Y). It estimates the play axis (the image
11    axis with the larger shuttle spread) and the net position (median shuttle coord on
12    that axis ? the shuttle clusters at the two ends and crosses the net region often,
13    so the median sits near the net), then counts sign changes of (coord - net) across
14    the visible points inside a candidate window, with a deadband to ignore jitter.
15
16    Gate A in the over-fire fix spectrum (B = player-stance state machine, future).
17    Pure functions only ? no I/O, unit-tested; intended as a post-filter on the
18    windows from trajectory_to_action_windows, or to fold into it later.
19    """
20    from __future__ import annotations
21
22    from dataclasses import dataclass
23    from typing import List, Sequence, Tuple
24
25    Interval = Tuple[float, float]
26
27
28    @dataclass
29    class GatedWindow:
30        start: float
31        end: float
32        crossings: int
33        visible_points: int
34        kept: bool
35
36
37    def _spread(vals: Sequence[float]) -> float:
38        """Robust spread = p95 - p5 (ignores a few outlier detections)."""
39        if len(vals) < 2:
40            return 0.0
41        s = sorted(vals)
42        lo = s[max(0, int(0.05 * (len(s) - 1)))]
43        hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]
44        return hi - lo
45
46

```

```

47 def _median(vals: Sequence[float]) -> float:
48     if not vals:
49         return 0.0
50     s = sorted(vals)
51     n = len(s)
52     return s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2])
53
54
55 def estimate_play_axis(points) -> Tuple[str, float, float]:
56     """Return (axis 'x'/'y', net_value, span) from visible points.
57
58     Play axis = the image axis along which the shuttle travels most (larger spread).
59     For a baseline camera that's Y (players top/bottom); for a side camera, X.
60     """
61     xs = [p.x for p in points if p.visible]
62     ys = [p.y for p in points if p.visible]
63     sx, sy = _spread(xs), _spread(ys)
64     if sx >= sy:
65         return "x", _median(xs), sx
66     return "y", _median(ys), sy
67
68
69 def count_net_crossings(window_points, axis: str, net: float, deadband: float) -> int:
70     """Count sign changes of (coord - net) across visible points, ignoring the
71     deadband zone near the net (so jitter at the net doesn't inflate the count)."""
72     crossings = 0
73     last_side = None
74     for p in window_points:
75         if not p.visible:
76             continue
77         v = (p.x if axis == "x" else p.y) - net
78         if abs(v) < deadband:
79             continue
80         side = v > 0
81         if last_side is not None and side != last_side:
82             crossings += 1
83         last_side = side
84     return crossings
85
86
87 def _points_in(points, fps: float, start: float, end: float):
88     return [p for p in points if start * fps <= p.frame <= end * fps]
89
90
91 def gate_windows(points, windows: List[Interval], fps: float,
92                 min_crossings: int = 2, deadband_frac: float = 0.06) ->
List[GatedWindow]:
93     """Annotate each (start,end) window with its shuttle net-crossing count and a
94     keep decision (crossings >= min_crossings). Deadband is a fraction of the play
95     axis span. Net/axis are estimated once over the whole trajectory (stable)."""
96     axis, net, span = estimate_play_axis(points)
97     deadband = deadband_frac * span
98     out: List[GatedWindow] = []
99     for (s, e) in windows:
100         wp = _points_in(points, fps, s, e)
101         c = count_net_crossings(wp, axis, net, deadband)
102         vis = sum(1 for p in wp if p.visible)
103         out.append(GatedWindow(start=s, end=e, crossings=c, visible_points=vis,
104                               kept=c >= min_crossings))
105     return out
106
107
108 def apply_gate(points, windows: List[Interval], fps: float,
109               min_crossings: int = 2, deadband_frac: float = 0.06) -> List[Interval]:
110     """Convenience: return only the windows that pass the crossing gate."""

```

```

111         return [(g.start, g.end) for g in
112                 gate_windows(points, windows, fps, min_crossings, deadband_frac) if g.kept]
113

```

27. user (2026-06-10T06:01:29.817Z)

```

46:- **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss strict
temporal boundaries**
58:- **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits
structured timestamped
72:- **Model/framework: UNCHANGED.** One shared TAL head + one Qwen3-VL base, fine-tuned across
sports (sport = an input
99:## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
105:labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg · TAL
ActivityNet-JSON · VLM clip?QA),
120: per sport.** Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).
122:### M0.4 ? Gen-0 TAL harness (**FIRST-CLASS ACTION ITEM** per owner) · *greenlight item*
131:## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
132:Run the A/B; if the learned Gen-0 clears the ship-gate with per-video calibration, it
becomes the rally/highlight

```

28. user (2026-06-10T06:01:30.526Z)

```

tests\test_eval_annotations.py
tests\test_eval_batch.py
tests\test_eval_classifier.py
tests\test_eval_metrics.py
tests\test_eval_training.py
tests\__pycache__\test_eval_annotations.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_eval_batch.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_eval_classifier.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_eval_metrics.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_eval_training.cpython-313-pytest-9.0.3.pyc
tests\test_eval_harness.py
tests\test_eval_regression.py
tests\__pycache__\test_eval_harness.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_eval_regression.cpython-313-pytest-9.0.3.pyc
tests\test_eval_partial_labels.py
tests\__pycache__\test_eval_partial_labels.cpython-313-pytest-9.0.3.pyc
tests\__pycache__\test_eval_annotations.cpython-313.pyc
tests\__pycache__\test_eval_batch.cpython-313.pyc
tests\__pycache__\test_eval_classifier.cpython-313.pyc
tests\__pycache__\test_eval_harness.cpython-313.pyc
tests\__pycache__\test_eval_metrics.cpython-313.pyc
tests\__pycache__\test_eval_partial_labels.cpython-313.pyc
tests\__pycache__\test_eval_regression.cpython-313.pyc
tests\__pycache__\test_eval_training.cpython-313.pyc

```

29. user (2026-06-10T06:01:35.857Z)

```

90     - **On-device (deferred) = the only place a 2nd language appears, and it doesn't dictate
our dev language:** train in
91     Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
92     runs the artifact with zero Python. **The one concrete constraint now: train with
export-clean ops** so the compression
93     target stays reachable without re-architecting. (On-device base stays on the
Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
94     - Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/PyO3 ? only then a
95     separate service. **None exists today.**
96
97     ---
98

```

```

99     ## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
100
101     ### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) .
    *greenlight item*
102     Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index
    schema (the conversational seam)
103     ? same row, wired to a store. Fields: `{video_id(MD5), rally_ordinal, sport, start, end,
    ending_reason/fault_tags,
104     shot_count, label_type, source(human|pseudo|vendor), consent/training_opt_in, split(BY
    MATCH), trajectory_ref,
105     labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg · TAL
    ActivityNet-JSON · VLM clip?QA),
106     one scorer spine (`metrics.match_intervals`). Authored in `sports-data-collector`
    (system-of-record; extends PR #13).
107     **Reserve fault/event taxonomy now.** Acceptance: round-trip test; split-by-match
    enforced; **no Gemini-sourced rows in any training view**.
108
109     ### M0.2 ? Grow the corpus · **DONE (n=6), running** · *owner action + my scoring loop*
110     6 distinct matches labeled (3 singles incl. Mahadevpura, 1 doubles, 2 multi-court); per-
    match + LOVO recorded; the
111     "contrast" data-quality metric validated. Owner adding 1 more varied multi-court-club
    match ? margin. **Bottleneck =
112     data + calibration, not method.** Keep growing per new sport.
113
114     ### M0.3 ? Compliance cleanup (**hard blocker ? do before any training run**) .
    *greenlight item*
115     - **(i) PURGE the Gemini-derived TRAINING labels** in `training.label_candidates` (over
    Gemini CSVs) ? a **live
116     no-paid-LLM-training violation** that bakes in the moment a training run precedes it.
    Retarget to human + open-teacher
117     (own-model/WASB) labels; Gemini = eval/reference/fallback only.
118     - **(ii) WASB C3 ? first-class action item (owner):** the academic-data checkpoint
    provenance is a **commercial-ship
119     blocker** that a clean head does NOT launder. Resolve via **own-trained weights** or a
    clean detector swap; **multiplies
120     per sport.** Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).
121
122     ### M0.4 ? Gen-0 TAL harness (**FIRST-CLASS ACTION ITEM** per owner) · *greenlight item*
123     Productionize `rally_seq_proto.py` ? a **one-command train?eval?ship-gate** harness over
    **ActionFormer (MIT, 1:1
124     rally=instance)** and/or **ASFormer (MIT, TAS)** (use `sj-li/MS-TCN2` if MS-TCN++, *not*
    the Commons-Clause repo),
125     reusing `metrics.match_intervals`. **Define the ship-gate target up front** (open item):
    a FLOOR of "beat the honest n=6
126     heuristic LOVO **0.480**" is necessary but not sufficient ? set an explicit **F1@tIoU**
    target for "very high P/R" (e.g.
127     F1@tIoU=0.5 for highlights, tighter for precise rally cuts) before declaring victory.
    Trains on the GPU/WSL box.
128
129     ---
130
131     ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
132     Run the A/B; if the learned Gen-0 clears the ship-gate with per-video calibration, it
    becomes the rally/highlight
133     extractor (local, MIT). If not, the gap (more data/features) is the next signal ? no
    overfitting to declare victory.
134
135     ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + healthy corpus +
    explicit go)
136     - **Gen-1:** concatenate fully-local cue channels onto the head. ? **Audio is NOT a
    promising cue here** ? our E1 probe
137     (PR #35) found it weak (~2.2× discrimination, AUC ~0.6) and the foreground-audio idea
    also failed this session; keep it
138     parked, not "in reserve." Pose/court is the more credible #2 cue (license/cost-vet
    first).

```

```

139     - **Gen-2 (VLM):** ms-swift fine-tune of **Qwen3-VL** (SFT cold-start ? GRPO/RFT with a
temporal-IoU-vs-golden reward,
140     our own CoT supervision). **Two-stage** (VLM proposes, head refines). Cloud train on
Modal (no-access) / RunPod-Secure.
141     - **Hard pre-serving gates (unmeasured, load-bearing):** the **$/call crossover** (self-
hosted VLM vs Gemini) *** a real
142     VRAM/token + cost benchmark** on rented A100/L40S (ms-swift multimodal-GRPO examples
use 6?8 GPUs ? **budget is the
143     largest open risk and is currently unquantified**). Don't stand up an owned serving
stack until these clear.
144
145     ---
146
147     ## Decisions / status
148     | # | item | status |
149     |---|---|---|
150     | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4
harness** | awaiting owner |
151     | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |
152     | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ?
|
153     | 4 | Collector = corpus + event-store system-of-record | **owner: confirm?** |
154     | 5 | Rename repo off "badminton" | **owner action** ? |
155     | 6 | Conversational-QA: build seam now, punt model | **rec accepted?** |
156
157     ## Open risks / unknowns (none block Phase-0 research)
158     Budget/cost to reach Gen-2 **unquantified** (largest risk) · ship-gate "very high P/R"
target undefined · per-sport clean
159     trajectory detectors not yet identified · base-model pretraining provenance unauditale
(risk-acceptance) · real VLM

```

30. user (2026-06-10T06:01:36.909Z)

```

tests\test_calibration.py
tests\test_calibrate_wasb.py
tests\test_calibrate_local.py

```

31. user (2026-06-10T06:04:21.782Z)

```

198-         segments_data = []
199-         for segment in ai_segments:
200-             try:
201-                 start = float(segment["start_time"])
202-                 end = float(segment["end_time"])
203-                 dur = end - start
204-                 shot_count = segment.get("shuttle_exchange_count")
205-                 try:
206-                     shot_count = int(shot_count)
207-                 except (ValueError, TypeError):
208-                     shot_count = max(3, int(dur / 0.8))
209-                 metadata = {"shot_count": shot_count,
210-                             "shot_count_confidence": segment.get("shot_count_confidence",
"Unknown"),
211-                             "detector": f"wasb-hybrid-{model_name}" }
212-                 segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
213-                 if segment_id:
214-                     candidate_id = segment.get("_candidate_id")
215-                     if candidate_id is not None:
216-                         self.db.link_candidate_to_segment(candidate_id, segment_id)

```

32. user (2026-06-10T06:04:23.562Z)

```

10:     def __init__(self, output_dir: str, end_padding: float = 1.0, begin_padding: float =

```

```

0.5):
12:         self.end_padding = end_padding
13:         self.begin_padding = begin_padding
67:         into a seamless highlights file, applying end_padding to prevent abrupt endings.
90:         logger.info(f"Trimming {len(segments)} segments
(begin_padding={self.begin_padding}s, end_padding={self.end_padding}s)...")
125:         padded_start = max(0.0, start - self.begin_padding)
126:         padded_end = end + self.end_padding
131:         logger.info(f"{segment_label}: Padded end ({end +
self.end_padding:.2f}s) exceeds video duration. ")

```

33. user (2026-06-10T06:04:23.734Z)

```

1      """Rally-detection evaluation harness.
2
3      Scores the pipeline's detected rallies against hand-annotated `[start, end)`
4      ground-truth rally timestamps, using temporal interval matching. See
5      docs/EVALUATION.md for the annotation format and CLI usage.
6
7      - `metrics`      ? pure interval-matching math (IoU, matching, P/R/F1, boundary error).
8      - `annotations` ? load ground-truth rally CSVs.
9      - `harness`      ? read predictions from the DB and score them.
10     """
11

```

34. user (2026-06-10T06:04:32.445Z)

```

83-
84-
85:class StitchingConfig(BaseModel):
86:    begin_padding: float = 0.5
87:    end_padding: float = 0.5
88-    use_cache: bool = False
89-    min_shot_count: int = 1

```

35. user (2026-06-10T06:04:37.880Z)

```

..... [ 80%]
..... [100%]
90 passed in 2.03s

```

36. user (2026-06-10T06:04:54.018Z)

```

iou=0 edge: matches = 1 (should be 0; pred and gt do not overlap)
score: P=1.00 R=1.00 F1=1.00
greedy ex: TP = 1 matches: [(0, 0, 0.9)]

```

37. user (2026-06-10T06:05:08.462Z)

```

all-tied AUC: 0.75 (midrank-correct value = 0.5)
reordered   : 0.25

```

38. user (2026-06-10T06:05:26.517Z)

```

.gitignore:39:output/    output/regression_baseline.json
.gitignore:39:output/    output/sports_indexer.db
Binary file .gitignore matches

```

39. user (2026-06-10T06:05:27.579Z)

No matches found

40. user (2026-06-10T06:05:33.721Z)

.github\workflows\ci.yml

41. user (2026-06-10T06:05:34.830Z)

Found 6 files
docs\CODE_MAP.md
run_regression.py
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\DEFERRED_HARDENING_PLAN.md
tests\test_eval_regression.py
docs\GOLDEN_SET_IMPLEMENTATION.md

42. user (2026-06-10T06:05:40.390Z)

```
1      name: CI
2
3      on:
4        push:
5          branches: [master]
6        pull_request:
7
8      permissions:
9        contents: read
10
11     concurrency:
12       group: ci-${{ github.ref }}
13       cancel-in-progress: true
14
15     jobs:
16       # Fast guard against the 7fbb0 class of breakage: a SyntaxError or
17       # module-level import error shipping while the suite stays green.
18       # Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
19       # on machines without the GPU stack (test_import_smoke stubs what's absent).
20       import-smoke:
21         name: Syntax sweep + import smoke (no GPU stack)
22         runs-on: ubuntu-latest
23         steps:
24           - uses: actions/checkout@v4
25           - uses: actions/setup-python@v5
26             with:
27               python-version: "3.13"
28           - name: Install minimal deps
29             run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
30             genai
31           - name: Run import smoke tests
32             run: python -m pytest tests/test_import_smoke.py -v
33
34       full-suite:
35         name: Full test suite (offline, mocked)
36         runs-on: ubuntu-latest
37         steps:
38           - uses: actions/checkout@v4
39           - uses: actions/setup-python@v5
40             with:
41               python-version: "3.13"
42               cache: pip
43           - name: System libs for opencv-python
44             run: |
45               sudo apt-get update
46               sudo apt-get install -y libgl1 libglu1-mesa-dev || sudo apt-get install -y
47               libgl1 libglu1-mesa-dev
48           - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
49             run: python -m pip install torch torchvision --index-url
50             https://download.pytorch.org/whl/cpu
51           - name: Install requirements
```

```
49         run: python -m pip install -r requirements.txt pytest
50     # obsreport (private, soft dep) is absent here: reporting tests skip via
51     # pytest.importorskip ? run_all_tests.py prints which.
52     - name: Run full suite
53       run: python run_all_tests.py
54
```

43. user (2026-06-10T06:05:49.737Z)

```
docs\archives\quality-iterations\README.md:20:[Omitted long matching line]
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:35:per-video best-fit:
adarsh 0.777 | kushagra 0.163 (ceiling ? NO config beats it)
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:63:shifts across footage;
fixed by more videos / per-video calibration / a scale-invariant sequence model).
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:85:| mean held-out F1 |
0.544 ± 0.278 | ? | 0.615 |
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:94: rose because more
data closed its calibration gap ? Kushagra's held-out F1 went 0.049 ? 0.417 just
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:105:- Heuristic (current
pipeline): mean held-out F1 0.544 ± 0.278 ? the ±0.278 is the footage-fragility
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:107:- Learned prototype
(owned-model seed): mean held-out F1 0.615, per-frame AUC 0.85?0.91, +0.24
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:109: fps-norm + per-video
calibration still on the table) ? the trend, not the absolute, is the signal.
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:160: ranging 6?96 rallies;
the per-video learned-vs-heuristic gap on the hard videos is the real signal, and it's
decisive.
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:163:Honest cross-video LOVO
mean, heuristic vs learned, by corpus size:
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:167:| heuristic
(`calibrate_local`) | 0.443 | 0.544 | 0.463 | 0.480 ± 0.242 |
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:170:- The learned model has
led since n=3 and holds ~+0.10 (n=6: learned 0.583 vs heuristic 0.480). It stays robust as
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:250:- Keep growing the
corpus (more varied matches tomorrow) ? re-run LOVO + prototype; per-video threshold
calibration
```

44. assistant (2026-06-10T06:07:54.679Z)

I have everything I need. Compiling the audit now.

45. user (2026-06-10T06:09:54.257Z)

Structured output provided successfully

46. assistant (2026-06-10T06:10:10.100Z)

Audit complete. Structured output delivered with 14 findings (2 high, 7 medium, 4 low, 1 info) and 9 evidenced strengths, including three empirically verified bugs: the `iou_threshold=0.0` perfect-score boundary bug in `match_intervals`, the order-dependent tie handling in `rally_seq_proto.roc_auc`, and a greedy-vs-optimal matching example that halves TP count.